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Huawei Certified Kunpeng Application Developer Tutorial Series

HCIA-Kunpeng Application Developer

Student Manual

Version: 2.0



Huawei Technologies Co., Ltd.

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Huawei Certification System Introduction:

Huawei Certification is a certification system built by Huawei based on the "Platform + Ecosystem" strategy and centered around the new ICT technology architecture of "Cloud-Pipe-Device" collaboration. It covers the entire field of ICT (Information and Communications Technology) and includes three types of certifications: ICT infrastructure certification, basic software and hardware certification, and cloud platform and cloud service certification.

Based on the learning and advancement needs of ICT professionals, Huawei certification is divided into three levels: Engineer, Senior Engineer, and Expert.

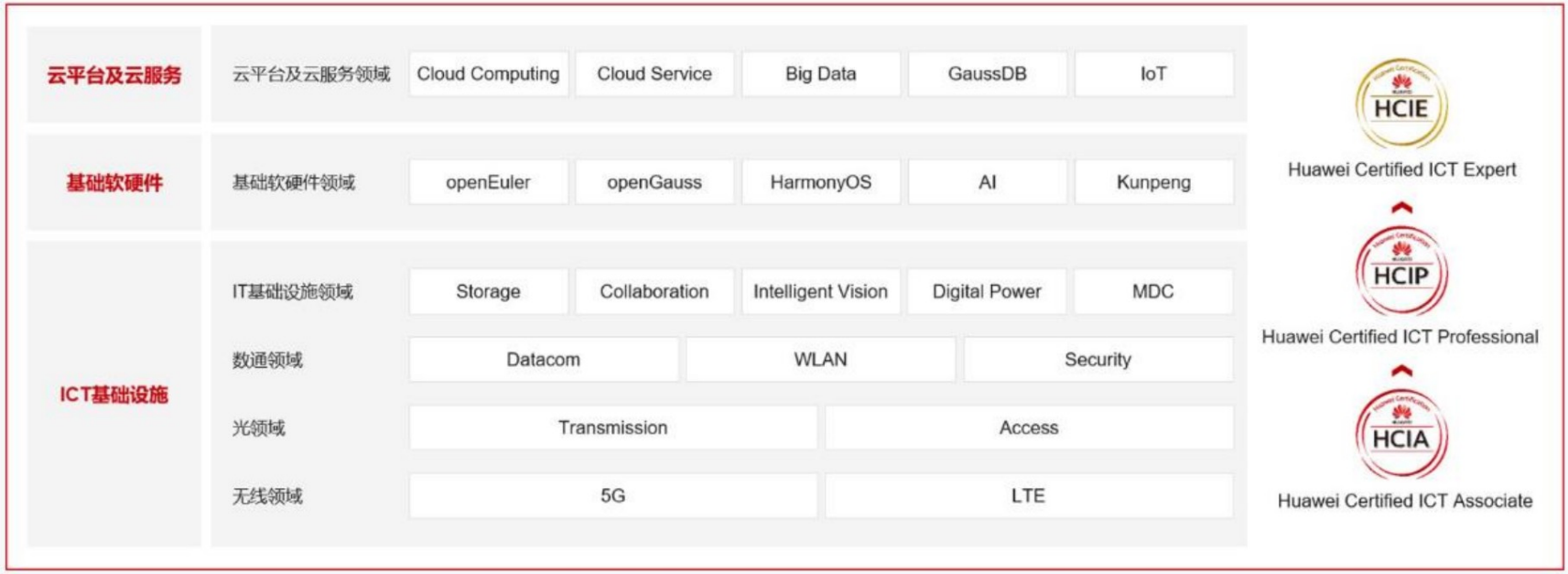
Huawei certifications cover the entire ICT field, align with the technological trend of ICT convergence, and are committed to providing a leading talent development system and certification standards to cultivate new ICT talents for the digital age and build a healthy ICT talent ecosystem.

The HCIA-Kunpeng Application Developer (Huawei Certified ICT Associate - Kunpeng Application Developer) program aims to cultivate and certify engineers who can use the Huawei Kunpeng computing platform for business deployment and migration, performance testing and optimization, and the design and planning of common solutions.

By obtaining the HCIA-Kunpeng Application Developer V2.0 certification, you will master the methods of using and maintaining the Kunpeng computing platform, possess the ability to manage the entire lifecycle of applications on the Kunpeng computing platform, and be competent for application development and operation and maintenance positions on the Kunpeng platform.

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Foreword

Introduction

This book is intended for HCIA-Kunpeng Application Developer certification trainees. It is suitable for students preparing for the HCIA-Kunpeng Application Developer exam or readers who wish to learn about the Kunpeng ecosystem, application migration principles, migration tool usage, Kunpeng application optimization methods, Huawei Kunpeng solutions, the Kunpeng community, and other related technical knowledge.

Content Description

This student manual contains 6 chapters, starting with an introduction to the Kunpeng system, and then introducing the Kunpeng ecosystem, Kunpeng application porting, application performance testing and optimization, comprehensive experiments on the Kunpeng instant messaging platform, Huawei Kunpeng solutions, and related content of the Kunpeng community.

Chapter One introduces the Kunpeng system, specifically including an introduction to the Kunpeng ecosystem, the Huawei Kunpeng processor, and Taishan services. Introduction to the server, the openEuler operating system, and the openGauss database.

Chapter Two covers Kunpeng application porting, specifically including software migration principles and processes, Kunpeng migration guidelines, and container migration guidelines. Guide, common migration issues and solutions, Kunpeng developer support suite.

Chapter Three covers application performance testing and optimization, specifically including software performance testing, Kunpeng general performance optimization methods, and Kunpeng performance... Analysis tools.

Chapter Four is a comprehensive experiment on the Kunpeng Instant Messaging Platform, which includes an introduction to the Kunpeng Instant Messaging Platform, deployment of the basic experimental environment, migration of Kunpeng code, setup of the Kunpeng Instant Messaging Platform, and data processing based on Hive.

Chapter Five covers the Huawei Kunpeng solution, including a comprehensive overview of the Huawei Kunpeng solution and general solutions for Huawei Kunpeng. case.

Chapter Six introduces the various sections of the Kunpeng Community, including the positioning and role of the Kunpeng Community, introductions to the various sections of the Kunpeng Community, the Kunpeng Development Kit (Devkit), the Kunpeng Application Enablement Kit (Boostkit), the Kunpeng Developer Learning Column, the Kunpeng Ecosystem Partner Network and Partner Program, and support and information.

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1 Introduction to the Kunpeng System

1.1 Introduction to the Kunpeng Ecosystem

1.1.1 Development Trends of the Computing Industry

1.1.1.1 The Challenges of Big Data Technology to Computing Power

The computing industry is the foundation of IT technology and the driving force behind every industrial revolution. From cloud computing, big data, and artificial intelligence to blockchain, edge computing, and the Internet of Things, all rely on the support of powerful computing capabilities.

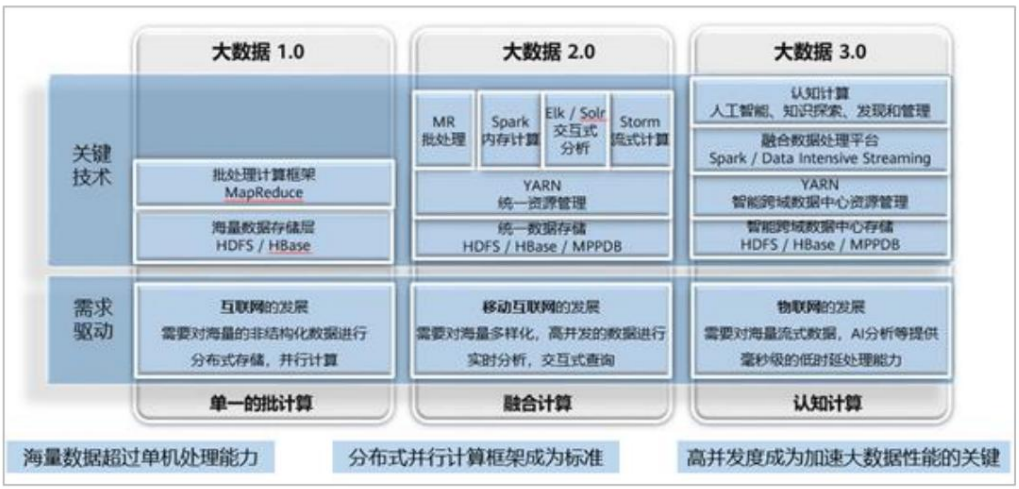
We will now focus on the three stages of big data and examine in detail the challenges that big data technology poses to computing power:

Big Data 1.0 Era: The main demand driver at this stage is the development of the Internet. Compared with traditional enterprise electronic office, a large amount of unstructured data has emerged, which puts forward high requirements for distributed storage and parallel technology of data.

Google is a typical representative of the traditional Internet, having published important big data papers such as GFS, MR, and BigTable, thus ushering in the first year of big data.

The Big Data 2.0 Era: With the booming development of the mobile internet, internet applications are far richer than those of the traditional internet (PC access), and the structure of data has also changed significantly. This has led to a substantial increase in the demands for real-time processing capabilities and high concurrency. Mobile banking is a prime example. Modern mobile banking apps offer a wide range of content, including not only traditional online transactions but also news, entertainment, and shopping. Therefore, the data processing capabilities typically require the combined use of multiple technologies, such as batch processing, in-memory computing, interactive queries, and streaming real-time processing. This has driven the development of technologies like MapReduce, Spark, Solr, Storm, and YARN.

The Big Data 3.0 Era: With the development of the Internet of Things, the future will be a completely digital world where everything will be connected to the network. The key is to connect these things and discover the underlying patterns, thereby creating greater value for human life and production. This is the big data processing of the artificial intelligence era.



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图1-1 Three stages of big data development

1.1.1.2 The Challenges of Artificial Intelligence to Computing Power

Current artificial intelligence refers more to intelligent individuals, specifically those capable of making single-point decisions intelligently through their own continuous learning capabilities.

The three conditions for the AI industry to explode are: algorithms, computing power, and data.

Machine learning requires vast amounts of historical data, making data collection ubiquitous and its growth exponential. Currently, both storage and communication are centralized, making it impossible for a single, massive point of contact to support such a large scale in the future. Storage and communication capabilities will inevitably become bottlenecks.

Computing costs are also a major pain point in the artificial intelligence industry. Current AI companies are making huge investments in hardware, and AI has a very high demand for computing power. Therefore, the requirements for high-performance computing and customized deep learning chips are very high, which means that many companies have to spend a lot of money to buy computing power and build many computing centers, resulting in a great waste of resources.

The evolution of AI computing trends also faces significant challenges: as the required accuracy of models increases, the computational demands also rise. Future algorithm development will pose even greater challenges to overall computing needs, making it crucial to improve the performance and efficiency of the entire AI computing system.

Furthermore, the failure of Moore's Law has led to a bottleneck in CPU performance improvements. Intel announced the official discontinuation of the "Tick-Tock" processor development model, shifting its future development cycle from two to three years. The improvement in single-CPU performance is slowing, traditional servers are struggling to meet the demands of parallel computing power, and server CPU shipments have stagnated.

Overall, AI faces enormous computational challenges, making it particularly important to improve the performance and efficiency of AI computing systems, which requires comprehensive consideration from a system perspective.

1.1.1.3 The diversity of applications and data necessitates new computing architectures.

In the future, the amount of information will be enormous, computing will be ubiquitous, and computing applications will be diverse, ranging from robotic vacuum cleaners to smartphones, smart homes, IoT, and autonomous driving. This diversity of scenarios leads to a diversity of data, including numbers, text, images, videos, graphics, structured data, and unstructured data.

Integer calculations are suitable for scenarios such as text processing, storage, and big data; floating-point calculations are suitable for scenarios such as scientific computing and video processing.

1.1.1.4 Emerging computing industries call for new computing power

The computing industry is currently showing two major trends:

1. Mobile smart devices are replacing traditional PCs: Global shipments of traditional PCs reached 250 million units in 2018, marking the seventh consecutive year of decline. Conversely, global shipments of mobile smart devices exceeded 1.6 billion units in 2018. This has led to a shift in computing from x86 to ARM; and applications are moving from PC applications to mobile applications and then to cloud-based mobile applications.
2. The world is entering an era of the Internet of Things: In 2018, the number of connected devices worldwide exceeded 23 billion; by 2025, this number is expected to surpass 100 billion, generating massive amounts of data. For example, autonomous driving generates 64TB of data daily, and XX Safe City generates 1000TB of data daily.

The computational power requirements for processing massive amounts of data:

1. Real-time intelligent processing at the edge requires AI computing power;
2. Data centers analyze, process, and store massive amounts of data, requiring high concurrency, high performance, and high throughput computing power.

Diverse computing methods meet computing power requirements

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As just mentioned, computing has entered an era of diversity, which will bring about a large demand for heterogeneous computing. No single computing architecture can meet the processing needs of all scenarios and all data types. For example, we see various CPUs, DSPs, GPUs, AI chips, etc.

The simultaneous presence of FPGAs and other components offers the following advantages:

1. Heterogeneous computing, where multiple computing architectures coexist, is the optimal solution.

2. Heterogeneous computing, where multiple computing architectures coexist, will reshape Moore's Law, thereby better meeting the diverse needs of various services and data.

According to diverse needs.

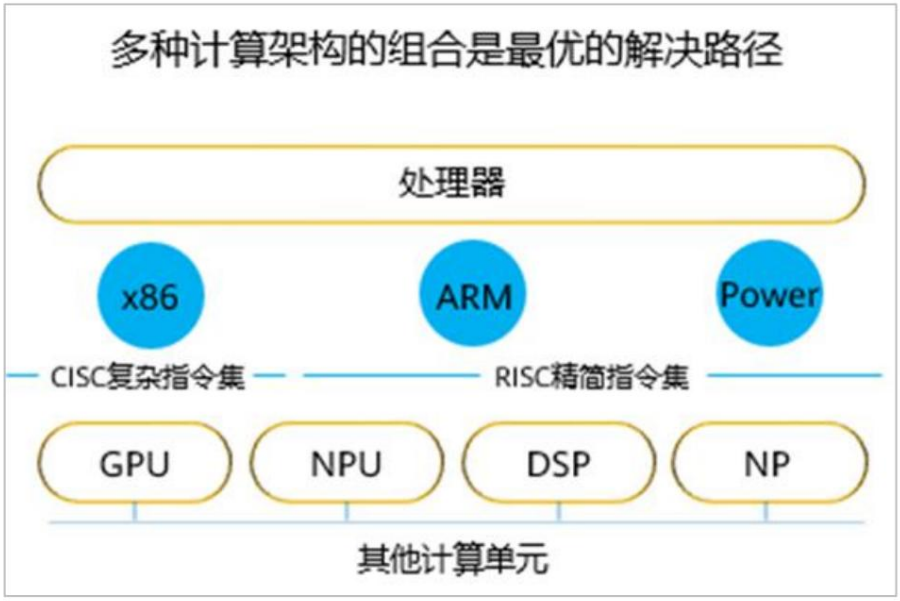
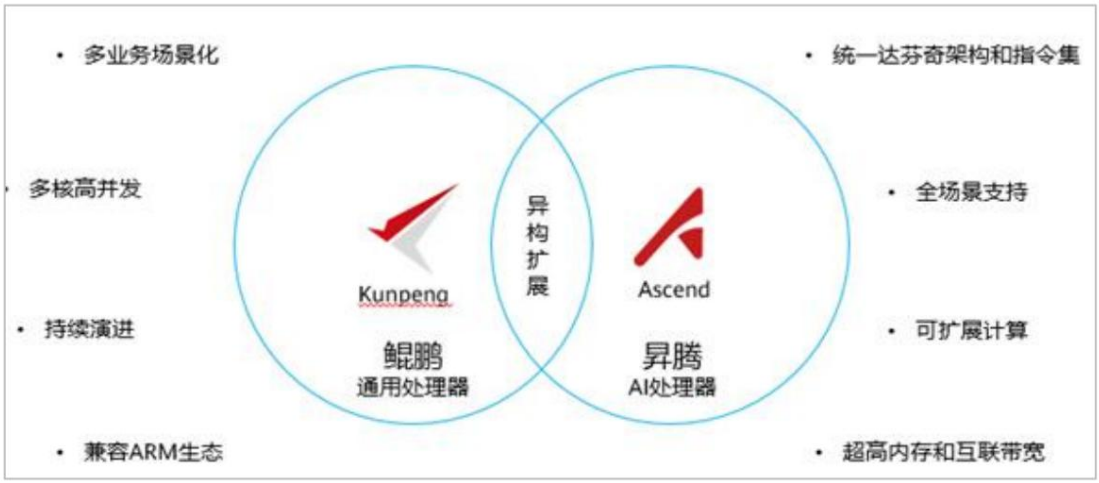


图1-2 Diversified computing power

Huawei is focusing on combining Kunpeng and Ascend technologies to create core computing power.



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图1-3 Kunpeng + Ascension

1.1.2 Introduction to Kunpeng

1.1.2.1 What is Kunpeng?

In the previous section, we mentioned that Huawei's main strategy is to combine Kunpeng and Ascend to improve computing power. So what exactly is Kunpeng? The following diagram will provide a clearer picture:

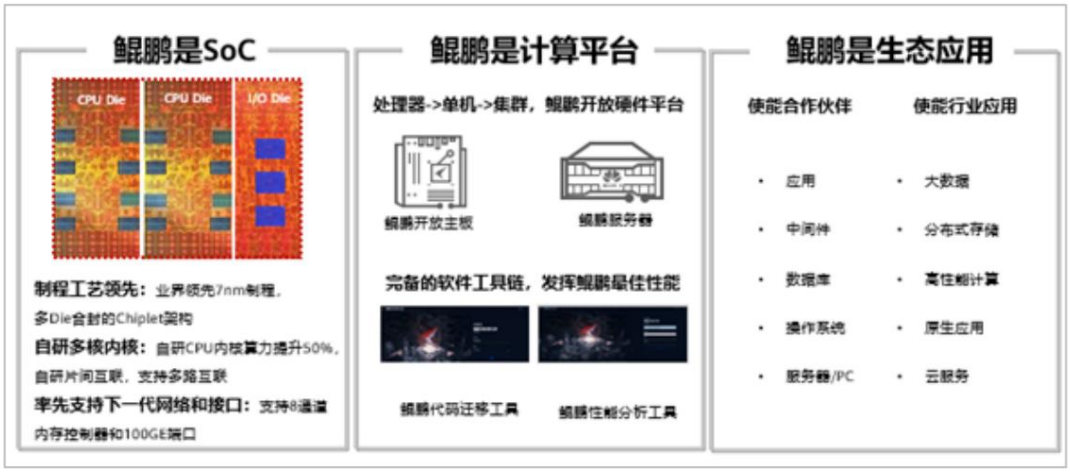


图1-4 What is Kunpeng?

It can be seen that Kunpeng is not only a chip, but also a computing platform, and can also be an ecosystem application.

1.1.3 Overview of the Kunpeng Computing Industry

1.1.3.1 The computing industry market with a scale exceeding one trillion yuan

New applications, new technologies, and new computing architectures, along with billions of connections and explosive data growth, will reshape the ICT industry landscape, spawn new computing industry chains, and give rise to new manufacturers and new ecosystems.

Software : Operating systems and virtualization software, databases, middleware, big data platforms, enterprise application software, cloud services, and data center management services;

Hardware : Servers and components, enterprise storage devices;

Cloud services and data center management services.

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图1-5 Calculate industry scale

1.1.3.2 Overview of the Kunpeng Computing Industry

The Kunpeng computing industry is a full-stack IT infrastructure, industry applications, and services built on the Kunpeng processor, including PCs, servers, storage, operating systems, middleware, virtualization, databases, cloud services, industry applications, and consulting and management services.



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图1-6 Kunpeng computing industry

1.1.3.3 Kunpeng Processor - Committed to Continuous Innovation

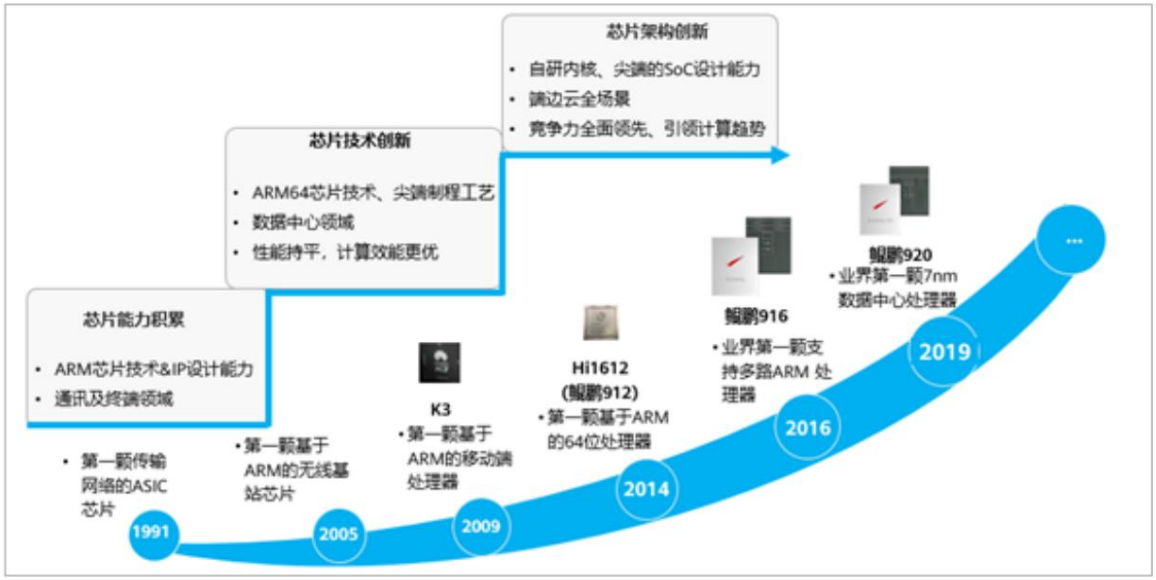


图1-7 Kunpeng processor evolution

Huawei has been exploring the field of chips and processors for more than 20 years.

In 1991, the first ASIC chip for transmission networks was developed.

In 2005, the first wireless base station chip based on the ARM architecture was developed.

In 2009, the K3 smartphone CPU was launched, the predecessor of the Kirin chip. Today, the Kirin chip has brought mobile phones into the smart era.

In 2016, the first generation of Kunpeng processors was launched, namely the Kunpeng 916 processor, which is designed for data center applications. Server CPUs, the predecessors of Kunpeng chips, were also launched.

In 2019, the Kunpeng 920 was launched, which was the industry's first 7nm data center processor.

Huawei invests more than 10% of its annual sales revenue in R&D, with more than one-third of that investment in chips/SoC (System on a Chip). This has enabled Huawei's chip R&D capabilities to achieve a leapfrog improvement from initial technology accumulation to technological innovation and then to architectural innovation.

In the future, the Kunpeng processor will maintain its product strategy of long-term investment, comprehensive layout, forward compatibility, and continuous evolution, mass-producing one generation, developing another, and planning the next.

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1.1.3.4 Comprehensive Chip Innovation - Building a Fully Self-Developed Server TaiShan

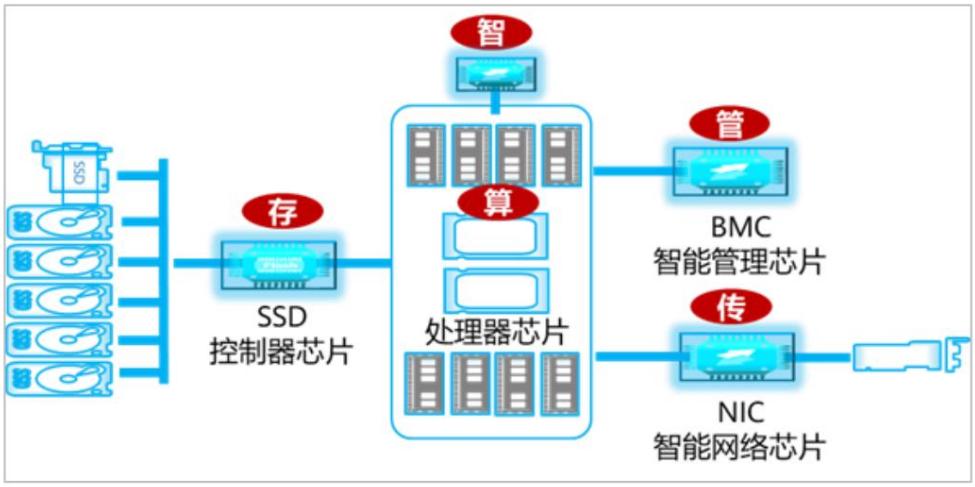


图1-8 Multifunctional Fusion Chip

It covers five aspects: computing, storage, transmission, management, and intelligence. Corresponding examples include the well-known Huawei Kunpeng 920, the first 7nm ARM server processor with 32/48/64 cores and a 2.6GHz clock speed; the Hi1812, an intelligent SSD controller chip that achieves intelligent acceleration through the integration of PCIe NVMe and SAS; the Hi1822, an intelligent converged network chip that accelerates protocols through the integration of Ethernet and FC; the Hi1710, an intelligent management chip with a built-in intelligent management engine, suitable for intelligent fault management scenarios; and the Ascend 310/910, artificial intelligence chips using the Da Vinci architecture to improve energy efficiency and performance.

1.1.3.5 Advantages of Kunpeng Computing Industry

Within the framework of the Kunpeng computing industry, upstream and downstream manufacturers in the industry chain can develop products and solutions with differentiated advantages based on the existing ecosystems of Kunpeng and Arm.

As a member of the Kunpeng computing industry, Huawei possesses key technologies in Arm64 processor cores, microarchitecture, and chip design, holds a permanent Armv8 architecture license, and has developed the Kunpeng series of processors based on this. Furthermore, Huawei's long-term investment and continuous innovation in the chip field have enabled it to create a comprehensive and differentiated chip system for computing, storage, transmission, management, and AI, capable of meeting the computing power foundation needs that support the continued development of the Kunpeng computing industry.

The primary task of the Kunpeng computing industry is to incubate and improve industry-specific applications based on the needs of users in various sectors, expand industry space, and find matching production factors and market opportunities for all manufacturers. Through commercial success, it aims to promote continuous innovation across all links of the industry chain, forming a spiral-like upward development path. As the Kunpeng computing industry grows and expands in the Chinese market, it will attract more domestic and international companies, ultimately becoming a computing industry with continuous innovation capabilities and global leading advantages.

1.1.3.6 Typical Application Scenarios of the Kunpeng Computing Industry

The growth and expansion of the Kunpeng computing industry is inseparable from the joint efforts of upstream and downstream partners in the industry. Over the past few years, Huawei has worked with ISVs from various industries to incubate and improve five major Kunpeng computing solutions, achieving large-scale commercial application in multiple industries.

For example, in big data scenarios, the Kunpeng 920 processor supports up to 64 cores. This multi-core advantage means that more data can be processed in parallel. Real-world test data shows that the Kunpeng processor's big data performance is more than 30% higher than that of traditional platforms.

In distributed storage scenarios, the Kunpeng processor improves the operating efficiency of distributed storage software through its built-in data compression engine and multi-core parallel computing. Compared with traditional processor algorithm solutions, it reduces compression time by 66% and achieves an IOPS performance improvement of more than 20%.

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From the initial design stage of the Kunpeng processor, Huawei fully considered how to meet the needs of distributed parallel computing. Building upon the inherent multi-core nature of Kunpeng, it enhances the design with high I/O bandwidth, high memory bandwidth, and high data throughput, better meeting the development trend of IT distributed evolution.



图1-9 Typical application scenarios

1.1.4 Introduction to the Kunpeng Ecosystem

1.1.4.1 Panoramic View of the Kunpeng Computing Industry Ecosystem



图1-10 Kunpeng Ecosystem Panorama

Huawei is working with industry technology vendors to create an open and inclusive Kunpeng computing industry ecosystem. Together with partners, Huawei provides targeted solutions for the industry, provides a platform for developers to exchange and compete through community building and developer competitions, and actively carries out cooperation with universities to cultivate talent.

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1.1.4.2 Industry Ecosystem - Overview of the Kunpeng Computing Industry Development



图1-11 Kunpeng Industrial Ecosystem

The goal of the Kunpeng computing industry is to establish a comprehensive system of developers and industry talent. Through industry alliances, open-source communities, the Kunpeng Innovation Center flagship store, and industry standards organizations, we will improve the industry chain, connect the entire industry stack, and make the Kunpeng ecosystem the first choice for developers and users.

1.1.4.3 University Cooperation - Joining hands with universities to cultivate Kunpeng computing talents



图1-12 University Cooperation Program

The development of every emerging industry is inseparable from talent cultivation. Huawei has launched the "1234 Kunpeng University Talent Program" for Chinese universities to promote the cultivation of talent in the computing industry.

1. Investing 10 Million Yuan: In 2019, Huawei will invest 10 million yuan in software through the Ministry of Education's Industry-University Collaborative Education Project.
   Hardware resources: Collaborate with university experts to develop experimental resources for courses such as computer architecture, operating systems, databases, and artificial intelligence;

2. Serving two major groups: Teachers and students. Specifically, for teachers, jointly develop no fewer than 20 high-quality courses.
   The program includes courses and digital teaching materials, as well as teacher training and technical seminars related to the Kunpeng ecosystem; for students, it launches innovation training camps and elite talent programs, and regularly holds various innovation competitions.

3. Collaboration with three types of organizations: Huawei will collaborate with three types of educational expert organizations—the Teaching Steering Committee (Computer Science and Technology Teaching Steering Committee, Software Engineering Teaching Steering Committee, etc.), the Alliance (National Exemplary Software College Alliance, Information Technology New Engineering Industry-University-Research Alliance, etc.), and the Academic Societies (China Computer Federation, China Association of Higher Education, etc.)—to formulate talent training plans for key core technology fields.

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The plan includes developing and improving professional standards, designing and developing industry-academia collaborative courses, building and promoting open-source communities for the computing industry, and exploring new models for talent cultivation cooperation in the Kunpeng computing industry.

4. Cultivate four types of talent: based on four Huawei certification standards: "Kunpeng Application Developer Certification", "GaussDB Data..." The "Database Certification," "Intelligent Computing Certification," and "Artificial Intelligence Certification" support Chinese universities in cultivating talent needed for the Kunpeng ecosystem and fostering talent for the computing industry.

1.1.4.4 Partner Ecosystem - Kunpeng Partner Program

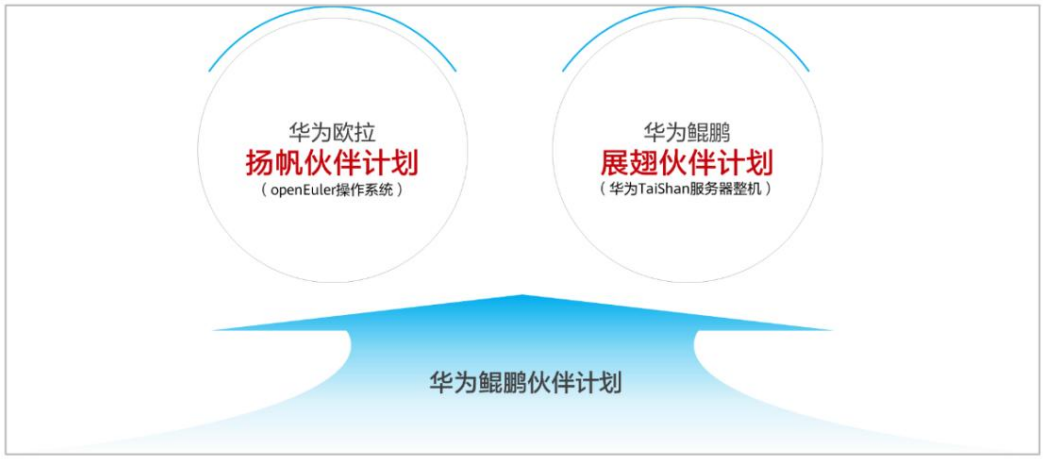


图1-13 Kunpeng Partner Program

Euler Sailing Partner Program: The Euler Sailing Partner Program (Huawei) is an ecosystem partner program launched by Huawei around openEuler. It aims to ensure the healthy and orderly development of the openEuler ecosystem, accelerate the development of products and solutions by ecosystem partners based on the openEuler series of products and components, and enable the development and commercial success of the partner ecosystem. By joining us, you can gain:

1. openEuler professional certification training services and exam vouchers.

2. Kunpeng Ecosystem Innovation Center Partner Resources.

3. Partner resources of the Euler Ecosystem Innovation Center.

4. Marketing materials and exhibitions may use Huawei Partner Level logos and certificates. 5. Priority access to participation and brand exposure in openEuler industry marketing events.

6. Obtain priority in project recommendations in market opportunities led by Huawei or Huawei sales partners.

Kunpeng Soaring Partner Program: The Kunpeng Soaring Partner Program is an ecosystem partner program launched by Huawei around the Kunpeng computing platform. It aims to help partners integrate the full stack of technologies and brand resources of the Kunpeng computing industry, build product and solution competitiveness, connect customers with the Kunpeng ecosystem, and create unlimited business opportunities. By joining this program, partners can receive:

1. Partner technical documentation download.

2. Resources for building the Kunpeng Ecosystem Innovation Center.

3. Huawei Partner Level Logo and Certificates may be used in marketing materials and exhibitions.

4. Support partner marketing activities.

5. Obtain priority in project recommendations in market opportunities led by Huawei or Huawei sales partners.

6. Provide equivalent technology certification support services.

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1.1.4.5 Community Ecosystem - Building a One-Stop Cloud-Based Kunpeng Zone



图1-14 Community Ecology

1.1.4.6 Developer Ecosystem - Fertile Soil Project 2.0

The Fertile Soil Program mainly provides services such as training and empowerment, Fertile Soil Universities, Huawei certification, Kunpeng Innovation Center flagship stores, joint marketing, innovation funds, and Fertile Soil Factories. The program plans to invest US$1.5 billion in the industrial ecosystem development from 2019 to 2024.

1.2 Introduction to Huawei Kunpeng Processor

1.2.1 Introduction to Processor Architecture

The traditional CISC (Complex Instruction Set Computer) architecture is extremely complex in hardware due to its large instruction set, variable instruction length, and varying instruction execution cycles. This poses significant challenges to chip design, development, and cost reduction.

As computer technology advances, new and complex instruction sets are constantly being introduced, leading to increasingly complex computer architectures to support these new instructions. However, the frequency of use varies significantly among the various instructions in the CISC instruction set. Approximately 20% of the instructions are used repeatedly, accounting for 80% of the entire program code. The remaining 80% are used infrequently, representing only 20% of the program design. Clearly, this structure is not very reasonable.

In response to these obvious weaknesses, the University of California, Berkeley proposed the concept of RISC (Reduced Instruction Set Computer) in 1979. RISC is not simply about reducing instructions, but focuses on how to make the computer's structure simpler and more reasonable to improve the computing speed.

RISC architecture prioritizes the use of the most frequently used simple instructions and avoids complex instructions; it fixes the instruction length and reduces the types of instruction formats and addressing modes; and it relies primarily on control logic, using little or no microcode control to achieve the above objectives.

Comparing the x86 and ARM architectures, it can be seen that the ARM architecture has better concurrency performance.

Currently, over 90% of mobile devices use ARM architecture processors. With the development of IoT, AI, and cloud computing, ARM's dominant position in the terminal market will drive its entry into the data center market, becoming the next rapidly growing market segment.

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1.2.2 Overview of Huawei Kunpeng Processor

Chips are the foundation for building various computing products, enabling upper-layer software and applications, and driving the sustainable innovation and development of the entire industry chain. As the foundation of the Kunpeng computing industry, Huawei will continue to invest heavily in the Kunpeng processor, adhering to an evolutionary pace of mass-producing one generation, developing another, and planning the next. This will implement the fundamental strategy of "long-term investment, comprehensive layout, backward compatibility, and continuous evolution," efficiently meeting the market's demand for new computing power by providing the industry with a chip family centered on the Kunpeng series processors and corresponding products. The Huawei Kunpeng processor is an enterprise-level series of processors independently developed by Huawei based on the ARM architecture, encompassing five product system architectures: computing, storage, transmission, management, and intelligence.

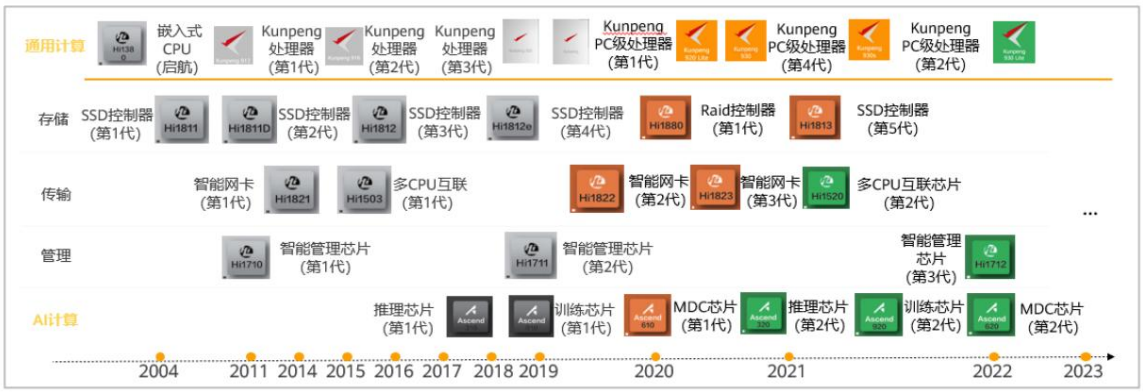


图1-15 Huawei Kunpeng processor evolution

Features of Huawei Kunpeng processor architecture:

Advantages:

1. Utilizing the ARM architecture, it achieves the same functionality and performance with a smaller chip area, lower power consumption, higher integration, and more hardware...
   CPU cores offer better concurrency performance.

2. Supports 64-bit instruction sets, ensuring good compatibility with various application scenarios from IoT and terminals to the cloud.

3. Extensive use of registers, with most data operations performed in registers, resulting in faster instruction execution.

4. It adopts the RISC instruction set, with fixed instruction length, flexible and simple addressing modes, and high execution efficiency.

ÿ Disadvantages:

1. It is still a new entrant in the data center field, and its ecosystem is still developing.

1.2.3 Huawei Kunpeng Processor Models and Specifications

Huawei began developing its own chips based on ARM technology in 2004. In the field of general-purpose computing processors, it released the Kunpeng 912 processor in 2014, the Kunpeng 916 processor in 2016, and the Kunpeng 920 processor in January 2019. The Kunpeng 920 processor is the industry's first data center-grade ARM architecture processor using a 7nm process.

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图1-16 Kunpeng processor models

1.2.4 Technological Innovation of Huawei Kunpeng Processor

ÿ Kernel entirely self-developed

ÿMemory /Network Interface & I/O Protocols

Leading manufacturing process

ÿReliability Improvement

1.3 Introduction to Taishan Server

1.3.1 Introduction to General Server Knowledge

1.3.1.1 What is a server?

A server is a type of computer that provides shared resources (including querying, storage, and computing) to client computers (such as PCs) in a network environment under the control of a network operating system. Its high performance is mainly reflected in its high-speed CPU computing power, long-term reliable operation, and powerful I/O external data throughput capabilities.

1.3.1.2 Server Development History

The world's earliest computer was the abacus, invented in China 2,600 years ago .

ÿ In 1964, IBM launched the first mainframe – the System/360, becoming a true server. ÿ In 1989, Compaq launched the world's first IA architecture server: the System Pro, using the Intel 486 microprocessor. ÿ In the 1990s, minicomputers emerged to cater to the low-cost needs of small and medium-sized enterprises.

The core component of a server , the CPU, has consistently followed Moore's Law, with the vast majority of servers using an x86-based architecture.

Intel processor

1.3.1.3 Server Composition

A server mainly consists of three major components: CPU, memory, and hard drive, along with basic hardware such as power supply, motherboard, and chassis, to provide information services.

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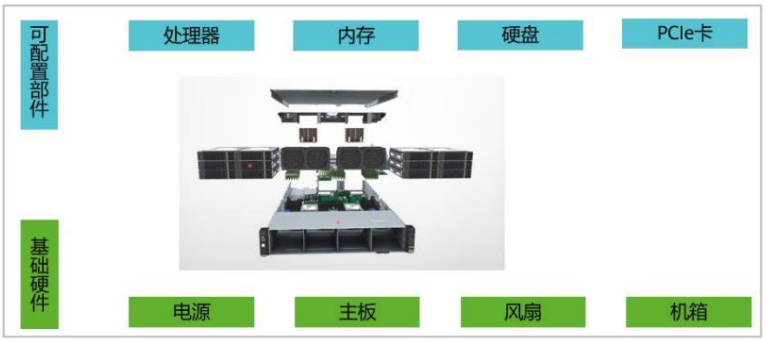


图1-17 Server composition

1.3.1.4 Mainstream Server Types

ÿClassified by CPU instruction set

ARM servers, x86 servers

Classified by server form factor

Tower, rack, blade, high-density

ÿClassified by load type

Database server, application server, web service, access server, file server

ÿClassified by application level

Entry-level servers, workgroup servers, departmental servers, enterprise servers

1.3.2 Taishan Server Overview

1.3.2.1 Building overall computing power based on Huawei Kunpeng processor

In 2016, the first-generation Taishan 100 server was launched, based on the Kunpeng 916 processor. It uses Huawei's Kunpeng processor as the foundation to build the overall computing power of the machine.

Launched in 2019, the Taishan 200 server is based on the Kunpeng 920 processor and remains a flagship product in the market.

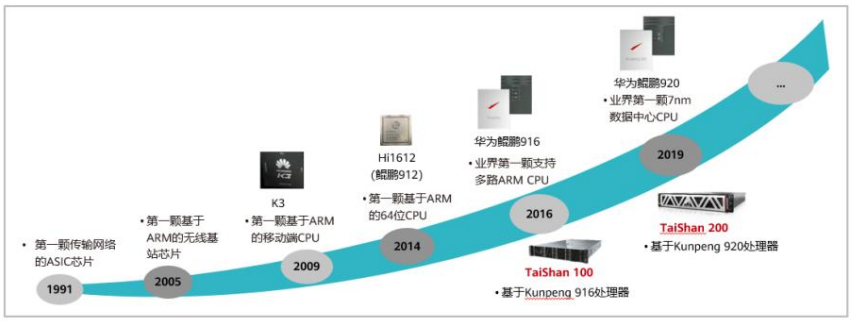


图1-18 **Taishan** Server Evolution

Taishan servers have the following characteristics:

High -performance computing: Providing high-performance Huawei Kunpeng processors, TaiShan servers, and solutions compatible with ARM architecture.

The plan aims to bring high-performance computing to data centers.

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ÿSafe and reliable: Huawei Kunpeng processor is based on a self-developed kernel, and TaiShan server computing chips are all self-developed.

ÿOpen Ecosystem: Provides open platform functionality, supporting mainstream software and hardware in the industry; builds the Kunpeng ecosystem, collaborating with developers, partners, and...

Industry organizations work together to build a new foundation for intelligent computing

1.3.3 Introduction to Taishan 200 Rack Server

Based on the Kunpeng processor, Huawei has created the TaiShan server, a benchmark for Kunpeng computing systems. The TaiShan 200 server fully utilizes engineering technologies such as liquid cooling, high-speed interconnect, reliability design, and quality control, making it a premium product and setting an industry benchmark for the application of Kunpeng processors in data center servers.

TaiShan servers currently available for commercial use include the 2280 Balanced, 5280 Storage, and X6000 High-Density models.

1.4 Introduction to the openEuler Operating System

1.4.1 Introduction to the Linux Operating System

1.4.1.1 Introduction to Operating Systems

An operating system is a program that manages computer hardware and software resources; it is also the kernel and foundation of a computer system. The operating system is responsible for fundamental tasks such as managing and allocating memory, determining the priority of system resource supply and demand, controlling input and output devices, operating networks, and managing file systems. The operating system also provides an interface for users to interact with the system. From small smart wearable devices and TV boxes to large computers, servers, and routing and switching equipment, each has its own operating system.

1.4.1.2 Origins of the Linux Operating System

On October 5, 1991, Linus Torvalds officially announced Linux at the comp.os.minix news organization .

The kernel was born and made open source.

Thanks to the efforts of participants in the open-source community, the Linux kernel version 1.0 was officially released in March 1994.

Linux is a free and open-source Unix-like operating system, based on POSIX and Unix.

A user-friendly, multitasking operating system that supports multithreading and multiCPU.

The components of a Linux operating system:

The components of a Linux operating system: the Linux kernel and applications.

Common Linux operating systems: Red Hat, CentOS, Fedora, Kaili Linux, Debian, Ubuntu

Deepin, NeoKylin, etc.

1.4.1.3 Introduction to CentOS

CentOS Linux distribution was a stable, predictable, manageable, and reproducible platform compiled from the source code released by Red Hat Enterprise Linux (RHEL) under the open-source license (mostly GPL). On December 8, 2020, CentOS officially announced the discontinuation of the CentOS Linux project, leaving current CentOS users facing questions about how to update, maintain, and migrate their systems after the project's discontinuation.

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Note: TaiShan servers support Cent OS 7.4 for ARM and above, but not all servers are compatible with these versions. Please check the Smart Computing Product Compatibility Query Assistant for specific compatibility information.

1.4.1.4 Introduction to Debian

Debian is the most GNU-compliant Linux distribution to date. It offers support for nearly 100,000 different open-source software programs, enjoys high recognition and usage rates, provides excellent support for various kernel architectures, boasts strong stability and security, and offers free technical support.

Due to its excellent stability and low resource consumption, Debian is widely used in virtual private servers.

1.4.1.5 Introduction to Ubuntu

Ubuntu is a derivative of Debian, and is an open-source and free operating system primarily focused on desktop applications.
Ubuntu focuses on system deployment on servers, cloud computing, and even some mobile devices running Ubuntu Linux.

Ubuntu has a large community, and users can easily get help from the community.

Note: TaiShan servers support Ubuntu 16.04.3 LTS ARM and above, but not all servers are compatible with these versions. Please check the Intelligent Computing Product Compatibility Query Assistant for specific compatibility information.

1.4.1.6 Introduction to Deepin

Deepin is a Linux distribution developed by Wuhan Deepin Technology Co., Ltd. Deepin features proprietary software including Deepin Software Center, Deepin Screenshot, Deepin Music Player, and Deepin Video Player, all using the proprietary deepinUI. Deepin focuses on providing users with the ultimate experience for daily office

work, study, life, and entertainment, and is suitable for laptops, desktop computers, and all-in-one PCs.

Note: TaiShan servers support deepin V15 for ARM and above, but not all servers are compatible with these versions. Please check the Intelligent Computing Product Compatibility Query Assistant for specific compatibility information.

1.4.1.7 Introduction to NeoKylin

NeoKylin is a Linux distribution released after the merger of NeoKylin and GalaxyKylin. The NeoKylin operating system uses an enhanced Linux kernel and is divided into desktop, general, advanced and security versions to meet the requirements of different customers. It has been widely used in industries such as energy, finance, transportation, government and central enterprises.

Note: TaiShan servers support NeoKylin Server v5.0 U5 for ARM and above, but not all servers are compatible with these versions. Please check the Intelligent Computing Product Compatibility Query Assistant for specific compatibility information.

表1-1 Commonly used **operating systems** in various industries

industry	Common **OS**
Telecom operators	CentOS, etc.
Banks, securities,	CentOS, NeoKirin, GalaxyKirin, etc.
Internet companies	customize OS, Debian, etc.

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Government, state-owned enterprises	CentOS, NeoKylin, GalaxyKylin, Tongxin, etc.
Power and energy:	CentOS, NeoKylin, GalaxyKylin, Ubuntu, etc.
Transportation systems	such as CentOS and Kylin OS
Anping	CentOS, etc.
Education, Healthcare	CentOS, Kylin OS, etc.

1.4.1.8 Huawei Kunpeng Processor OS Compatibility

表1-2 Kunpeng processor **OS** compatibility

	Common **OS**
Kunpeng	deepin
	ubuntu
	Galaxy Kirin
	suse
	Debian
	CentOS
	Zhongbiao Qilin

This section lists some operating systems compatible with the Huawei Kunpeng processor. Currently, the Kunpeng processor only supports Linux-type operating systems. The above lists some of the operating systems supported by the Kunpeng processor. For a more detailed list, please refer to the "Intelligent Computing Product Compatibility Query Assistant ": http://support.huawei.com/onlinetoolweb/ftca/index?serise=9.

1.4.2 Introduction to the openEuler Operating System

1.4.2.1 Overview of openEuler

openEuler is derived from EulerOS, a server operating system developed and used by Huawei since 2010. EulerOS is one of the Linux distributions and is named after the famous mathematician Leonhard Euler.

In September 2019, EulerOS was officially open source and named openEuler.

openEuler is an open-source operating system. Currently, the openEuler kernel is derived from Linux, supports Kunpeng and various other processors, and can fully unleash the potential of computing chips. It is a high-efficiency, stable, and secure open-source operating system built by global open-source contributors. openEuler is suitable for applications such as databases, big data, cloud computing, and artificial intelligence. At the same time, openEuler is also the name of a global open-source operating system community.

1.4.2.2 openEuler Technical Features - A-Tune Resource Tuning Automation

A-Tune is an AI-powered automatic tuning system that uses a non-intrusive system profiling approach to identify business needs and match the best resource model, responding in real time to changes in business characteristics.

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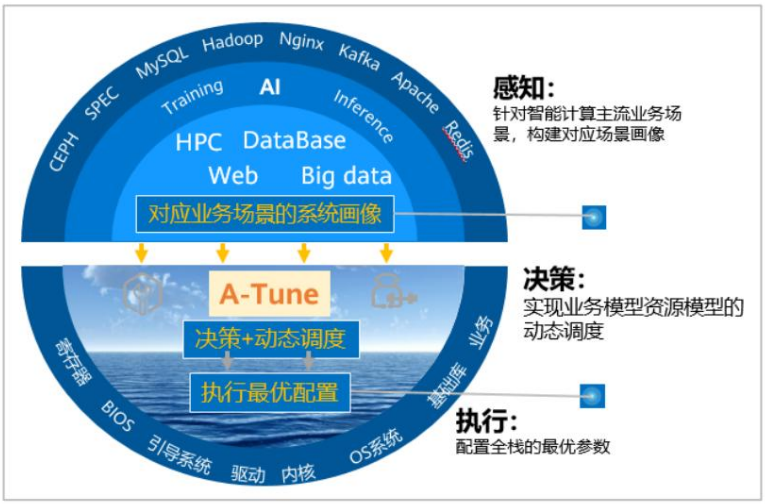


图1-19 A-Tune Resource Tuning Automation

A -Tune categorizes business workloads based on system profiles, identifies mainstream industry scenarios for Kunpeng, and enables resource model building and...

The system self-optimizes to leverage the computing power of Kunpeng.

ÿBased on machine learning, a system profile is created. By combining clustering and classification through multiple iterations, the business load is accurately categorized, covering the mainstream computing scenarios of current ICT businesses.

ÿBased on business characteristics, complete system resource modeling, rationally optimize system resource and parameter configurations, and maximize the value of system capabilities.

Great transformation.

1.4.2.3 openEuler Technical Features - iSulad Container Solution

iSula, the general-purpose container engine, is a new container solution that provides a unified architecture to meet the diverse needs of the CT and IT domains. Compared to Docker, written in Golang, lightweight containers are characterized by their lightness, agility, efficiency, and speed. They are not limited by hardware specifications and architecture, have lower noise overhead, and can be applied to a wider range of fields. iSula is the foundational platform for container execution provided in openEuler-20.03-LTS.

Containers can be divided into three models based on their deployment model: application containers, security containers, and system containers.

* Application Container: The most widely used container type, where only customer-defined applications run. openEuler-20.03-LTS integrates Moby version 18.09, with bug fixes and stability enhancements. * Secure Container: Essentially a virtual machine, but with an interface encapsulated as a container interface. openEuler-20.03-LTS integrates...

Kata-container version 1.7, based on the community version, has been strengthened in terms of stability and reliability, added support for heterogeneous computing, and is more suitable for production environments with Host CGroup configuration and DPDK/SPDK high-performance network and storage support.

ÿSystem Containers: Essentially containers, but used in the same way as virtual machines; they focus on solving the bulky issues of virtual machines.

The iSula container platform supports the creation of system containers and enables dynamic adjustment of devices and running resources within system containers, while providing superior user namespace isolation.

1.4.2.4 Kunpeng Hardware-Related Features – MPAM

MPAM (Memory Partitioning and Monitoring) is suitable for controlling/monitoring the allocation and use of L3 cache and memory bandwidth, and is comparable to the RDT features of high-end x86 chips.

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In openEuler version 21.03, resource management is also supported through Arm64 architecture cache QoS and memory bandwidth control technology.

1.4.3 Introduction to Embedded OS

1.4.3.1 Overview of Embedded Operating Systems

An embedded operating system (EOS) is an operating system used in embedded systems. It is a versatile type of system software that typically includes hardware-related low-level driver software, a system kernel, device driver interfaces, communication protocols, a graphical user interface, and a standardized browser. The embedded operating system is responsible for allocating all software and hardware resources, scheduling tasks, and controlling and coordinating concurrent activities within the embedded system. It must reflect the characteristics of the system it inhabits and be able to achieve the required functions by loading and unloading certain modules. Currently, widely used operating systems in the embedded field include: embedded real-time operating systems µC/OS-II, embedded Linux, Windows Embedded, VxWorks, and Android and iOS used in smartphones and tablets.

Currently, embedded operating systems in industries such as manufacturing, healthcare, and military/aerospace are basically monopolized by foreign companies. Although China has initially developed embedded software and hardware capabilities, they are characterized by being "small, scattered, and weak," lacking sufficient technical level and market competitiveness, and lacking a complete technical system.

1.5 Introduction to the openGauss Database

1.5.1 Database Overview

A database is a "repository for organizing, storing, and managing data according to a data structure." It is a large collection of organized, shareable, and uniformly managed data that is stored long-term in a computer.

relational database

The introduction of the relational model was a landmark event in the history of database development. The tremendous success of relational theory research and the development of relational database management systems further promoted the development of relational databases.

Main application scenarios of relational databases

ÿ Online Transaction Processing (OLTP): OLTP is the primary application of traditional relational databases, a system with very high transactional requirements. These are typically highly available online systems, primarily handling small transactions and queries, mainly basic, routine transaction processing, such as banking transactions. OLTP systems emphasize database memory efficiency; when evaluating such systems, the number of transactions executed per second and the number of SQL statements executed per second are generally considered. Emphasis is placed on command rate, bind variables, and concurrent operations across various memory metrics.

Online Analytical Processing (OLAP): OLAP is the primary application of data warehouse systems, supporting complex analytical operations, focusing on decision support, and providing intuitive and easy-to-understand query results. In such systems, the execution volume of statements is not a key performance indicator, as the execution time of a single statement can be very long, and the amount of data read can be enormous. OLAP systems emphasize data analysis, and performance evaluation criteria are often based on disk subsystem throughput (bandwidth), SQL execution time, disk I/O, and partitioning.

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1.5.2 Introduction to openGauss Database

openGauss is a world-leading enterprise-grade open-source relational database co-developed with partners. It offers extreme performance for multi-core processors, end-to-end business and data security, AI-based optimization, and efficient operation and maintenance capabilities. openGauss is fully user-friendly and open, released under the Magnolia Permissive License v2, and deeply integrates Huawei's years of R&D experience in the database field, continuously building competitive features to meet enterprise-level scenario needs.

1.5.2.1 Development History of openGauss

The development of openGauss can be divided into the following four stages:

ÿInternal use: From 2001 to 2011, it was used internally in multiple Huawei products and was referred to as an enterprise-level in-memory database.

ÿProductization : From 2011 to 2019, we launched official products for commercial use, such as the core data warehouse of Bank G and the commercial DWS Huawei Cloud; the core business system of Bank Z replaced the commercial database; and supported more than 40 main products within the company. More than 30,000 sets of these products were commercially deployed in more than 70 operators worldwide, serving more than 2 billion people globally.

ÿCloud & Open Source: On May 15, 2019, GaussDB was globally released, aiming to build a partner ecosystem, be compatible with mainstream industry ecosystems, and complete integration with industries such as finance; on June 30, 2020, openGauss was open-sourced. ÿEcosystem Building: Since 2021,

after being open-sourced, Huawei's main focus with openGauss has been sharing enterprise-level data management capabilities.

We will leverage our strength to guide ecological development and promote the advancement of database education.

1.5.2.2 openGauss Features

High performance: It provides concurrency control technology for multi-core architectures combined with Kunpeng hardware optimization, achieving a performance of 1.5 million tpmc in TPCC Benchmark under two Kunpeng paths; in response to the current trend of multi-core hardware architecture, it adopts NUMA-Aware data structures in key kernel structures; and provides SQL-bypass intelligent fast engine technology.

High Availability: Supports multiple deployment modes including primary/standby synchronous, asynchronous, and cascaded standby; data page CRC verification, with damaged data pages automatically repaired by the standby machine; standby machines can recover in parallel and become the primary service provider within 10 seconds.

High security: Supports fully encrypted computing, access control, encrypted authentication, database auditing, dynamic data masking and other security features, providing comprehensive end-to-end data security protection.

ÿEasy Operation and Maintenance: AI-based intelligent parameter tuning and index recommendation, providing AI automatic parameter recommendations; slow SQL diagnosis, multi-dimensional performance self-monitoring view, and real-time control over system performance; provides online self-learning SQL time prediction.

ÿFully Open: Adopts the Magnolia Permissive License, allowing free modification, use, and referencing of the code; database kernel capabilities.
Fully open; offering a wealth of partner certifications, training systems, and university courses.

1.5.2.3 SQL Definition

SQL (Structured Query Language) is a purpose-specific programming language used to manage relational database management systems or to perform stream processing in relational streaming data management systems.

SQL is based on relational algebra and tuple relational calculus, and includes a data definition language and a data manipulation language. The scope of SQL includes data insertion, querying, updating, and deleting; database schema creation and modification; and data access control.

openGauss supports the standard SQL92/SQL99/SQL2003/SQL2011 specifications, supports GBK and UTF-8 character sets, supports standard SQL functions and analytic functions, and supports stored procedures.

Introduction to SQL Syntax

ÿ DDL (Data Definition Language)

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Used to define or modify objects in a database. Examples include: tables, indexes, views, databases, sequences, users, roles, tablespaces, stored procedures, etc.

ÿ DML (Data Manipulation Language) is used to manipulate data in database

tables, such as inserting, updating, and deleting data.

ÿ DCL (Data Control Language) is used to set or change database

transactions, authorize operations (user or role authorization, permission revocation, role creation, role deletion, etc.), lock tables (supporting SHARE and EXCLUSIVE locking modes), and shut down the system.

ÿ DQL (Data Query Language)

It is used to query data within a database, such as retrieving data or merging the result sets of multiple select statements.

1.5.3 Introduction to Application Development

1.5.3.1 ODBC

Open Database Connectivity (ODBC) was developed to solve the problem of data sharing between heterogeneous databases. It has now become a major part of WOSA (The Windows Open System Architecture) and a database access interface standard based on the Windows environment.

ODBC provides a unified interface for accessing heterogeneous databases, allowing applications to access data managed by different DBMSs using SQL as the data access standard. It enables applications to directly manipulate data in the database, eliminating the need to change data as the database changes. ODBC can access database files on various computers, and even non-database objects such as Excel spreadsheets and ASCII data files.

1.5.3.2 JDBC

Java Database Connectivity (JDBC) is an application programming interface in the Java language that specifies how client programs access databases, providing methods such as querying and updating data in the database.
JDBC is also a trademark of Sun Microsystems. The JDBC we usually refer to is for relational databases.

1.5.3.3 Psycopg Psycopg

is a Python language interface for the PostgreSQL database; its main

advantage is its support for the full Python DBAPI 2.0. It is designed for heavily multithreaded applications; the Psycopg

distribution includes Zpsycopgda and the Zope database adapter.

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2 Kunpeng application porting

2.1 Software Migration Principles and Process

2.1.1 Software Migration Principles

In Chapter 1, we learned about Kunpeng hardware and the Kunpeng ecosystem. This Kunpeng application developer course aims to help developers join the Kunpeng ecosystem. Besides native development on Kunpeng, the focus is on migrating mature x86 applications to the Kunpeng platform and optimizing them for compatibility. Therefore, this chapter on Kunpeng application porting focuses on how to migrate applications originally running on the x86 platform to the Kunpeng platform and ensure their proper functioning.

Before discussing migration in the data center field, let's first take a common example: a Windows computer with an x86 architecture and an Android phone with an ARM architecture. For instance, transferring an executable program from a Windows system, such as putty.exe, to a phone is also a case of x86 to ARM migration. Obviously, putty, which runs normally on an x86 computer, will not run on a phone. Have you ever considered the underlying reason?

To answer this question, let's first look at the computer's technology stack and the execution process of a program.

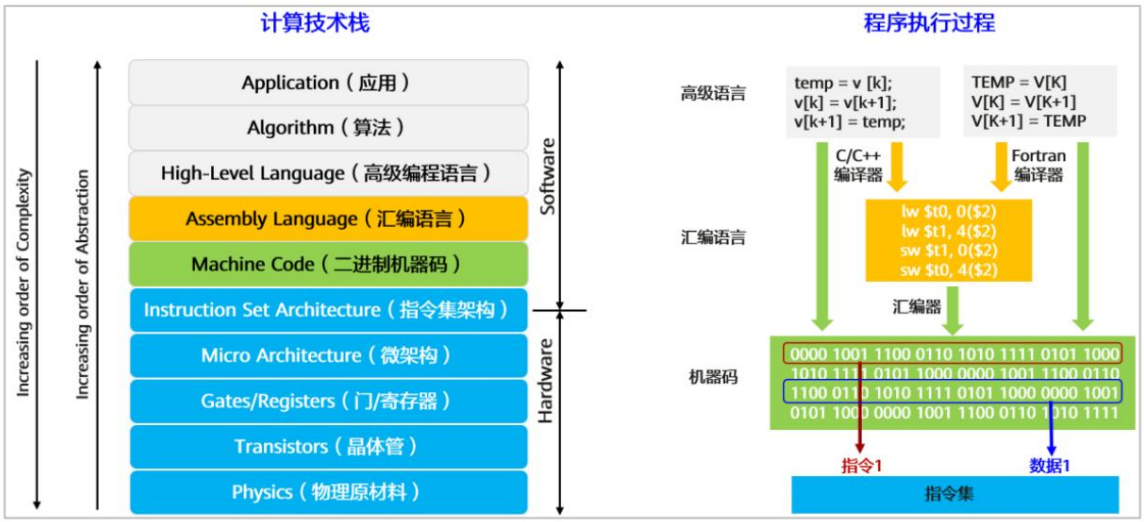


图2-1 Computing technology stack and program execution process

The diagram on the upper left shows the computing technology stack. At the hardware level, physical materials and transistors are used to implement gate circuits and registers, which then form the CPU's microarchitecture. At the software level, applications complete business functions through specific software algorithms. These programs are typically developed using high-level languages such as C/C++/Java/Go/Python. What directly interacts with the computer is binary machine code. A program exists on disk as a collection of instructions and data, forming binary machine code—what we commonly call a binary file. The CPU's instruction set is the interface between hardware and software, occupying a pivotal position. The instruction set guides the hardware implementation, and applications drive the hardware to perform calculations using the instructions defined within it.

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The diagram on the upper right illustrates the program execution process. High-level languages need to be compiled into assembly language, and then the assembler converts them into binary machine code according to the CPU instruction set. Therefore, there is a one-to-one correspondence between instruction sets and low-level languages; different instruction sets correspond to different machine code and assembly languages.

Returning to our example, the putty.exe file on a Windows computer is binary machine code, and it is machine code compiled based on the x86 instruction set. When this binary file is transferred to a mobile phone, the ARM-based mobile phone cannot recognize it, so it cannot run.

Next, let's return to the data center field and take a few simple C programs as an example to see the differences in instructions between the Kunpeng processor and the x86 processor.

程序代码（C/C++）：

```
int main()
{
    int a = 1;
    int b = 2;
    int c = 0;
    c = a + b;
    return c;
}
```

编译

鲲鹏处理器指令

指令	汇编代码	说明
b9400fe1	ldr x1, [sp,#12]	从内存将变量a的值放入寄存器x1
b9400be0	ldr x0, [sp,#8]	从内存将变量b的值放入寄存器x0
0b000020	add x0, x1, x0	将x1(a)中的值加上x0(b)的值放入x0寄存器
b90007e0	str x0, [sp,#4]	将x0寄存器的值存入内存（变量c）

x86处理器指令

指令	汇编代码	说明
8b 55 fc	mov -0x4(%rbp),%edx	从内存将变量a的值放入寄存器edx
8b 45 f8	mov -0x8(%rbp),%eax	从内存将变量b的值放入寄存器eax
01 d0	add %edx,%eax	将edx(a)中的值加上eax(b)的值放入eax寄存器
89 45 f4	mov %eax,-0xc(%rbp)	将eax寄存器的值存入内存（变量c）

图2-2 Instruction differences

As shown in the diagram, a simple line of C/C++ code, `c = b + a`, is compiled with significantly different instructions on the Kunpeng processor and x86 processor platforms. On the Kunpeng processor, two `ldr` instructions load data from memory into a register, one `add` instruction performs the addition, and finally, one `str` instruction stores the data back into memory. On the x86 processor, three `mov` instructions and one `add` instruction are used. The two `mov` instructions load data from memory into a register, the one `add` instruction performs the addition, and finally, the one `mov` instruction stores the calculated data into the memory location corresponding to variable `c`. This diagram clearly shows that the mnemonics and register representations for the same function differ between the Kunpeng and x86 processors in their assembly code. At the machine code level, x86 processors use variable-length instructions, up to 15 bytes, while Kunpeng processor ARM instructions are fixed-length at 4 bytes.

The compiler compiles the source code into an executable program (an .exe file on Windows and a binary .bin file on Linux). This program file has its own structure, but it mainly consists of a set of instructions and data. When the program is executed, the CPU loads the instructions and data from the program into the cache, and then fetches the instructions one by one and executes them.

Therefore, different CPUs use different instruction sets. Thus, the same source code and the same function require different instructions to be executed on different CPU platforms. This translation from source code to instructions is what we usually call compilation, done by a compiler. Therefore, when porting a program to different CPU platforms, if the CPU's instruction set is different, it is necessary to re-translate the source code according to the target platform's instruction set.

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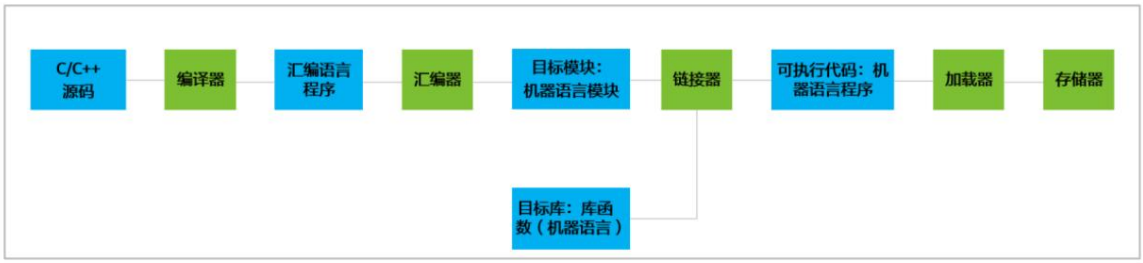


图2-3 Source code compilation process

Compiled languages require a compiler to generate an executable program before it can be loaded into memory for execution. We often use `gcc test.c -o test` to accomplish this in one step, but it actually involves multiple stages.

Step 1: Preprocessing. The source code is read, and pseudo-instructions (instructions starting with #, i.e., macros) and special symbols are "replaced." This process generates an output file without macro definitions, conditional compilation directives, or special symbols. This file has the same meaning as the unprocessed source file; it's still a C file, but its content is different. You can use `gcc -E test.c -o test.i` to perform only the preprocessing operation to obtain the precompiled file. Step 2: Compilation. The code is compiled into assembly code. You can use `gcc -S test.i -o test.s` to compile the precompiled file and output the assembly file.

The third step is assembly. The assembly code file is assembled into an object file using an assembler. This can be done by using `gcc -c test.s -o test.o`.

Step 4: Linking. This step links the target file with the necessary additional target files (static and dynamic libraries, and the C standard input/output library) to generate the executable file test: gcc test.o -o test.

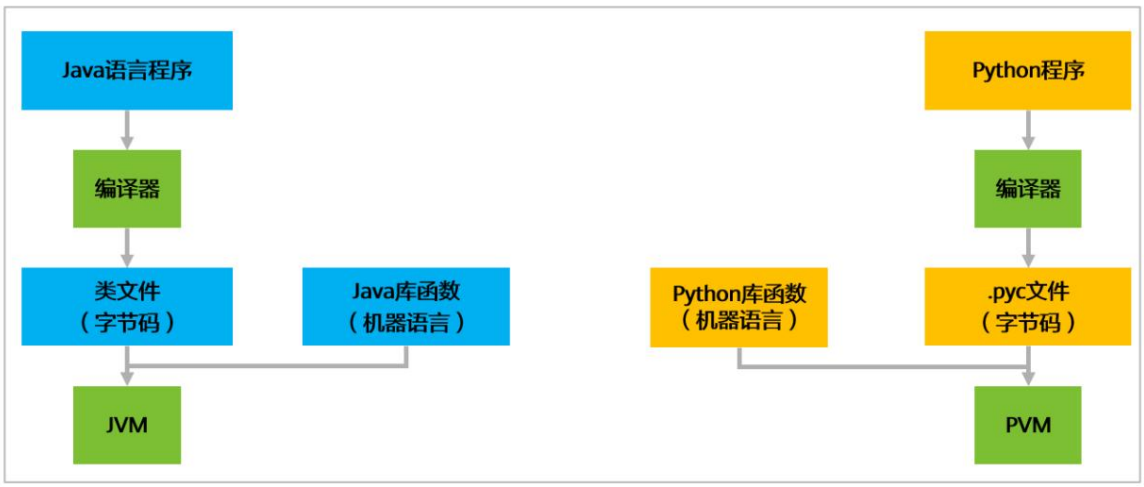


图2-4 Execution process comparison

Interpreted languages require initial compilation to obtain bytecode files before execution, and then are interpreted and executed by the corresponding language's virtual machine.

Compiled and interpreted language programs employ different migration methods depending on their execution processes.

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2.1.2 Software Migration Process

Software migration can be broadly divided into five phases: technical analysis, compilation migration, functional verification, performance tuning, and large-scale commercial deployment. Chapter Two focuses on the first two phases. Each phase is explored from three aspects: software porting, migration environment, and management.

The first phase involves technical analysis. From a software porting perspective, this requires a comprehensive analysis of the hardware and software technology stack and the development of a migration strategy. The hardware and software technology stack can be broadly divided into two parts: the runtime environment and the business software. The runtime environment includes the migration environment. We can obtain a Kunpeng test environment by acquiring a Taishan server test prototype or applying for remote server resources, which will be discussed in detail later. The runtime environment also includes drivers, operating systems, compilers, JDK/PVM, etc., all of which need to be replaced with environments that support the Kunpeng and Taishan versions. The operating systems supported by the Kunpeng hardware environment can be found at this link: https://info.support.huawei.com/computing/ftca/kunpeng. The other part, business software, mainly includes commercial... For commercial software, including self-developed and open-source software, we cannot obtain the source code. We need to contact the corresponding vendors to obtain the corresponding ARM version. If a compatible version cannot be obtained, we need to replace it with other similar software, such as replacing Oracle database with a domestic Gauss series database. The similarity between self-developed and open-source software is that we can obtain the source code. Therefore, we can recompile or re-interpret and execute based on the different processes of compiled and interpreted code. Note the replacement of dependency packages. In addition, most open-source software websites already have ARM versions available for direct download and use. Next, let's look at the different ways to obtain the migration environment in the first stage. The first method is the most cost-effective: purchasing a Kunpeng cloud server from Huawei Cloud to build a development environment. The disadvantage is that servers on public clouds are essentially virtual machines, which have some performance loss. For business scenarios with high performance requirements, cloud servers may not be sufficient, or the rental fee may be high. The second method is to purchase a physical server to build a local environment, solving the problem of insufficient cloud performance. However, the cost and technical requirements are correspondingly higher. The third method is to apply for a remote server through the Kunpeng Innovation Center flagship store; you need to contact your local innovation center for support. The fourth method is to apply for remote lab resources through the Kunpeng Community. The remote labs provide a pre-installed "one-stop" development kit for the Kunpeng development environment, including code migration tools, compilers, performance analysis tools (which will be discussed later), which developers can use flexibly. The drawback is that appointments are only available by time slot. From a management perspective, the first phase requires developing a migration plan and strategy and forming a migration team. The migration team needs a pyramid-shaped capability structure, with the most basic being operational capabilities, such as familiarity with server operations, shell scripts, and common Linux commands. Additionally, the migration process involves modifying and recompiling the source code, so a certain level of development ability is required. At the top level is performance tuning capability, as many factors affect performance, such as the application itself, the high-level language it uses, thread parallelism, lock operations, etc. The application's runtime environment also affects performance, mainly CPU, memory, disk, network, and JVM. Performance tuning engineers not only need to be familiar with business code but also with hardware, and be able to use tools to test performance and find bottlenecks for parameter tuning and optimization, so the technical requirements are very high. In terms of management, after forming the migration team, a migration plan can be developed, outlining the detailed work content for each of the five phases.

Phase two involves performing the compilation migration. We'll start by migrating compiled language programs, such as C/C++ code. First, obtain the source code. For self-developed software, directly connect to your own code repository; for open-source software, obtain the source code through GitHub, Gitee, or third-party open-source communities. Then, prepare the compilation environment, installing compilers such as gcc. Next, you can use pre-compiled code to generate a Makefile, and then execute the compilation using `make` or `cmake`. It's important to note that compiled language programs often contain... For SO libraries, we need to replace these dependent libraries with ARM versions or recompile them based on their source code. Finally, the executable program compiled on the Kunpeng environment can run normally. This process represents the ideal situation. For example, for simple code like "helloworld," the source code doesn't involve low-level languages and isn't strongly dependent on hardware; replacing the dependent libraries and recompiling will allow it to run normally. However, for code in real-world business applications, which is often more complex, different CPU platforms require different source code representations. In such cases, recompiling without modifying the code will result in errors. We need to modify the corresponding source code based on the errors until successful compilation. Here are some common issues that you can address by modifying the corresponding source code before compiling to avoid errors. For example, regarding data type definitions, if data is defined as `char` in x86 code, the system defaults to signed.

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On the Kunpeng platform, `char` is an unsigned type by default. Therefore, when migrating code from x86 to Kunpeng, you need to manually redefine the `char` type as `signed char`, or use compiler options to replace all `char` characters with signed `char` during compilation. Additionally, `builtin` functions represent functions with the same functionality differently on x86 and Kunpeng platforms. In this case, you need to consult the manual to replace the functions for successful compilation on the Kunpeng platform. Besides source code, compiler options and macros also need to be replaced. These modifications seem cumbersome, but we will introduce tools later to simplify the migration process. Having covered the migration of compiled language programs, let's look at interpreted language programs such as Java and Python. When porting Java programs, we need to prepare the migration environment, including installing the ARM version of the JDK, configuring environment variables, and then initially compiling to obtain bytecode, which is then interpreted and executed by the JVM. Note that most of the dependent libraries in the JAR package are compiled language programs, which need to be replaced with ARM versions and repackaged, or the source code of the dependent libraries needs to be obtained, recompiled, and then repackaged. Therefore, porting Java programs inevitably requires the use of build tools, such as Maven. When migrating Java programs using Maven, Maven first queries the local repository for dependency packages. If the package is not found locally, it will retrieve it from a remote repository. The dependencies retrieved here are architecture-independent. During the build process, if the dependency library is found to be an x86 version, the source code will be retrieved, and the program will be recompiled and rebuilt. Migrating Python programs is largely the same as migrating Java, with the main differences being environment preparation and dependency library porting. Other languages like Go and PHP also require different porting operations depending on whether they are compiled or interpreted languages. When setting up the program runtime environment, you can choose to directly install the interpreter software that comes with the Kunpeng platform operating system, or download the ARM version package from the official website and extract it, or compile and migrate the program yourself from the x86 source code. When migrating, if commercial closed-source software is involved, such as Oracle commercial databases, one approach is to contact the vendor to inquire whether there is an ARM version for replacement deployment. If not, you need to consider replacing it with other database software, such as domestic ones like GaussDB and DM, which both have ARM versions, or you can choose an open-source database.

2.1.3 Typical Software Migration Cases

A partner in the industry migrated its core application system from a 2.4GHz, 48-core x86+Debian OS environment to a 2.6GHz, 96-core Kunpeng platform. The operating system received 30 compatibility patches. The business system contained over 520 source code files in C++, Java, assembly, and other languages. The migration involved porting 55 dependency libraries, encountering 39 compilation issues, and requiring the modification of 195 lines of code. The porting was completed in 5 person-months, resulting in a 56% performance improvement. Environment preparation took 1 person-month, code modification, compilation, and porting took 2 person-months, and post-migration performance optimization took 2 person-months.

2.2 Kunpeng Migration Guide

2.2.1 Introduction to Kunpeng Code Migration Tool

We've already learned about the migration principles and process. Modifying the source code before recompiling can be challenging and tedious. To support the Kunpeng ecosystem, the Kunpeng community provides tools to simplify the migration process. The Kunpeng Code Migration Tool, provided by Huawei's Kunpeng DevKit, simplifies the migration of customer applications to Kunpeng 916/920-based servers. This tool only supports scanning, analyzing, and migrating x86 Linux software to Kunpeng Linux; it does not support scanning, analyzing, and migrating Windows software code. This tool allows for easy migration of x86 Linux software to the Kunpeng Linux platform. It offers three main functions: software migration assessment (analyzing non-source packages and providing a migration report), source code migration (analyzing source packages and providing porting suggestions), and package refactoring (refactoring x86 packages into Kunpeng packages). These three functions basically cover all the needs of the migration process. Two additional enhanced functions are available: specialized software migration and Kunpeng affinity analysis. When migrating source files, the source code migration tool can be used...

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The tool analyzes source code and generates an analysis report, including the number of lines of code to be modified or replaced, the number of dependent files, and the estimated migration workload. Furthermore, it provides specific modification suggestions and even a one-click replacement button for the source code that needs to be modified or replaced, allowing migration engineers to quickly modify the source code, save, and recompile for verification. When migrating non-source code files, such as binary packages or Java/Python programs, the software migration assessment function can analyze the package and generate an analysis report. This report primarily shows the number of dependent libraries involved in the package, which have existing ARM versions that can be downloaded and used directly from the analysis report interface, and which dependent libraries need to be compiled and replaced manually. Afterwards, the package refactoring function can be used to refactor the dependent libraries that need to be replaced.

When using code migration tools, the toolkit needs to be installed on the server, and then accessed via the web or used as an IDE plugin. If you choose to install the CLI using `./install cli`, then it can only be used via the command line. Commonly used IDE tools include VS Code, IntelliJ supports the installation and configuration of the Kunpeng code migration tool plugin.

2.2.2 Migration of Compiled Language Applications

Applications developed using compiled languages, such as C/C++, are compiled into executable programs. The instructions used by these executables are CPU architecture-dependent. Therefore, C/C++ applications compiled for x86 architecture cannot run directly on TaiShan servers or Huawei Kunpeng cloud servers and require porting and compilation. The application porting process includes source code analysis, modifying the source code according to recommendations, compilation and installation, and testing and verification. For specific migration procedures and operations, please refer to the Megahit source code migration case in the experimental manual.

As a compiled language, Fortran's migration process also follows the steps of source code analysis, porting suggestions, compilation and installation, and testing and verification. Here is a sample Fortran source code package: https://gitlab.com/DL_POLY_Classic/dl_poly/- /archive/RELEASE-1-10/dl_poly-RELEASE-1-10.tar.gz. Upload this source code package to the server. In the `sourcecode` directory, navigate to the extracted directory, copy the Makefile, and change its owner. Execute the source code analysis scan. Select the source path containing the Makefile as the root path for scanning, and check "Fortran" for the type. Then view the scan results. It shows one source file that needs migration, involving 10 lines of code. Click "Source Code Migration" to view the highlighted portion of the original source code in the lower right corner of the image. Follow the suggestions to make the necessary modifications, save the file, and then run the verification.

2.2.3 Migration of Interpreted Language Applications

Applications developed using interpreted languages, such as Java and Python, are CPU architecture independent. Porting these applications to TaiShan servers or Huawei Kunpeng cloud servers requires no modification or recompilation; they can be deployed and run in the same way as x86 applications. Java application JAR files may contain SO library files developed in C/C++. These SO libraries need to be ported and compiled, and the resulting JAR file needs to be repackaged. The specific porting method is the same as for C/C++ applications. The general process is as follows: upload the .jar file, create a new package analysis task to identify the dependency library files that need to be migrated; configure the Maven environment, recompile and repackage; unzip the JAR file, check the new SO library to verify successful migration, and then deploy and run the application.

2.2.4 ExaGear, a binary instruction translation tool

ExaGear is Huawei's self-developed dynamic binary translation tool. By translating x86 application instructions into ARM64 instructions at runtime and executing them, it enables Linux x86 applications to run on ARM64 servers without recompilation. This helps customers quickly migrate Linux x86 applications without source code to ARM servers, ensuring stable and reliable operation. ExaGear utilizes dynamic binary translation technology combined with dynamic binary optimization capabilities to stably support existing x86 and ARM32 services running on the Kunpeng platform without source code. It is suitable for a wide range of deployment scenarios, including direct deployment to the operating system and deployment on compatibility servers.

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Deploying containers within the Kunpeng platform, or even within ExaGear, can shield underlying platform differences without requiring source code, enabling smooth application migration at low cost and unleashing the powerful computing capabilities of the Kunpeng platform.

During the migration process, user businesses are often categorized into three types based on their importance: businesses without source code, non-critical businesses with source code, and critical businesses with source code. **Industry-related commercial software without source code:** This type of software only has binary code, no source code, but has significant influence within the industry. In this scenario, ExaGear's dynamic binary translation technology is needed to ensure a smooth migration. **Non-critical businesses with source code:** These have low usage frequency and are not performance-sensitive. For this part of the business, customers can use ExaGear to perform dynamic binary translation, allowing existing business applications to run directly on the Kunpeng platform without code porting. This process saves a significant amount of manpower and time required for porting or optimizing source code, and avoids introducing additional stability risks due to code porting. **Performance-sensitive critical businesses with source code:** It is recommended to complete the migration through manual code porting and performance optimization to achieve optimal performance expectations.

2.3 Container Migration Guide

2.3.1 Introduction to Container-Related Concepts

Containers are a lightweight, portable, and self-contained software packaging technology that allows applications to run in the same way almost anywhere. Developers can create and test containers on their own laptops, and then run them on virtual machines, physical servers, or public cloud hosts in production systems without any modifications. The most obvious difference between containers and virtual machines, as seen in this architecture diagram, is that each virtual machine has its own guest OS, while all containers share the host OS kernel. In terms of specifications, containers start faster, have higher deployment density, lighter images, and better performance than virtual machines, but their security is inferior.

Docker is the most popular container technology and is now "almost" the de facto container standard. Docker containers are created from Docker images. The relationship between containers and images is similar to that between objects and classes in object-oriented programming. An image is a multi-layered file system; when an image is run as a container, a writable container layer is added. The copy-on-write feature ensures the sharing of images. New content is added only at the container layer, and content that needs to be modified is copied to the container layer without affecting the underlying image file. Similarly, when a file needs to be deleted, the underlying image content is not actually deleted; instead, a corresponding whiteout file is added at the container layer to indicate that the content has been erased. The Docker architecture consists of three parts: client, engine service, and image repository. Users enter commands on the client, and the Docker daemon executes the commands and interacts with the repository based on the commands. For example, executing the `docker run` command runs an image to create a container. Images can be pulled from a local or remote repository using the `docker pull` command, built by specifying a Dockerfile script file using the `docker build` command, and packaged into an image using the `docker commit` command. A Dockerfile is a text-based configuration file that contains all the instructions needed to create an image. Based on the instructions in a Dockerfile, users can quickly create custom images.

The Dockerfile configuration file describes the contents of each layer of an image. The `docker build` command can be used to build an image based on the Dockerfile. The first line of the Dockerfile specifies the first layer of the image: the base image, which is usually FROM a base operating system image or a scratch empty image.

2.3.2 Introduction to the iSula Container Engine

iSula is a container technology besides Docker, and the openEuler operating system supports both Docker and iSula. Compared to Docker, the iSula general-purpose container engine is a new container solution, providing a unified architectural design to meet the different needs of the CT and IT domains. Compared to Docker, which is written in Golang, iSula's lightweight containers are implemented in C/C++, featuring lightness, flexibility, efficiency, and speed. They are not limited by hardware specifications or architecture, have lower underlying overhead, and can be applied to a wider range of fields. Meanwhile...

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To cater to different use cases, iSula offers various container types, including: general-purpose containers suitable for most common scenarios; secure containers suitable for strong isolation and multi-tenancy scenarios; and system containers suitable for scenarios where systemd is used to manage business processes within containers. iSula separates the container runtime and build sides; the build side requires the separate installation of the isula-build tool. This separation design meets the needs of some users who require a lighter footprint. iSulad is a container runtime engine based on the OCI standard, emphasizing simplicity, robustness, and lightweight design. As a daemon, iSulad provides container lifecycle management services, including image transfer and storage, container execution and monitoring management, container resource management, and networking. iSulad provides a CLI command-line interface similar to Docker, allowing for container management; and it provides a gRPC API compliant with the CRI interface standard, which Kubernetes can call according to the CRI interface protocol.

2.3.3 Container Migration Background and Principles

The runtime environment for a business application can be bare metal, a virtual machine, or a container. So how do you migrate container applications across platforms? If the architecture is the same, migration can be smooth by simply packaging the container into an image. You can directly obtain the image on the new platform, run the image, and run the same application. However, for different architectures, x86 container images obtained from a repository may not run successfully on an ARM platform, requiring the reconstruction of images and containers for the new architecture platform.

2.3.4 Main Container Migration Process

For example, to migrate a Redis container application, directly obtaining an x86 Redis container image will not work on the Kunpeng platform. We can obtain a Kunpeng platform Redis container application by writing a Dockerfile configuration file, building a new Redis image, and running it. The configuration steps for running a Redis container application on the Kunpeng platform using iSula are: install the iSulad container engine and build tools, create a Dockerfile, build the image, run the container, and verify Redis. For detailed operation procedures, please refer to the experimental manual.

2.4 Common Migration Problems and Solutions

2.4.1 Common Compilation Issues

The following are some common compilation errors that often occur when migrating x86 source code to the Kunpeng platform:

The default sign of the `char` data type is inconsistent. `char` variables have different default signs across different CPU architectures: `signed char` on x86 architecture and `unsigned char` on ARM64 platform. When porting, `char` variables need to be specified as `signed char`. This can be achieved by adding the `-fsigned-char` option to the compiler to specify `char` as a signed number on the ARM64 platform, or by directly declaring the `char` type as `signed char`. The code uses assembly instructions, which are strongly dependent on the CPU instruction set. Assembly instructions used on x86 architecture CPUs need to be modified to the instructions for the Kunpeng processor platform to compile successfully and achieve functional replacement.

-m64 is a compilation option for x86 64-bit applications. The m64 option sets int to 32 bits and long and pointers to 64 bits.
This code is generated for AMD's x86 64 architecture and is not supported on the Kunpeng processor platform. To resolve this, set the compilation options for the Kunpeng processor platform to -mabi=lp64 and recompile.

2.4.2 Common Functional Issues

The following are some functional issues that may arise when migrating x86 source code to the Kunpeng platform:

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When converting C/C++ double-precision floating-point numbers to integers, if the value exceeds the integer's range, the Kunpeng processor behaves differently than the x86 platform. The x86 instruction set defines an indefinite integer value (64-bit: 0x80000000000000000) for conversion from floating-point to integer. While x86 platforms generally adhere to this principle, different results occur when converting from double to unsigned integer. Kunpeng's handling is very clear and simple: in the event of overflow or underflow, it preserves the maximum or minimum value that the integer can represent.

An exception occurred when performing atomic operations on variables within a structure. The Kunpeng processor uses instructions like `ldaxr` and `stlxr` for atomic operations and locking, which require variable addresses to be aligned to the variable's length. Otherwise, executing the instructions will trigger an exception, leading to a coredump. This is generally because the code forces byte alignment on the structure, causing variable addresses to be out of alignment. Performing atomic operations or locking on these variables will then cause problems. Searching for the keyword `#pragma pack` (this macro changes the compiler's default alignment) in the code will identify structures that use byte alignment. If variables in this structure are used as input parameters for atomic operations, spinlocks, mutexes, semaphores, or read-write locks, the code needs to be modified to ensure these variables are aligned to their length.

2.4.3 Common Tool Issues

Different analysis tasks in code migration tools require placing the software packages to be analyzed in different server paths. Software migration assessment needs to be in the `package` directory, source code analysis in the `sourcecode` directory, and software package refactoring in the `packagerebuild` directory. A Maven software repository compilation error:

"Dependency not found." This requires modifying the `conf/settings.xml` file in the Maven installation path and setting the Maven remote repository path in the `mirror` tag.

2.5 Kunpeng Developer Support Kit

2.5.1 Kunpeng Development Kit (DevKit)

On the Kunpeng Community homepage, find the Developer section and select the technology topic: Kunpeng Development Kit. You'll find a series of tools that simplify native development or migration from x86 to Kunpeng, as well as post-migration testing and optimization. The Kunpeng Development Kit caters to the entire R&D workflow, providing enabling tools covering code migration, development debugging, compilation, testing, optimization, and diagnostics. It improves application migration and optimization efficiency, accelerates native development, and facilitates developers in quickly creating Kunpeng-friendly, high-performance software. In addition to the previously mentioned code migration tools and dynamic binary translation tools, the Kunpeng Development Kit also includes the Kunpeng Development Framework plugin, the Kunpeng Compiler and Debug Plugin, the Bisheng Compiler, the Bisheng JDK, GCC for openEuler, cloud testing services, and Kunpeng performance analysis tools.

The Kunpeng Development Framework Plugin is a tool that enables users to develop on Kunpeng servers. It fully leverages the various computing power and superior performance of third-party components on the Kunpeng platform, providing capabilities such as Kunpeng engineering wizards, heuristic programming, and code affinity checks. It allows for one-click import of Kunpeng acceleration libraries and rapid construction of Kunpeng application software frameworks, helping developers to more easily develop Kunpeng applications and enabling efficient innovation.

The Kunpeng compiler and debug plugin is a one-click, plug-and-play solution that supports multiple debugging methods for both CPU and GPU applications, as well as remote debugging capabilities on the Kunpeng platform. Users can set breakpoints, view threads, function stacks, register information, and variable information during operation. It supports breakpoint execution and single-step debugging of assembly instructions and provides a graphical interface, significantly improving debugging efficiency.

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Bisheng's compiler is developed based on the open-source LLVM, with optimizations and improvements. It also supports the Fortran language frontend and is a high-performance compiler for the Kunpeng platform. In addition to general LLVM functions and optimizations, it has performed in-depth optimizations on key technologies in the mid-end and back-end, and integrated the Auto-tuner feature to support automatic compiler tuning.

Bisheng JDK is a high-performance, production-ready OpenJDK distribution developed based on OpenJDK. Running on multiple Huawei products, Bisheng JDK has accumulated extensive experience with various use cases and feedback from Java developers, resolving numerous issues encountered in actual business operations and optimizing performance on the ARM architecture. Bisheng JDK achieves even better performance in big data and other scenarios. Bisheng JDK 8 is compatible with the Java SE standard and currently supports... Linux/aarch64 and Linux/x86_64 platforms. Bisheng JDK is also a downstream application of OpenJDK and will continue to provide stable support. Contribute to the OpenJDK community.

GCC for openEuler is a compiler toolchain (including compiler, assembler, and linker) developed based on the open-source GCC. It is released as open source by the openEuler community and its binary package is provided free of charge through the Kunpeng community. It supports the aarch64 processor architecture.

Cloud testing services provide developers with compatibility testing, reliability testing, security testing, functional testing, and performance testing services based on the Kunpeng platform, helping you quickly identify and locate problems in your application during runtime.

The performance analysis tool supports system performance analysis, Java performance analysis, and system diagnostics on the Kunpeng platform. It provides a comprehensive overview of the system and performance data collection and analysis capabilities for common application scenarios, and offers optimization suggestions based on an expert tuning system. It also provides a tuning assistant to guide users in quickly optimizing system performance.

2.5.2 Kunpeng Application Enablement Suite BoostKit

The Kunpeng Application Enablement Suite helps us improve the performance of our business applications on the Kunpeng platform, sharing best practices across 8 major scenarios. It conducts full-stack optimization from hardware and basic software to applications (including algorithm optimization, hardware-software collaboration, and original technologies of native architecture), and provides high-performance open-source components, basic acceleration packages, and application acceleration packages to enable ultimate application performance.

Kunpeng BoostKit Big Data Enabler Suite: https://www.hikunpeng.com/zh/developer/boostkit/big-data.

Kunpeng BoostKit Distributed Storage Enablement Suite: https://www.hikunpeng.com/zh/developer/boostkit/sds.

Kunpeng BoostKit Database Enablement Kit: https://www.hikunpeng.com/zh/developer/boostkit/database.

Kunpeng BoostKit Virtualization Enablement Kit: https://www.hikunpeng.com/zh/developer/boostkit/virtualization.

Kunpeng BoostKit ARM Native Enablement Kit: https://www.hikunpeng.com/zh/developer/boostkit/arm-native.

Kunpeng BoostKit Web Enabler: https://www.hikunpeng.com/zh/developer/boostkit/web. Kunpeng BoostKit CDN Enabler: https://www.hikunpeng.com/zh/developer/boostkit/cdn. Kunpeng HPC: https://www.hikunpeng.com/zh/developer/hpc.

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3 Application performance testing and tuning

3.1 Software Performance Testing

While both software performance testing and software development are related to software, their purposes and content differ significantly. Let's examine the differences between the two:

ÿDifferent Purposes: The purpose of software development is to develop software with specific functions and ensure its correctness. And reliability. The purpose of software performance testing is to test the software's performance under various conditions in order to determine whether the software can run efficiently and stably in real-world applications.

ÿDifferent Content: Software development mainly includes requirements analysis, design, coding, and testing. Software performance testing, on the other hand, mainly includes response time testing, load testing, concurrency testing, and security testing.

ÿDifferent testing processes: The testing process in software development is development-driven, with testing being one component. In contrast, the testing process in software performance testing... Performance testing is the primary focus and the core component of the entire process. Software performance testing needs to be conducted after software development is complete to test and verify the software's performance.

The purpose of software development is to develop software that meets user needs, while the purpose of software performance testing is to test the performance of software under various conditions to ensure that the software can run efficiently and stably.

3.1.1 Performance Testing Methods and Tools

After understanding the differences between software performance testing and software development, let's discuss the purpose of software performance testing.

The purpose of software performance testing, simply put, can be listed as follows:

1. Assess software system capabilities: Verify the relationship between software system load and response time to see if it meets requirements, and assist development managers in making decisions.

2. Assess the resilience of the software system.

3. Identify whether the software system has any unknown bugs under stress, including but not limited to:

- Under pressure, the software responds very slowly and cannot provide service.
- Under pressure, the system program exits, and data is lost.
- Once the pressure decreases, the program cannot heal itself, and the service cannot be restored.

4. Assess the durability of the software system: Verify the stability and reliability of the software system under long-term stress operation.

In the actual testing phase, what methodologies guide software performance testing?

The software performance testing process typically includes stages such as test plan development, test environment setup, test design, test execution, and test reporting.

During the test planning phase, it is necessary to determine the test scope, test objectives, test methods, and test timing.

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During the test environment setup phase, it is necessary to create the test environment, including hardware and software configurations.

During the test design phase, test plans and test cases need to be designed.

During the test execution phase, test cases need to be executed and test results recorded.

During the test report phase, it is necessary to analyze and summarize the test results and report them to relevant personnel.

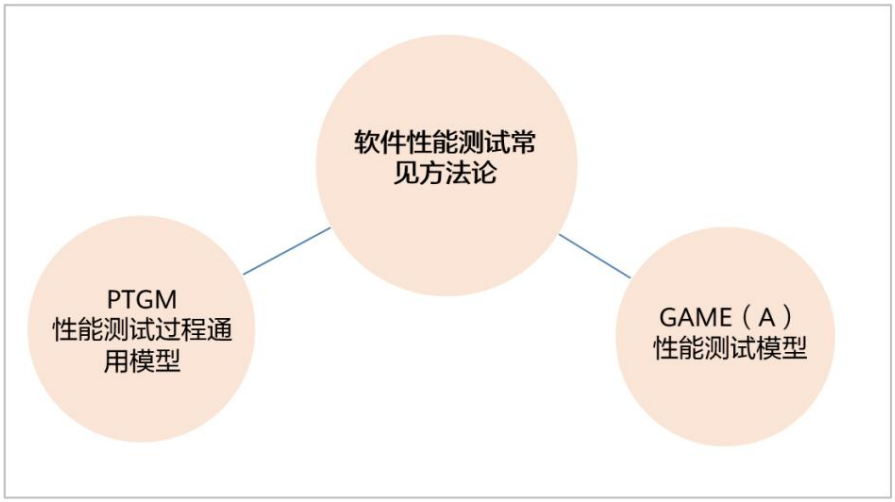


图3-1 Performance Testing Methodology

Over the past 40 years of software engineering development, systematic software testing methodologies such as PTGM and GAME(A) have been accumulated.

PTGM (Performance Testing General Model) is a general performance testing method that consists of the following 5 phases:

1. Pre-test preparation: Verification of basic system functions, formation of test team, confirmation of test tool requirements, and performance pre-testing.

2. Introduction of testing tools: tool selection, tool application skills training, and determination of the tool application process.

3. Test Plan: Performance testing domain analysis, user activity profiling and business modeling, performance target determination, and test schedule planning.

4. Test design and development: test environment design, test scenario design, test case design, script and auxiliary tool development.

5. Test execution and management: Set up the test environment, deploy test scripts and test scenarios, execute tests and record results.

The performance testing GAME(A) model includes the following five key components:

Goal: Define the objective of this test and select a test design strategy.

Analysis: Analyze performance requirements and system architecture. Metrics: Define

scenarios, transactions, and pass/fail criteria for virtual users. Execution: Prepare the environment, data, and scripts; run scenarios and

monitors; and generate reports.

Adjustment: Modifications to the application and optimization of middleware.

Having learned about the system's software testing methodology, we will now introduce a powerful software testing tool - LoadRunner.

LoadRunner is a commonly used performance testing tool that simulates user behavior under different loads to test software performance. LoadRunner supports different types of testing, including web testing, database testing, ERP testing, and more.

LoadRunner uses scripts for testing. These scripts can be recorded using the LoadRunner recorder or written manually. After testing, LoadRunner generates a detailed test report, including test results, performance metrics, and issue classifications.

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Performance testing using LoadRunner can identify software performance bottlenecks and provide a basis for software optimization and improvement.

3.1.2 Common Software Performance Testing Cases

The previous sections introduced the basic methodology of software performance testing and a system tool called LoadRunner. This section will introduce common software performance testing cases.



图3-2 Common performance testing tools

As shown in the figure, common performance testing tools were detected from the internet, and their usage frequency was plotted as a word cloud. The larger the font size of the testing tool in the figure, the higher its frequency of use. Let's list some of the most frequently used tools:

1. LoadRunner

2. JMeter

3. What are the

uses of tools like MySQLSLAP? How are they used in real-world scenarios? Let's introduce them below.

LoadRunner is a web server performance testing tool developed by HP. It is a large and feature-rich system that can analyze server performance using various transport protocols.

JMeter is a powerful and easy-to-use open-source performance testing tool for the Java platform.

MySQLslap is a benchmarking tool that comes with MySQL. Its advantages include simple syntax for querying data and its flexibility and ease of use. MySQLslap provides intuitive verification of MySQL performance before and after optimization. It is recommended that system administrators and DBAs familiarize themselves with common stress testing tools in order to handle issues such as the user traffic limit supported by online databases and their stress resistance.

In practice, test scenarios are usually divided into coarse-grained categories, such as web servers and databases.

Web server

Common web servers include Nginx and Tomcat.

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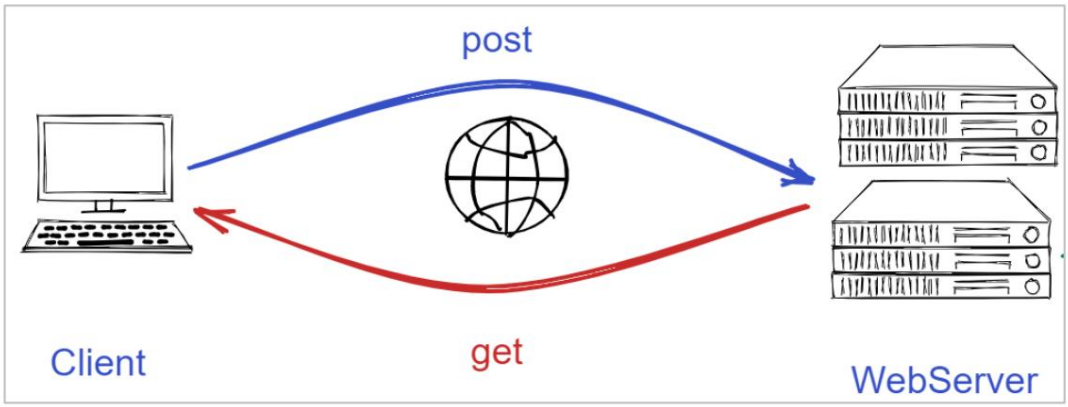


图3-3 **web** server

Nginx is a high-performance web and reverse proxy server, and a very popular open-source software. Nginx is designed for high concurrency, high performance, low memory consumption, and high reliability. Nginx supports multiple protocols, such as HTTP, etc.

HTTPS, SMTP, POP3, and IMAP are among the protocols supported. Nginx's reverse proxy functionality can forward requests to backend application servers for load balancing and high availability. Nginx can also function as a static resource server, providing fast access to static files. Nginx is highly flexible in its configuration, allowing for various customizations through configuration files. Its high performance and flexibility make it a leader in the web server and reverse proxy server field.

Tomcat is a popular Java web application server developed and maintained by the Apache Software Foundation. Tomcat's primary function is to provide a Servlet and JSP container for Java web applications. It can handle HTTP requests, forward them to the backend Java web application, and then return the response to the client. Tomcat also supports SSL/TLS encryption and various authentication methods to enhance the security of web applications. Tomcat can also be integrated with other web servers such as...

Used in conjunction with the Apache HTTP Server, it provides more efficient and secure web services.

Web server testing methods and metrics

For the two open-source web server tools mentioned above, Wrk can be used for performance testing. Wrk is an open-source HTTP performance testing tool that can simulate multiple concurrent users accessing a web application and evaluate its performance. Wrk's main features are high concurrency, high performance, and ease of use. Wrk uses Lua scripts for testing, allowing users to write their own test scripts to suit different testing needs. Wrk's test results include metrics such as request throughput, latency, and error rate, helping users identify performance bottlenecks in their web applications and implement optimizations and improvements. Due to its efficiency and ease of use, Wrk has been widely used in the field of web application performance testing.

When testing an HTTP server with wrk, the following parameters need to be considered:

1. Concurrency (-c): The number of concurrent users simulated. It is generally recommended to set it to twice the number of CPU cores on the machine.

2. Number of connections (-t): The number of connections to simulate. It is generally recommended to set it to twice the number of concurrent connections.

3. Duration (-d): The duration of the test. It is generally recommended to set it to at least 1 minute.

4. Request rate (-R): The number of requests sent per second. It is generally recommended to adjust this according to the actual situation.

5. Test Scripts: Write test scripts based on actual conditions to simulate real business scenarios.

By setting the above parameters, a comprehensive, effective, and efficient performance test of the HTTP server can be performed, providing a basis for optimization and improvement.

The test results of wrk can be evaluated from the following metrics:

1. Request rate: The number of requests sent per second. The higher this metric, the better.

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2. Throughput: The number of requests processed per second; the higher this metric, the better.

3. Latency: The response time of a request; the lower this metric, the better.

4. Error rate: The error rate of requests; the lower this metric, the better.

By comparing the performance of different parameters in the test results, performance bottlenecks can be identified and optimized or improved.

ÿDatabase

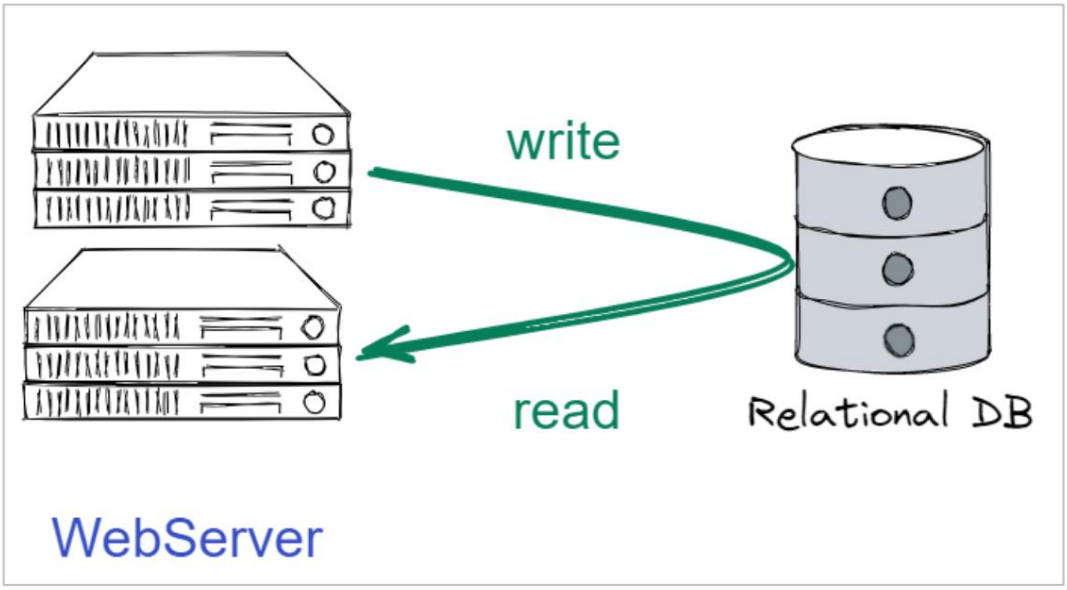


图3-4 database

A database is a software system used to store and manage data. It can store various types of data, such as text, numbers, images, audio, and video. A database typically consists of one or more tables, each containing various data records. Each record contains multiple attributes, such as name, age, and address. Databases also provide query functions so users can retrieve, sort, and filter data. Common database software includes Oracle, MySQL, Microsoft SQL Server, and PostgreSQL. Databases are widely used in various applications, such as enterprise management, e-commerce, social media, and games.

MySQL is a popular open-source relational database management system (RDBMS) maintained and developed by Oracle Corporation.
MySQL supports multiple operating systems, such as Linux, Windows, and macOS, as well as multiple programming languages, such as...
PHP, Java, Python, and C++, among others. MySQL is a mature, stable, and high-performance database system with the following characteristics:

Open source and free: MySQL is open source software, which can be used and modified for free. High

performance and high reliability: MySQL supports various optimization techniques, such as indexes, partitioning, and caching, to improve database performance and reliability.

Secure and easy to use: MySQL provides a variety of security features, such as encryption, access control, backup and recovery, to protect the database. At the same time, MySQL also provides a variety of easy-to-use tools and interfaces, such as MySQL Workbench, phpMyAdmin, and JDBC.

MySQL is widely used in various application scenarios, such as web applications, enterprise management systems, and e-commerce systems. MySQL has become one of the most popular open-source databases in the world, with a large user base and community support.

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PostgreSQL, another commonly used open-source relational database, shares some similarities with MySQL in its use cases, but also has significant differences:

1. Development History: MySQL was originally developed by the Swedish company MySQL AB, later acquired by Sun Microsystems, and is now owned by Oracle. PostgreSQL, on the other hand, was originally developed by Canadian computer scientist Michael Stonebraker and others, and is now maintained and developed by the PostgreSQL Global Development Group.

2. Licensing: MySQL is licensed under the GPL and a commercial license, while pgsql is licensed under the BSD license. 3. Data Types: Both MySQL and pgsql support various data types, such as integers, floating-point numbers, dates, and times. However, pgsql supports more data types, such as arrays, ranges, network addresses, and JSON. 4. Performance: Both MySQL and pgsql perform well in terms of performance, but they optimize performance differently. MySQL uses techniques such as indexes, caching, and partitioning to improve performance, while pgsql uses techniques such as inheritance, views, and triggers.

5. Scalability: Both MySQL and PostgreSQL support horizontal and vertical scaling. However, PostgreSQL performs better in horizontal scaling, using techniques such as sharding and replication to achieve horizontal data scaling.

6. Compatibility: Both MySQL and PostgreSQL support the SQL standard, but their implementations differ in some aspects. For example, PostgreSQL supports more standard SQL features, such as window functions, recursive queries, and the WITH statement.

The prototype of the OpenGauss database is based on pgsql.

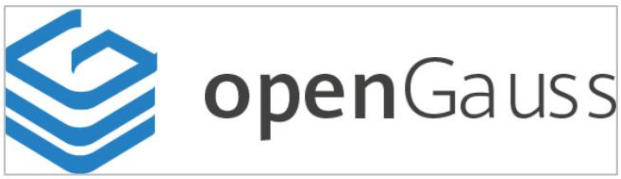


图3-5 OpenGauss database

Database testing methods and metrics

Sysbench is a popular open-source benchmarking tool that can simulate multiple concurrent users testing a system and evaluating its performance and stability. Sysbench supports various test modes, such as CPU, memory, disk, file I/O, and database. Among these, the database test mode is one of the most popular, simulating multiple concurrent users accessing a database and measuring metrics such as throughput, latency, and error rate.

sysbench supports various database systems, such as MySQL, PostgreSQL, MariaDB, and MongoDB. It uses Lua scripts for testing, allowing users to write their own test scripts to suit different testing needs. sysbench test results include metrics such as request throughput, latency, and error rate, helping users identify system performance bottlenecks and implement optimizations and improvements.

The test parameters for sysbench include: 1.

Concurrency (--threads): The number of concurrent users simulated. It is generally recommended to set it to twice the number of CPU cores on the machine.

2. Test time (--time): The duration of the test. It is generally recommended to set it to at least 1 minute.

3. Database size (--db-size): The size of the database used during testing. It is generally recommended to adjust this according to the actual situation.

4. Test Scripts: Write test scripts based on actual conditions to simulate real business scenarios.

By setting the above parameters, a comprehensive, effective, and efficient performance test of the system can be conducted, providing a basis for optimization and improvement.

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The sysbench test results include:

1. Throughput: The number of requests processed per second; the higher this metric, the better.

2. Latency: The response time of a request; the lower this metric, the better.

3. Error rate: The error rate of requests; the lower this metric, the better.

3.2 Kunpeng General Performance Tuning Methods

3.2.1 Performance Tuning

Chapter 2 details the five steps of migrating software from the x86 platform to the Kunpeng platform. Do you remember which step performance tuning was? The answer is the fourth step!



图3-6 Five stages of software migration

The reason why performance tuning is a necessary process when migrating software on the Kunpeng platform is that most programs have underlying differences in their operation on the x86 platform compared to the Kunpeng platform.

Can everyone recall what the specific differences were?

Software performance testing and tuning are two phases of the same topic. Only through testing can we understand the current performance status of the software. With clear business indicators, we can achieve the goal of software performance optimization through hardware and software tuning.

The following introduces general performance tuning methods for Kunpeng. These are called general methods because different software systems have inherent differences in code logic and method calls during development, leading to varying bottlenecks in actual operation. Mastering general tuning methods can solve most problems encountered. As for extreme code-level issues, such as deadlocks in Java programs, dedicated Kunpeng performance analysis tools can be used for later location and optimization.

The general performance tuning method unfolds from three resource levels: computing, storage, and network. The tuning logic is to first find out which resource is consuming the most resources, and whether it is normal or abnormal consumption. If it is normal consumption, then horizontal scaling needs to be considered to increase hardware service capacity; if it is abnormal consumption, then the specific cause needs to be found and analyzed, and the tuning should be completed through reasonable methods.

3.2.2 CPU and memory subsystem performance tuning

The performance optimization approach is as follows:

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If CPU utilization is low, it means that resources are not being fully utilized. You can use tools (such as perf) to see where the application is blocked, whether the network or application business logic is hibernating or waiting for signals. These optimization measures are described in other chapters.

If CPU utilization is high, you can choose better hardware, optimize hardware configuration parameters to adapt to business scenarios, or reduce CPU utilization by optimizing software.



图3-7 Kunpeng **920** processor

perf is a Linux system performance analysis tool that can analyze CPU, memory, disk, and other performance metrics. Specifically, perf list lists supported event types, perf stat counts the number and percentage of various events, perf record records event data, and perf report generates reports and displays performance data. For analyzing application bottlenecks, the perf record and perf report commands can be used. The specific steps are as follows: 1. Use the perf list command to list the supported event types and find the appropriate event type.

2. Use the `perf record` command to record event data, for example: `perf record -e sched:sched_stat_wait -g --sleep 10`. 3. If the application is blocked, use Ctrl+C to stop the `perf record`

command. 4. Use the `perf report` command to generate reports and display performance data, for example: `perf

report --sort=dso,comm,symbol`.

`top` is a commonly used Linux system monitoring tool that can display various system metrics and process information in real time. `top` can display the usage of system resources such as CPU, memory, disk, and network, as well as the CPU, memory, and I/O metrics of individual processes. By default, `top` sorts processes by CPU usage and updates the display in real time in an interactive manner. Users can use the `top` command to understand the system load and process execution status, and perform operations and adjustments using `top`'s interactive commands.

Common parameters for `top` include:

`d`: specifies the update interval for `top`. `p`: specifies

that `top` should only display information for the specified process ID.

u: Specifies that top will only display process information for the specified username.

c: Displays complete command-line information about the process.

H: Displays the thread information of the process.

In addition to commonly used parameters, top also supports a variety of interactive commands, such as:

k: Sends a signal to the specified process.

r: Modify the priority of the specified process.

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z: Suspends or resumes the specified process.

f: Toggles the display of process information fields.

o: Specifies the sorting field and sorting method.

q: Exit top. Through

interactive commands, users can manipulate and adjust processes to meet different needs. top is a very practical system monitoring tool that helps users understand the system's status and process operation, as well as perform system optimization and adjustments.

The output of `top` includes overall system load and the running status of individual processes. In the overall system load section, you can view information such as total system load, CPU utilization, memory utilization, and swap space utilization, and you can sort and filter as needed. In the process running status section, you can view information such as process PID, user, CPU utilization, memory utilization, and process status, and you can sort and filter as needed. Users can use the interactive commands of `top` to operate and adjust settings to meet different needs.

Besides `top`, Linux offers other system monitoring tools such as `vmstat`, `sar`, and `iostat`. These provide more detailed and comprehensive system performance information to help users understand the system's status and performance bottlenecks, enabling optimization and adjustments. Users can choose the appropriate system monitoring tool based on their needs and optimize and adjust according to the output to improve system performance and stability.

Chapter 1 introduced the multi-core features of the Kunpeng processor. In conjunction with the CPU and memory optimization in this section, it is necessary to understand NUMA.

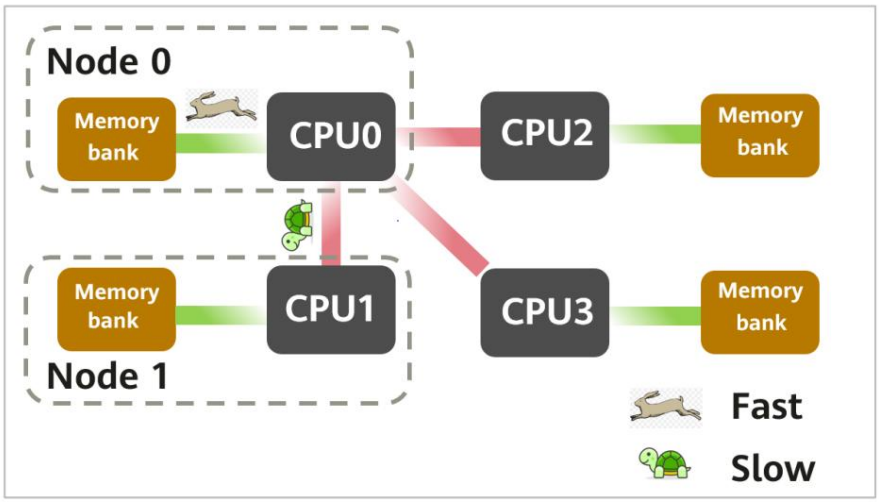


图3-8 **NUMA** architecture diagram

NUMA (Non-Uniform Memory Access) is a computer system architecture that divides memory into multiple regions and allocates each region to one or more processors. This architecture can improve system scalability and performance, especially in multiprocessor systems. Each processor can only access memory within its local region, but processors need to communicate and access memory in other regions via a high-speed interconnect network.

The advantage of NUMA architecture lies in its ability to distribute memory and processors across different nodes, avoiding the contention and bottleneck issues inherent in traditional shared memory architectures, thereby improving system efficiency and concurrency. Furthermore, NUMA architecture can dynamically adjust memory and processor allocation based on application access patterns to maximize system performance and resource utilization.

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However, NUMA architecture also has some drawbacks. First, because processors are distributed across different nodes, additional optimizations are needed to avoid issues such as memory fragmentation and data inconsistency. Second, the need for access via high-speed interconnect networks increases system latency and complexity. Finally, the design and implementation of NUMA architecture also requires consideration of factors such as the number of processors, memory size, and interconnect network bandwidth, which increases the difficulty of system design and management.

The processor's NUMA feature can be set via Numactl.

Numactl is a Linux system tool used to manage memory and processor allocation in NUMA architectures. Using Numactl commands, processes can be bound to specified NUMA nodes, and the processes' memory access methods and policies can be controlled. Numactl supports various operating modes and parameters, such as:

-hardware: Displays the system's NUMA architecture and node information.

-show: Displays the NUMA policy and memory allocation of the current process.

-cpubind: Binds the process to the specified CPU node. -membind: Allocates memory for the process to the specified NUMA node.

-interleave: Interleaves the process's memory allocation across all NUMA nodes.

-preferred: Allocates memory for the process on the specified NUMA node; if allocation is not possible, it will be allocated on another node.

By using Numactl commands, users can optimize application performance and resource utilization, especially in multiprocessor systems. Numactl is a very useful system tool that helps users understand the NUMA architecture and performance characteristics of their system, and to optimize and tune it.

3.2.3 Network Subsystem Performance Tuning

For developers, the network subsystem is an easily overlooked performance tuning module because there is a lot of hardware involved in the network. Taking Internet traffic as an example, let's first look at which hardware modules the computer needs to pass through:

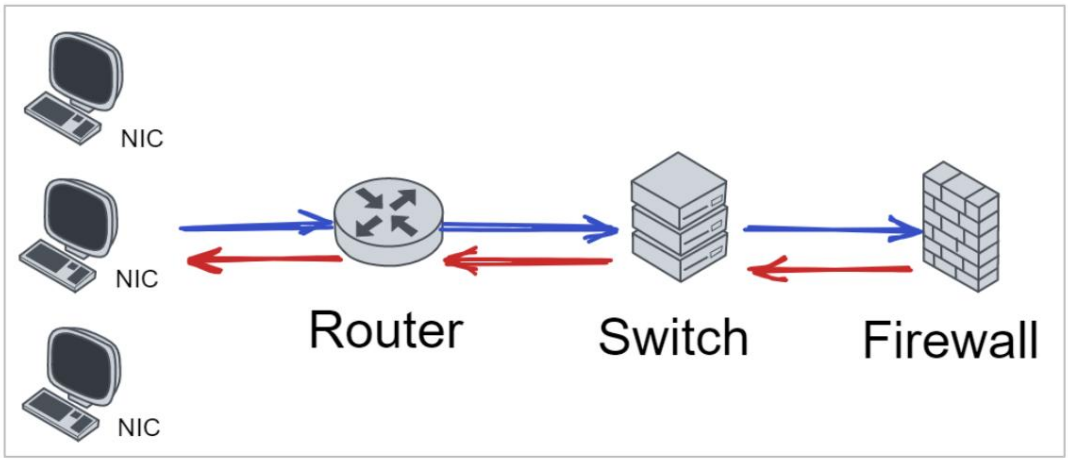


图3-9 External traffic access process of computer

ÿNetwork Interface Card (NIC): A NIC is the interface between a computer and a network, responsible for sending data from the computer to the network and transmitting data received from the network back to the computer. It typically includes an Ethernet port for connecting the computer to network devices such as switches and routers. A NIC can also support different...

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Network interface cards (NICs) employ physical layer protocols such as Ethernet, Wi-Fi, and Bluetooth to meet diverse network requirements. When calculating network traffic, the NIC serves as both the starting and ending point for data transmission and is crucial for performance optimization.

Router : A router is a device that connects different networks, responsible for data transmission and forwarding between them, routing data from source address to destination address. Routers typically have multiple network ports, allowing simultaneous connection to multiple networks, and support routing algorithms to select the optimal path for data transmission based on network topology and routing tables. In large enterprises and on the internet, routers are core components of network infrastructure, enabling efficient and reliable data transmission and network management.

Switch : A switch is a device that transmits and forwards data within a network, responsible for switching data from a source address to a destination address. Unlike a router, a switch does not perform routing; it only forwards data based on the destination address. Switches typically have multiple ports, allowing multiple computers and other network devices to be connected. They also support various flow control and priority settings to ensure network performance and quality of service.

Firewall : A firewall is a network security device responsible for detecting and blocking network attacks and malicious software. Firewalls protect network security. They can filter and audit network traffic according to rules and policies, preventing unauthorized access and data leaks. Firewalls typically include various types such as network perimeter firewalls, internal firewalls, and host firewalls. The appropriate firewall device can be selected based on different network environments and needs.

The above hardware modules work together to form the network infrastructure, enabling efficient, secure, and reliable data transmission and storage. In practical applications, different network environments and needs may require different combinations of hardware modules to meet varying performance and security requirements. Therefore, thorough planning and testing are essential when designing and implementing network infrastructure to ensure network stability and scalability.

From the computer's internal perspective, after a data packet enters the computer system from the network card, it goes through a series of steps before it can be processed into data.

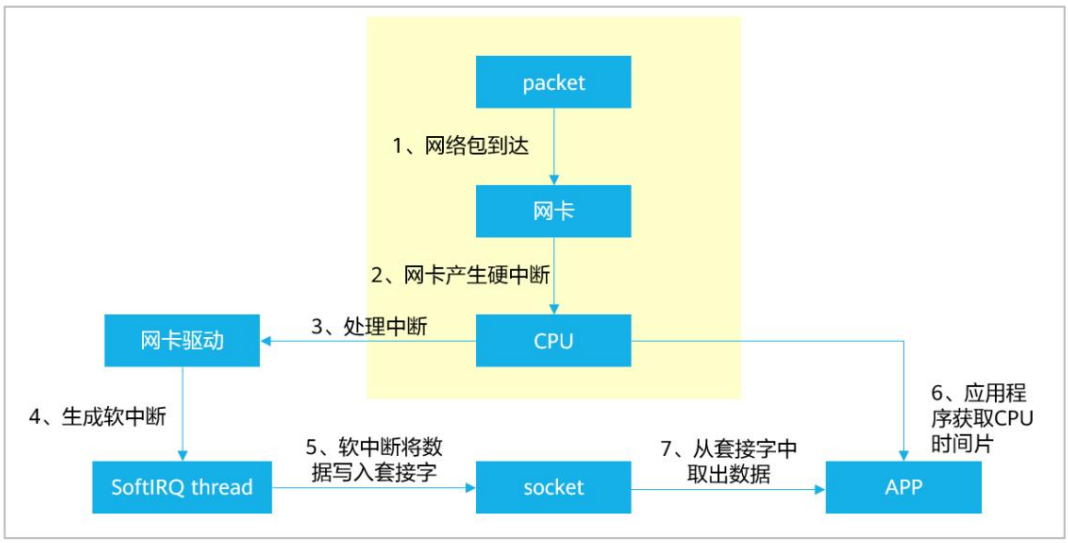


图3-10 Data packets enter the computer system from the network card

1. First, it enters the network protocol stack, which typically consists of different protocol layers, such as the physical layer, data link layer, network layer, transport layer, and application layer. Each protocol layer is responsible for different functions and processing flows, such as data encapsulation, routing, and error detection.

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2. Once a data packet passes through the network protocol stack, it is passed to the corresponding application for processing. The application can be a network service based on different protocols, such as a web server, email server, or FTP server. The application will perform different operations based on the protocol and content of the data packet, such as parsing HTTP requests or sending SMTP emails.

3. Applications typically use system calls to access underlying network interfaces and hardware devices. System calls allow applications to directly control network traffic and data transmission, such as opening sockets, sending data, and receiving data. System calls trigger the corresponding network driver, which is responsible for communication and data transmission with the network interface card (NIC) and other network devices. Network drivers are usually part of the operating system kernel and provide efficient and reliable data transmission services.

4. Finally, the data packets reach the network interface card (NIC), which converts them into electrical signals and sends them into the network. The NIC can also receive data transmitted from the network, convert it into data packets, and pass them through the processing flow described above.

In the above process, each step needs to process data and traffic packets efficiently and reliably to ensure network stability and performance. Therefore, the design and implementation of network infrastructure must fully consider the requirements and characteristics of each step.

Besides these aspects, other factors need to be considered during network traffic processing. For example, packet size, transmission speed, network topology, and security measures all affect network performance and stability. Furthermore, different types of network services place different demands on packet processing, such as real-time performance, reliability, and scalability. Therefore, when designing and implementing network infrastructure, it is necessary to comprehensively consider various factors to meet diverse performance and security requirements.

Having learned about the entire process of network traffic packets from external computer hardware devices to the network protocol stack inside the computer, we will now introduce the network management tool ethtool.

`ethtool` is a command-line tool used to query and modify network interface card parameters. It supports various network interface cards, such as Ethernet cards, wireless network cards, and FDDI cards, and can display and modify the attributes and status of various interface cards. Commonly used `ethtool` commands include:

```
ethtool -i <interface>
```

Displays the driver information and version number of the specified network interface card.

```
ethtool -p <interface> <duration>
```

Make the LED of the specified network interface card flash for a period of time so that the user can locate the interface card.

```
ethtool -S <interface>
```

Displays statistics and performance data for the specified network interface card, such as received and sent traffic, packet loss rate, and error rate.

```
ethtool -a <interface>
```

Displays the adaptive rate and duplex mode settings for the specified network interface card.

```
ethtool -c <interface>
```

Displays the RX/TX flow control settings for the specified network interface card.

```
ethtool -g <interface>
```

Displays the network interface card's buffer size and descriptor ring size.

```
ethtool -k <interface>
```

Displays the unloading/unloading characteristics of a specified network interface card, such as GRO, LRO, TSO, etc.

```
ethtool -m <interface>
```

Displays the optical module information of the specified network interface card, such as optical power, wavelength, and transmission distance.

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ethtool -r <interface>

Resets the RX and TX error counters of the specified network interface card.

By using the ethtool command, users can understand and adjust various attributes and statuses of network interface cards to optimize network performance and stability. ethtool is a very practical system tool, especially in network troubleshooting and performance analysis, helping users quickly locate and resolve network problems.

3.2.4 Disk I/O Subsystem Tuning

Having introduced the two operational-level optimization scenarios of computing and networking, we will now introduce the disk I/O subsystem.

First, it's important to understand why it's called the IO subsystem, or Input/Output system. The server's computing resources are mainly used to process code logic, which involves continuously processing raw data. Therefore, the path between data and CPU computing resources is another key factor affecting business operations.

Multi -level storage

Multi-level storage is a classic design approach in disk I/O subsystems.

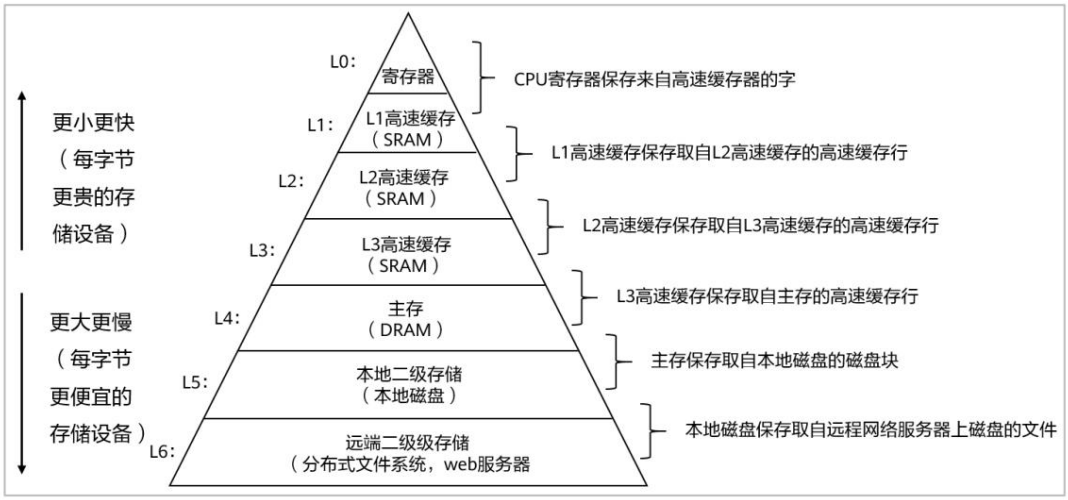


图3-11 Multi-level storage architecture

In multi-level storage, cache is the fastest storage medium. It is usually located inside the CPU or between the CPU and memory, and is used to cache frequently accessed data and instructions to improve CPU speed. Cache is divided into multiple levels such as L1, L2, and L3, each with different capacity and access speed. L1 cache has the smallest capacity and the fastest speed, while L3 cache has the largest capacity and the slowest speed, but is still much faster than memory.

Memory (RAM) is a fast, large-capacity storage medium typically used to store data and code for applications and the operating system. Memory access speed is much faster than disk access, but its capacity is smaller. Memory capacity and access speed also come in several tiers, for example...

DDR4, DDR3, DDR2, etc., with speed and capacity decreasing as the DDR level increases.

Disks are slower, larger-capacity storage media, typically used for long-term storage of data and files. Disks are generally divided into two types: hard disk drives (HDDs) and solid-state drives (SSDs). SSDs are much faster than HDDs, but are also more expensive. Disks also come in multiple tiers, such as SATA, SAS, and NVMe, with speed and capacity increasing with each tier.

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图3-12 Huawei **SSD** hard drive

The advantages of multi-tiered storage are improved data access speed and storage capacity, while reducing storage costs and energy consumption. Caches can accelerate CPU operation and improve system responsiveness; memory can cache application and operating system data, speeding up data access; and disks can store data and files long-term, providing substantial storage capacity. The disadvantages are the need to manage multiple storage tiers, involving data migration, backup, and recovery operations, requiring additional management and maintenance costs.

Multi-level storage is widely used in modern computer systems, especially in large databases, cloud computing, and big data analytics, to improve the efficiency and performance of data processing and analysis. In data centers, multi-level storage is typically used to store different types of data with varying access frequencies, achieving efficient and reliable data storage and access services. Furthermore, multi-level storage can enhance data reliability and availability through technologies such as data migration and backup, ensuring data security and integrity.

Multi-level storage is an effective storage technology that can be flexibly configured and managed according to different access needs and cost requirements to achieve efficient and reliable data storage and access services. With the continuous development of computer systems and the expansion of application scenarios, multi-level storage will play an increasingly important role, providing us with more efficient, secure, and reliable data storage and access services.

Besides the concept of multi-level storage, there are other factors that affect storage efficiency, such as: file system, I/O scheduling strategy, and buffer.

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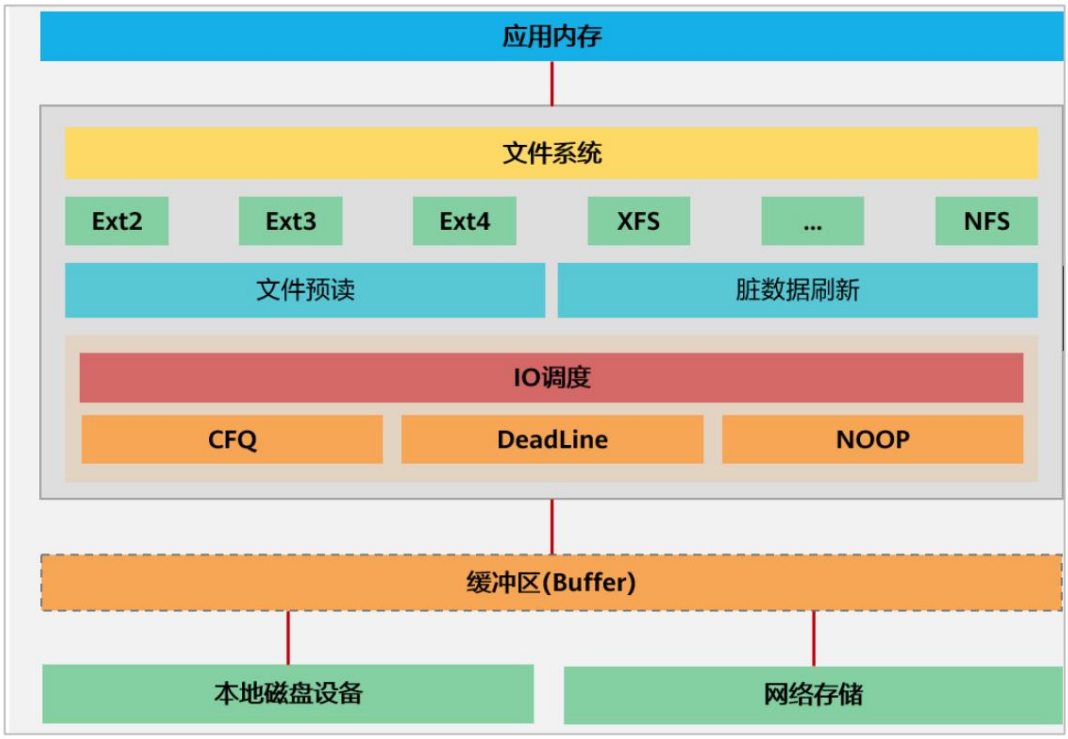


图3-13 Internal workings of the **I/O** subsystem

ÿFile System

A file system is a mechanism used by an operating system to manage files and directories. It defines how files and directories are organized, access permissions, and security policies. Different file systems have different characteristics and design goals, so choosing the right file system can have a significant impact on storage efficiency and performance.

Generally, the following factors should be considered when choosing a file system:

Storage media type: Different types of storage media (such as disks, SSDs, etc.) have different performance and characteristics, so it is necessary to select a file system that is suitable for the storage media to obtain the best performance and reliability.

File size and number: The design and implementation of a file system usually affect the size and number of files, so it is necessary to choose a suitable file system based on actual storage needs.

Concurrent access: The concurrent access capability of a file system usually affects storage efficiency and performance. Therefore, it is necessary to choose a file system that supports high concurrency access in order to improve storage efficiency and performance.

Security and data integrity: The security and data integrity of the file system are key factors in storage efficiency and performance. Therefore, it is necessary to choose a file system with good security and data integrity protection mechanisms to ensure the safety and integrity of data.

Compatibility and portability: The compatibility and portability of a file system often affect storage efficiency and performance. Therefore, it is necessary to choose a file system with good compatibility and portability to enable data sharing and transfer between multiple operating systems and platforms.

The choice of file system has a significant impact on storage efficiency and performance, and a suitable file system needs to be selected based on actual storage requirements and characteristics. At the same time, the design and implementation of the file system must also consider factors such as different storage media, concurrent access, security, and data integrity to improve storage efficiency and performance.

ÿ IO scheduling strategy

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Common I/O scheduling strategies include CFQ, Deadline, and NOOP.

CFQ (CFent-Queued) scheduling is a queue-based scheduling algorithm that schedules access requests according to process priority and I/O load to ensure fairness and stability. CFQ is suitable for multi-user and multi-tasking environments, effectively avoiding I/O blocking and I/O starvation. Deadline scheduling is a deadline-based scheduling algorithm that schedules access requests according to deadlines and I/O load to improve I/O response speed and throughput. Deadline scheduling is suitable for access requests requiring fast responses, effectively reducing I/O latency and improving system response speed. NOOP (First-In, First-Out) scheduling is a simple FIFO algorithm that does not perform any scheduling, simply processing I/O requests according to the first-in, first-out principle. NOOP is suitable for low-load and high-throughput environments, effectively improving I/O performance and throughput.

Choosing the right I/O scheduling strategy requires evaluation and testing based on actual storage needs and characteristics. Typically, the following factors need to be considered:

I/O access patterns and load characteristics: Different I/O access patterns and load characteristics have different impacts on different I/O scheduling strategies. It is necessary to select an appropriate I/O scheduling strategy based on the actual access patterns and load characteristics.

I/O latency and response speed: For access requests that require fast response, it is necessary to select an I/O scheduling strategy that can provide the best response speed and the minimum latency.

I/O Throughput and Performance: For access requests that require high throughput and high performance, it is necessary to select an I/O scheduling strategy that can provide the maximum throughput and the highest performance.

System load and fairness: In a multi-user and multi-tasking environment, it is necessary to select an IO scheduling strategy that can provide fairness and stability to avoid I/O blocking and I/O starvation.

Data reliability and security: When selecting an IO scheduling strategy, data reliability and security need to be considered to ensure data integrity and confidentiality.

Choosing the right I/O scheduling strategy can improve storage efficiency and performance, thereby meeting different storage needs and characteristics. In practical applications, different I/O scheduling strategies need to be evaluated and tested to select the optimal one. Simultaneously, system optimization and configuration are also required to maximize the performance and efficiency of the I/O scheduling strategy.

ÿ Buffer

A buffer is a mechanism for temporarily storing data. It typically sits between the data source and destination, buffering and regulating the data transmission rate to improve efficiency and reliability. A buffer can store a certain amount of data, and when the transmission rates between the source and destination are inconsistent, it can adjust the transmission rate to prevent data loss and retransmission. The size and number of buffers can be adjusted and configured according to actual transmission needs and performance requirements to maximize data transmission efficiency and reliability.

The size and number of buffers have a significant impact on storage efficiency. Generally, the buffer size should be large enough to prevent data loss and retransmission. If the buffer is too small, data loss and retransmission will occur when the data transfer rate exceeds the buffer's processing capacity, thus reducing data transfer efficiency and reliability. Additionally, the number of buffers should be sufficient to handle transfers between multiple data streams and multiple data sources. If the number of buffers is too small, data congestion and loss will occur, further reducing data transfer efficiency and reliability.

A buffer is an important mechanism for improving data transmission efficiency and reliability. It can adjust the data transmission rate and processing capacity to meet different transmission needs and performance requirements. The size and number of buffers can be adjusted and configured according to actual transmission needs and performance requirements to maximize data transmission efficiency and reliability.

So how should the disk I/O subsystem be designed to improve the read and write capabilities of business systems?

Specific strategies need to be analyzed in conjunction with specific scenarios. The following summarizes the main optimization ideas for disk I/O:

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1. Reduce disk read/write: Optimize applications to reduce disk read/write; optimize dirty data flushing strategies.

2. Reduce blocking: Utilize caching and disk prefetching; utilize asynchronous read/write mechanisms.

3. Improve file read/write efficiency: Choose a suitable file system; choose a suitable file read/write method (fopen/open and directio); choose a suitable disk scheduling method.

Finally, I'd like to introduce iostat, the most frequently used tool for troubleshooting disk I/O problems.

The `iostat` command is a performance analysis tool in Linux systems used to monitor and report system disk and CPU performance. It displays real-time disk I/O usage, CPU usage, and system load, helping administrators diagnose and optimize system performance issues. The output of the `iostat` command includes the following information:

CPU usage: Displays CPU utilization, context switch count, interrupt count, etc.

Disk I/O usage: Displays disk I/O throughput, response time, wait time, etc.

Network usage: Displays network throughput, error rate, packet loss rate, etc.

System load status: Displays the system load status, number of running processes, etc.

The usage of the iostat command is as follows:

```
iostat [-c] [-d] [-h] [-k] [-m] [-N] [-p] [-t] [-x] [interval [count]]
```

The parameter `interval` represents the output time interval in seconds; the parameter `count` represents the number of outputs; other parameters are explained in detail below:

- c: Displays CPU usage.

- d: Display disk I/O usage.

-h: Display the output in a human-readable format.

- k: Display output results in KB.

- m: Display the output in MB.

- N: Displays the usage of the NFS file system.

-p: Displays the disk I/O usage of the specified process.

- t: Displays the timestamp of the output results.

- x: Display extended disk I/O usage.

3.2.5 Application Tuning

The preceding sections introduced the main ideas and general methods of system optimization from the perspectives of computing, network, and storage resources.

Please remember that the above theories and methods are fundamental aspects that are involved in any system tuning and are not exclusive to the Kunpeng platform. However, due to the differences in instruction set architecture, the Kunpeng processor requires special consideration and rethinking of performance tuning when optimizing actual business systems.

Only by mastering the basic theories and continuously accumulating practical experience in actual business scenarios can one find the ultimate solution on the long road of performance optimization.

This section will introduce optimization methods for the Kunpeng platform from an application perspective. It mainly includes: compiler option optimization, Cacheline optimization, file buffering mechanism, Neon directive.

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Slightly different from resource-level optimization, application-level optimization will focus more on specific business scenarios, and the optimization methods used will be more refined. Please read carefully.

ÿ Compilation option optimization

Compilation options are a series of parameters provided by the compiler to control various behaviors and features during the compilation process. Compilation options can affect compilation speed, optimization level, and the quality of the generated code. Typically, compilation options are specified via command-line arguments. The specific content and meaning of compilation options are usually related to the compiler version and type. For example, the GCC compiler provides a large number of compilation options, including optimization options, debugging options, and warning options. Different compilers may offer different compilation options, so it is necessary to select the appropriate compilation options based on the actual compilation requirements and the compiler type.

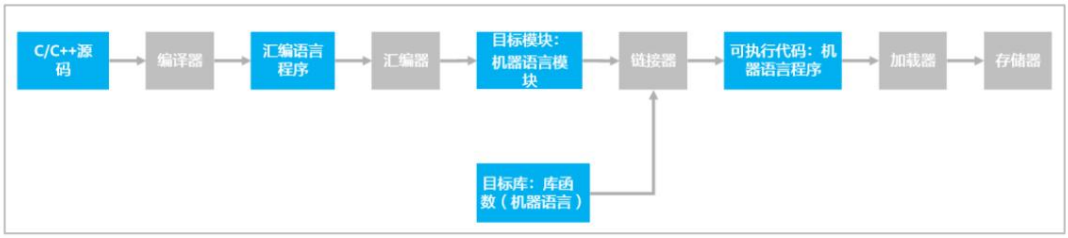


图3-14 From source code to executable program - compiled languages

Common compilation options include:

- Optimization options: These control the compiler's optimization level and strategies to improve the efficiency and performance of generated code.

- Warning options: Used to control the level and type of warnings from the compiler, helping developers to check and fix potential problems in the code.

- Debugging options: These control the debugging information and symbol table generated by the compiler, enabling source code-level debugging and tracing.

- Target options: Used to specify the target platform and architecture of the compiler to generate code compatible with the target platform.

- Library options: These specify the library files and paths that the compiler links to, so that functions and variables from the libraries can be used during the compilation process.

- Input/output options: Used to specify the compiler's input and output files, as well as the type and name of the generated object files.

Compiler options are a series of parameters provided by the compiler to control various behaviors and features during the compilation process. The specific content and meaning of compiler options are usually related to the compiler version and type, and appropriate compiler options need to be selected based on the actual compilation requirements and the compiler type.

On the Kunpeng platform, the BiSheng Compiler is provided, adapted to the underlying architecture. In addition to the compilation capabilities of open-source GCC, the BiSheng Compiler has also been specially developed with enhanced features adapted to the Kunpeng processor: it supports Kunpeng microarchitecture chip and instruction optimization, and provides higher performance than open-source LLVM through hardware and software collaboration; it also integrates Auto-tuner features to support automatic compiler tuning.

For more information about the Bisheng compiler, please refer to the Kunpeng Community documentation - Bisheng Compiler (https://www.hikunpeng.com/document/detail/zh/kunpengdevps/compiler/ug-bisheng/kunpengbisheng_06_0001.html).

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ÿ Cacheline optimization

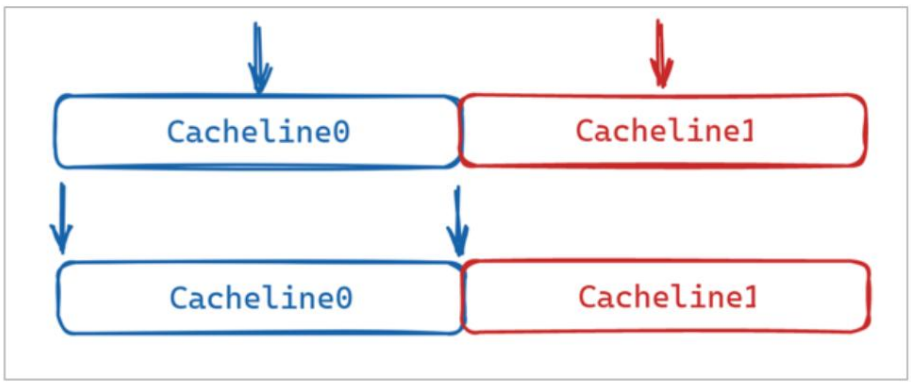


图3-15 **Cacheline** optimization

A cache line is a concept in computer architecture, a memory unit used to cache data accessed by the processor. A cache line typically consists of multiple bytes or words, usually 64 or 128 bytes in size. Cache lines were introduced to address the speed mismatch between the processor and main memory. When the processor accesses main memory, due to the slower speed of main memory, the processor needs to spend a considerable amount of time waiting for a response. Cache lines allow some main memory data to be cached in the processor cache, thereby improving data access speed and efficiency.

The size and number of cache lines have a significant impact on system performance and efficiency. Generally, the size of a cache line should be large enough to maximize caching efficiency and performance. If a cache line is too small, cache invalidation can occur, reducing data access speed and efficiency. Conversely, the number of cache lines should also be sufficient to cache more data and improve data access efficiency. If the number of cache lines is too small, cache contention and conflicts can occur, further reducing data access speed and efficiency.

Cache lines are an important concept in computer architecture, used to cache data accessed by the processor, improving data access speed and efficiency. The size and number of cache lines have a significant impact on system performance and efficiency, and need to be adjusted and configured according to actual access needs and performance requirements.

Cacheline alignment is an improvement made by the Kunpeng processor in terms of memory usage efficiency. Combined with the design and use of the new kernel of the openEuler operating system, it can optimize the execution efficiency of system programs in virtualization/Java memory-intensive application scenarios.

In most processors, data is transferred from the cache to main memory in cache lines (typically 64 bytes, sometimes called cache blocks; this article uses "Cache line" consistently). The CPU loads data from memory one cache line at a time, and writes data to memory one cache line at a time. Suppose the processor first accesses data object A, which is exactly 64 bytes in size. If the starting address of data object A is not aligned, meaning data object A occupies part of two different cache lines, the processor needs two memory accesses to access this data object, resulting in low efficiency. However, if data object A is memory aligned, meaning it resides within a single cache line, the processor only needs one memory access, significantly improving efficiency. The compiler can improve program performance by strategically arranging data objects to avoid unnecessarily spanning multiple cache lines, concentrating the same object within a single cache, and thus effectively utilizing the cache. The above purpose is achieved by allocating objects sequentially, that is, if the next object cannot be placed in the remaining part of the current cacheline, then those remaining parts are skipped and the object is allocated from the beginning of the next cacheline, or by allocating objects of the same size in the same storage area, and aligning all objects on multiples of the size boundary.

File caching mechanism selection

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The disk I/O subsystem introduced the concept of a cache and discussed the impact of cache size and quantity on data access. In addition, the choice of file caching mechanism directly affects file/data access efficiency; here we introduce two mechanisms: Clib Buffer and PageCache.

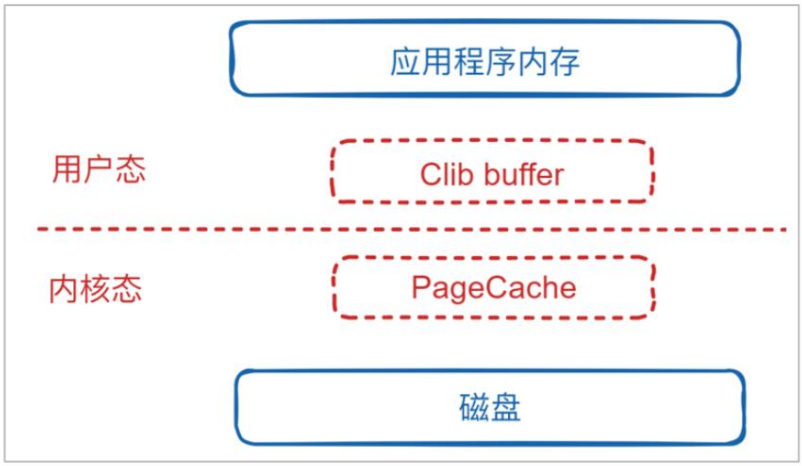


图3-16 File caching mechanism

Clib buffer: A data buffering mechanism in user space. When enabled, data copied from the application buffer to the clib buffer is not immediately synchronized to the kernel. Instead, it is synchronized to the kernel only when the buffer reaches a certain size or is actively triggered. When querying data, data is first queried from the clib buffer, thereby reducing the switching between user space and kernel space.

PageCache: A kernel-level file caching mechanism. When a user reads or writes a file, they first operate on the PageCache. The kernel will then synchronize the data to the disk according to the scheduling mechanism or when it is actively triggered by the application, thereby reducing disk access.

Understanding the impact of the two file mechanisms allows applications to match caching mechanisms and file read/write methods according to specific scenarios.

So what are the file read/write methods? Here, file read/write methods refer to the operating system level, specifically system call methods.

Sort by memory copy count: fopen > open > direct io ÿ mmap. Sort by system call count: open = direct io >

fopen > mmap.

Why is it important to understand the resource consumption of these system calls? Mainly, it's to make more appropriate matching decisions in specific business scenarios, thereby maximizing the efficiency of system resource utilization.

The `fread` and `fwrite` functions utilize the clib buffer caching mechanism, while `read` and `write` do not. Therefore, `fread` and `fwrite` involve an additional layer of memory copying—a copy from the application buffer to the clib buffer. However, `fread` and `fwrite` require fewer system calls than `read` and `write`. For operations involving large numbers of bytes read or written each time, memory copying consumes more resources than system calls; in this case, `read` and `write` can be used to reduce memory copying. For operations involving small numbers of bytes read or written each time, system calls consume more resources than memory copying; in this case, it is recommended to use `fread` and `fwrite` to reduce the number of system calls.

ÿ Neon command

Over the past 50+ years of development, Intel processors have incorporated SIMD technology to improve computing power per unit area. The most well-known implementation of SIMD is the SSE (Streaming SIMD Extensions) instruction set, which includes SSE, ...

There are multiple versions such as SSE2 and SSE3, with the latest being the AVX-512 instruction set. These instruction sets are widely used in Intel processors and can greatly improve the computing speed and efficiency of computers.

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SIMD (Single Instruction Multiple Data) is a parallel computing technique that improves data processing efficiency and performance by processing multiple data sets with a single instruction. SIMD typically uses vector registers to store multiple data sets and employs specific instructions to operate on these data sets, such as addition, subtraction, multiplication, and division.

In SIMD technology, each vector register contains multiple data elements, and operations on these elements can be performed simultaneously within a single clock cycle. This parallel computing approach makes SIMD technology excel in computationally intensive applications such as image processing, signal processing, and scientific computing.

The main advantage of SIMD technology is its ability to process multiple data sets within a single instruction cycle, significantly improving data processing efficiency and performance. Furthermore, SIMD technology reduces the number of instructions and the amount of data transmitted, thereby lowering power consumption and latency. Compared to traditional Single Instruction Single Data (SISD) computing, SIMD technology can greatly enhance the speed and efficiency of computer operations.

In practical applications, SIMD technology is widely used in multimedia fields, such as video encoding and decoding, audio processing, and image processing. Furthermore, SIMD technology is also used in scientific computing, data mining, and artificial intelligence.

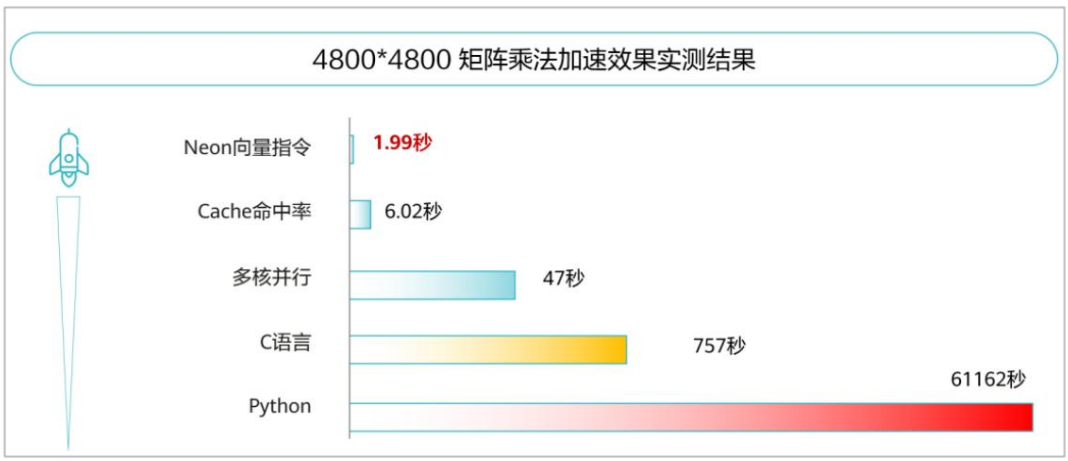
While SIMD technology offers numerous advantages, it also has some drawbacks. First, SIMD requires specific hardware support; if the processor does not support SIMD, it cannot be used. Second, using SIMD requires specific programming skills, placing a high demand on programmers. Furthermore, not all computational tasks are suitable for SIMD; for tasks with many branches or strong data dependencies, SIMD's effectiveness is not significant.

SIMD (Simultaneous Data Encryption) is a parallel computing technique that improves data processing efficiency and performance by processing multiple data points with a single instruction. SIMD has wide applications in multimedia, scientific computing, and data mining, but requires specific hardware support and programming skills.

The Kunpeng processor's multi-core feature inherently gives it the ability to process data in parallel. So, can the Kunpeng processor run SIMD instructions?

The answer is no! But Neon instructions can be executed.

Neon technology is a SIMD (Single Instruction Multiple Data) technology under the ARM architecture. It improves data processing efficiency and performance by processing multiple data elements with a single instruction. Neon technology was initially designed for multimedia processing in mobile devices, such as image and audio processing. The core of Neon technology is the SIMD register, which is 128 bits in size and can store multiple data elements. Neon technology can operate on multiple data elements simultaneously within a single clock cycle, thus achieving highly efficient data processing.



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图3-17 **Neon** command execution effect

Besides its application in multimedia processing on mobile devices, Neon technology is also widely used in computationally intensive applications such as gaming, virtual reality, image processing, and digital signal processing. The advantages of Neon technology lie in its low power consumption, high performance, and low latency, which can significantly improve device performance and efficiency.

Neon technology has the following advantages compared to other SIMD technologies:

Low power consumption: Neon technology employs a low power consumption design, which can save energy while ensuring high performance.

High performance: Neon technology can operate on multiple data simultaneously within a single clock cycle, thereby achieving efficient data processing.

Low latency: Neon technology features low latency, enabling it to respond quickly to user actions.

Easy to program: Neon technology uses an instruction set similar to the ARM instruction set, which makes it easy to program and optimize.

Neon technology is a SIMD technology under the ARM architecture that can improve data processing efficiency and performance by processing multiple data points with a single instruction. Neon technology features low power consumption, high performance, and low latency, and is widely used in computationally intensive applications such as multimedia processing, gaming, virtual reality, and image processing in mobile devices.

3.3 Kunpeng Performance Analysis Tool

The first few sections of this chapter mainly introduce the basic methods and commonly used tools for software performance testing. Focusing on the three resource levels of computing, network, and storage, it elaborates on the key theories of performance optimization and introduces the corresponding tool usage methods, aiming to establish a macro-level panoramic view of performance testing and optimization for students.

The last section of this chapter will introduce the Kunpeng Performance Analysis Tool, a toolbox for testing and tuning based on the Kunpeng platform.

3.3.1 Tool Introduction

The Kunpeng Performance Analysis Tool is a performance analysis software for Huawei Kunpeng servers. It helps users quickly locate system performance bottlenecks, optimize system performance, and improve system stability and reliability.

This tool has the following features:

It supports real-time monitoring and analysis of various performance parameters of the Kunpeng server system, including CPU utilization, memory utilization, disk I/O, network bandwidth, etc.

It supports real-time monitoring and analysis of the performance data of various applications in the system, including application response time, application concurrent request count, application error rate, etc.

It supports displaying performance data in chart form, making it convenient for users to perform visual analysis.

The Kunpeng Performance Analysis Tool is a highly practical performance analysis software that helps users quickly pinpoint system performance bottlenecks and improve system stability and reliability. It is an indispensable tool for Huawei Kunpeng server users.

The tool's use cases are mainly divided into: tuning assistant, system performance analysis, Java performance analysis, and system diagnosis. The following sections will elaborate on these four scenarios.

It should be noted that the Kunpeng performance analysis tool requires the underlying processor to be Kunpeng 916/920. There is no clear test data on the compatibility of other ARM processor versions.

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How the Kunpeng Performance Analysis Tool Works

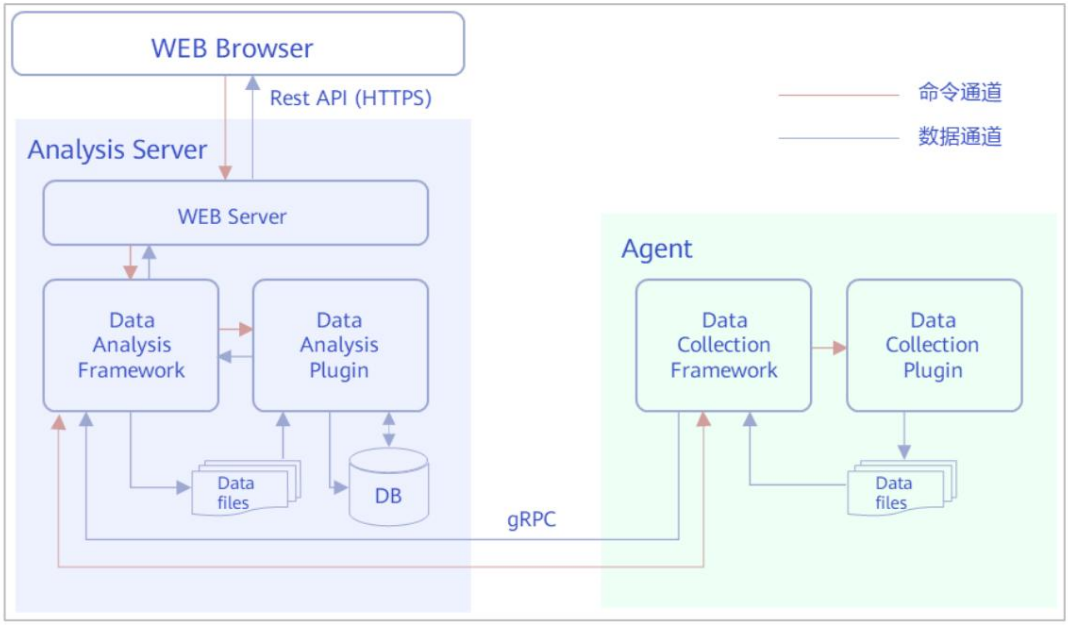


图3-18 Kunpeng Performance Analysis Tool Logical Model

The image above shows the latest logical model of the Kunpeng performance analysis tool.

Structurally, the tool is divided into an AnalysisServer and an Agent. The AnalysisServer performs performance data analysis and presents the results, while the Agent collects performance data.

The main components are described in the table below:

表3-1 Introduction to **the Analysis Server** module

Module Name	Function
Web Browser	Web browsers are used for interactive operations and data presentation.
Web Server	The web server receives requests from web browsers and triggers the Data Analysis Framework to perform specific business processing.
Data Analysis Framework	The main functions of the data analysis framework are: to instruct the Data Collection Framework to collect data and receive the collected data files; to call the corresponding Data Analysis Plugin to store and analyze the data files and save the analysis results; and to provide a channel for querying the analysis results to the web server.
Data Analysis Plugin	Data analysis plugins are available for different performance analysis functions. Their main functions include: preprocessing data files and importing them into the database; analyzing raw data to determine the most suitable data format for display and the relationships between data points; and providing optimization suggestions based on past project tuning experience. They also provide an interface for querying performance analysis results.

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表3-2 Introduction to the **Agent** Module

Module Name	Function
Data Collection Framework	The main function of the data acquisition framework is to receive acquisition notifications from the Data Analysis Framework, invoke the corresponding Data Collection Plugin to acquire data, and push the acquired data files to the Data Analysis Framework.
Data Collection Plugin	Data acquisition plugins are available for different performance analysis functions. Their main function is to collect specific performance data and save it to a file.

In practice, the Analysis Server and Agent tools can operate on different servers (Server A runs the performance analysis tool, Server B runs the business software), i.e., a distributed deployment, or they can operate on a single server (Server A runs the business software and also has the performance analysis tool installed), i.e., a single-machine deployment. The Server side is the brain of the analysis tool, while the Agent side is the plugin for collecting information.



图3-19 Distributed deployment

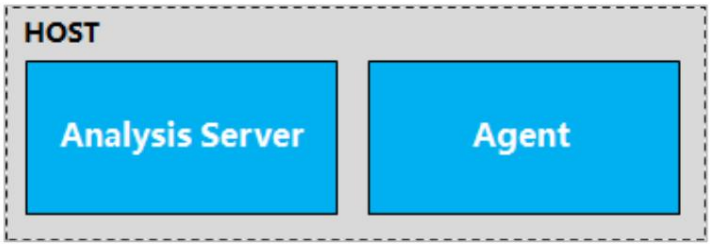


图3-20 Standalone deployment

3.3.2 Tuning Assistant

The tuning assistant tool systematically organizes and analyzes information such as performance indicators, hot functions, and system configurations to form a chain of system resource consumption. It guides users to analyze performance bottlenecks according to the optimization path and provides optimization suggestions and operation guidance for each optimization path, thereby achieving rapid tuning.

The main functions include: system configuration, hot functions, system performance, and process/thread performance.

ÿSystem Configuration

Collect and analyze system configuration information that is highly correlated with performance, and guide users to analyze related performance bottlenecks based on optimization paths.

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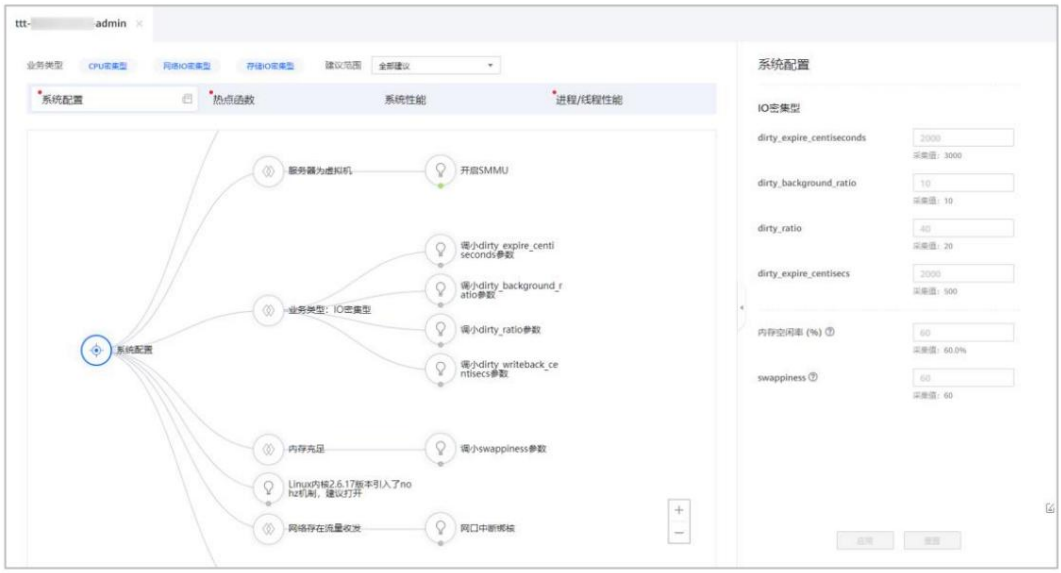


图3-21 System configuration optimization indicators

Optimization suggestions and operational instructions will be provided for each optimization path.

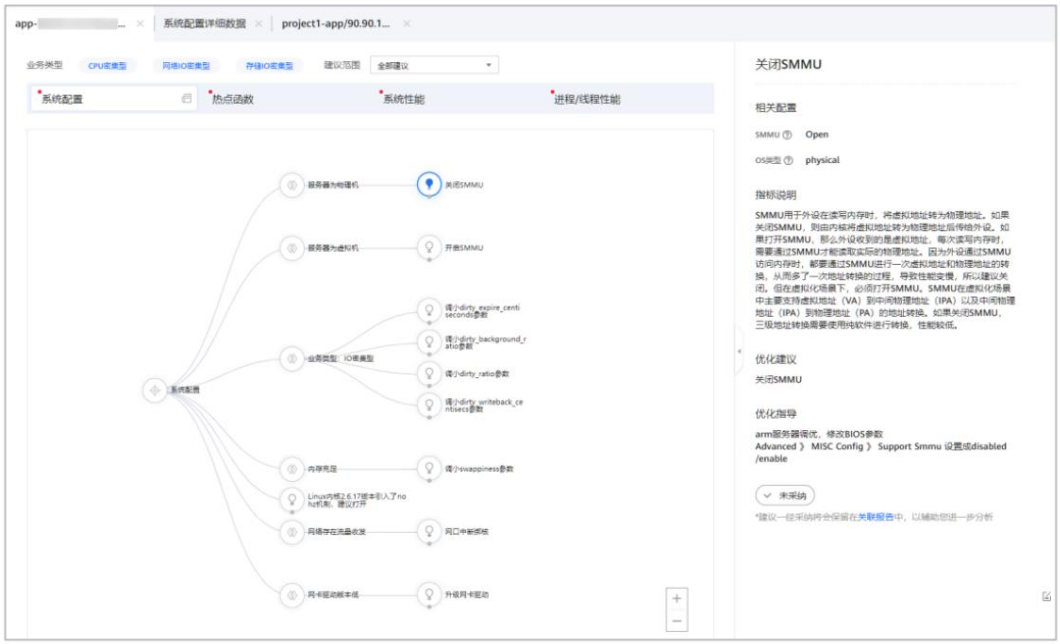


图3-22 Optimization suggestion topology tree diagram

Hotspot functions

Collect and analyze hotspot function information to guide users in analyzing related performance bottlenecks based on optimization paths.

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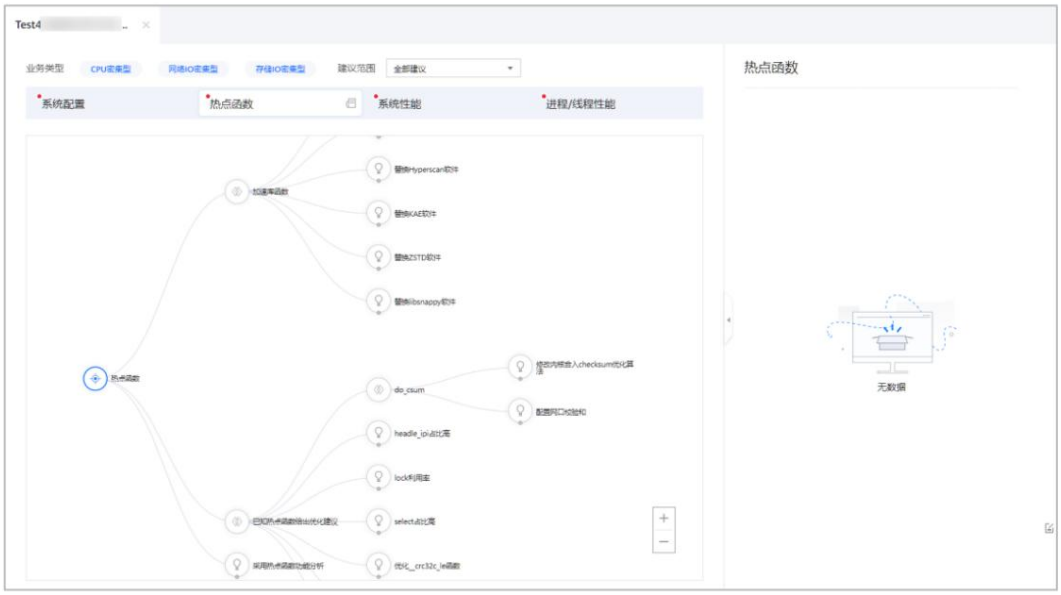


图3-23 View the hotspot function analysis results

Optimization suggestions and operational instructions will be provided for each optimization path.

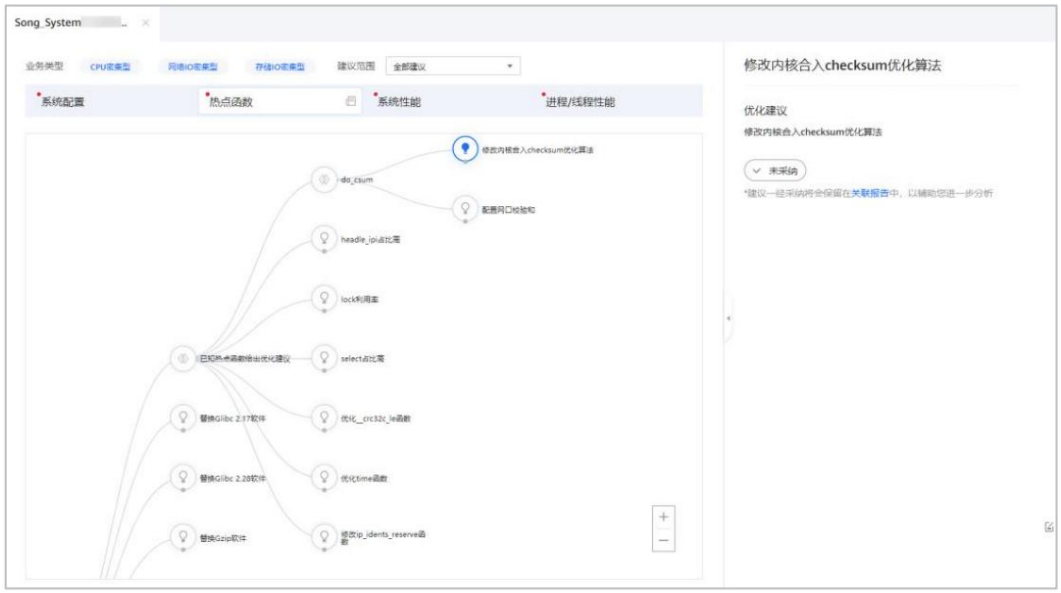


图3-24 Optimization suggestion topology tree diagram

3.3.3 System Performance Analysis

System performance analysis is a performance analysis tool for Kunpeng-based servers. It can collect performance data at various levels, including server processor hardware, operating system, processes/threads, and functions, analyze system performance indicators, locate system bottlenecks and hot functions, and provide optimization suggestions. This tool can help users quickly locate and resolve software performance issues.

Main functions include: panoramic analysis, OpenMP/MPI analysis, microarchitecture analysis, and memory access analysis. The commonly used panoramic analysis function is introduced below.

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By collecting system hardware and software configuration information, as well as the operating status of system CPU, GPU, memory, storage I/O, and network I/O resources (including RDMA RoCE), metrics such as their utilization, saturation, and error counts are obtained to identify system bottlenecks. For certain system metrics, optimization suggestions are provided based on existing benchmark values and optimization experience. The system also supports collecting partial performance data to control data volume and improve processing speed.

The system checks the hardware, system, and component configurations for big data and distributed storage scenarios, identifies non-optimal configurations, and provides typical hardware configurations and software version information.

ÿSystem Configuration

Displays CPU and subsystem component diagrams (shown in physical machine environment) and information on each configuration item, and provides optimization suggestions for some configuration items based on current experience.

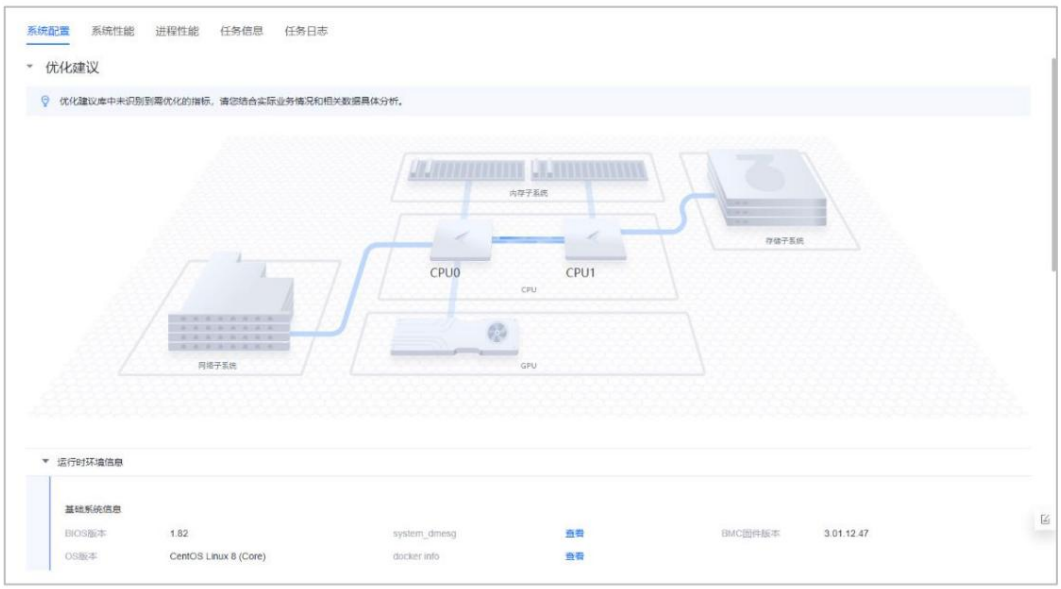


图3-25 System Configuration

PCIe Topology Information

Display detailed information about each device in a topology diagram format.

Hovering the mouse over a device icon in the topology diagram displays detailed device information and suggestions; clicking the corresponding device icon displays detailed information about that device.

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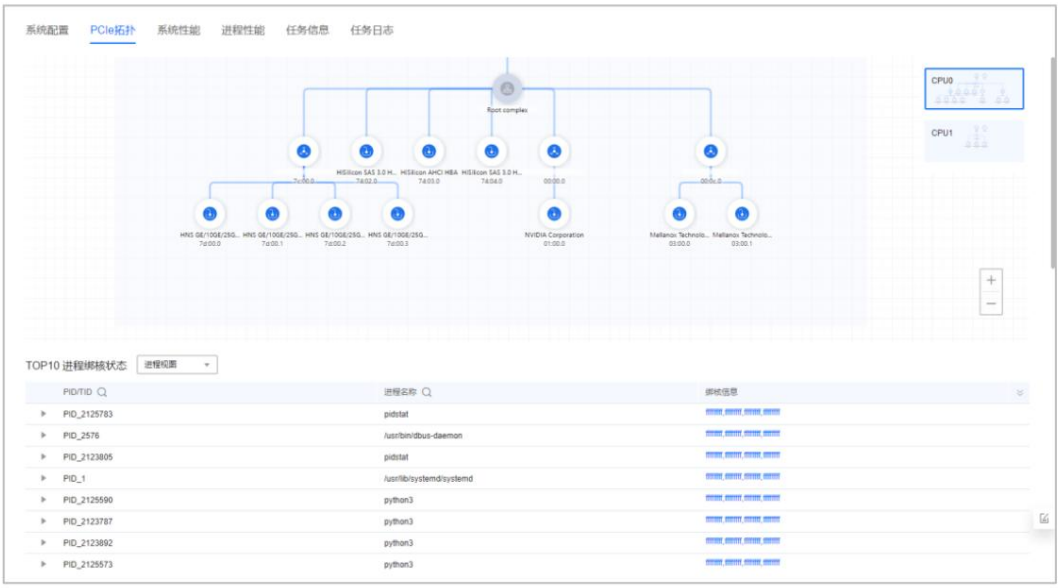


图3-26 **PCIe** Topology Information

ÿSystem Performance

The time-series data of each system performance indicator throughout the entire data acquisition process are displayed using a line graph.

The system displays optimization suggestions and average values for each performance metric in a list view. If the average value of a metric exceeds its baseline, it is highlighted in red, with the direction indicating whether it is greater than or less than the baseline. Optimization suggestions will bubble up when the mouse hovers over the average value.

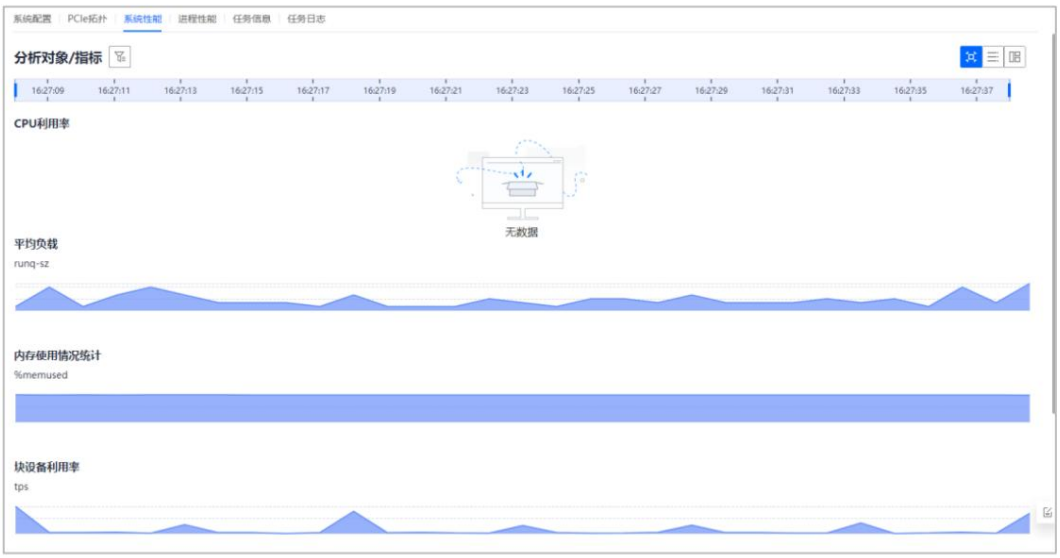


图3-27 System performance

Typical Configuration

The system checks the hardware, system, and component configurations for big data and distributed storage scenarios and displays non-optimal configurations. It also analyzes and provides typical hardware configurations and software version information.

Machine Translated by Google



图3-28 Typical configuration in big data scenarios

3.3.4 Java Performance Analysis

Java performance analysis is a performance analysis and optimization tool for Java programs running on Kunpeng servers. It can graphically display information such as the Java program's heap, threads, locks, and garbage collection, collect hot functions, locate program bottlenecks, and help users take targeted optimizations.

The Java performance analysis features supported are as follows:

ÿOnline Analysis

Online analytics encompasses dual analysis of the target JVM and the Java program. This includes internal Java Virtual Machine states such as the Heap, GC activity, and thread states, as well as performance analysis of the upper-level Java program, such as call chain analysis, hotspot functions, lock analysis, program thread states, and object creation distribution. JVM runtime data is acquired online via an agent for precise analysis.

The main analytical results include:

Overview

Displays the Java Virtual Machine system status online.

It displays the JVM's Heap size, GC activity, number of threads, number of classes loaded, and CPU utilization online.

[Hot Topic Analysis]

By analyzing PMU event sampling, we can identify hot calls that consume CPU resources within the target Java application and their call relationships, and trace hot methods in the Java/C++/OS Kernel and their associated assembly instructions.

[Memory Information]

By capturing heap snapshots, we can analyze the histogram distribution and dominant call relationships of the heap at a certain moment, trace the reference chain from each Java live object in the heap memory to the GC root, and help locate potential memory problems. By comparing and analyzing heap snapshots at different times, we can provide changes in heap usage and allocation, and help users discover abnormal situations in the allocation and use of heap memory.

Get the number and size of each object created in the Java heap, display the relevant memory usage and refresh it in real time.

By collecting various metrics of GC events, we can analyze GC performance issues.

[GC Information]

By capturing memory snapshots of the target Java application, the histogram distribution and dominant call relationships of the heap in the target Java application at a certain moment can be analyzed. Users can trace the reference chain from each surviving Java object in the heap memory to the GC root, which helps to locate potential memory problems.

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[IO Information]

Analyze data such as file I/O, socket I/O latency, and bandwidth consumption in the application online to identify hot I/O operations.

[Thread Information]

It can obtain the real-time active thread status and current thread dump in the current JVM, graphically display the thread locking status, and analyze thread deadlock situations.

[Database Information]

Monitor and analyze the database connection pool. Monitor the status of database connection pool connections, help users locate potential connection leaks, and provide optimization suggestions for inappropriate connection pool configurations.

Analyze frequently accessed SQL operations in JDBC. Record SQL call times, execution times, and stack traces in the application to help users identify the most time-consuming SQL operations.

Analyze NoSQL hotspot operations. Record the time, duration, and stack trace of NoSQL database access/operation calls in the application to help users identify the most time-consuming hotspot NoSQL operations.

[HTTP Information]

Record the HTTP request times and durations in the application to identify hot HTTP requests.

[SpringBoot Information]

With user authorization, third-party frameworks, such as Spring Boot Actuator, can be used to collect metric and performance data, which are more closely aligned with user business needs.

[Snapshot Information]

It supports generating snapshots during online analysis of heap, IO, and workload, and comparing these snapshots to help users discover trends in resource and business-related metrics, and locate potential resource leaks or performance degradation issues.

ÿSampling Analysis

By sampling, internal JVM activity/performance events are collected and analyzed offline through recording and playback. This method incurs minimal overhead on the system and has little impact on business operations, making it suitable for large Java programs.

The main analytical results include:

Overview

Displays the Java Virtual Machine system status.

The JVM's Heap usage, GC activity, IO consumption, and CPU utilization are displayed through sampling and playback.

[Thread dump and lock analysis results]

Analyze program thread states and locks. Obtain thread state changes and current thread dumps within the sampling time, graphically display thread locking states based on the thread dumps, and analyze thread deadlock situations.

Analyze the heap objects used by the analysis program. Obtain the number, size, and stack trace of each object in the Java heap during the sampling time.

Analyze and estimate the objects and duration of thread blocking.

[Method Sampling Analysis Results]

Analyze the proportion and location of CPU cycles in hot functions in Java and native code.

It supports viewing hot functions and their call stacks through flame graphs.

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[Memory Analysis Results]

By sampling objects with long retention periods, we can analyze potential heap memory leaks in Java applications and help users locate potential causes.

Based on the sampling and analysis records, reports and optimization suggestions will be generated regarding startup parameters and GC.

[IO Analysis Results]

Analyze the latency and bandwidth consumption of file I/O and socket I/O in the application to identify hot I/O operations.

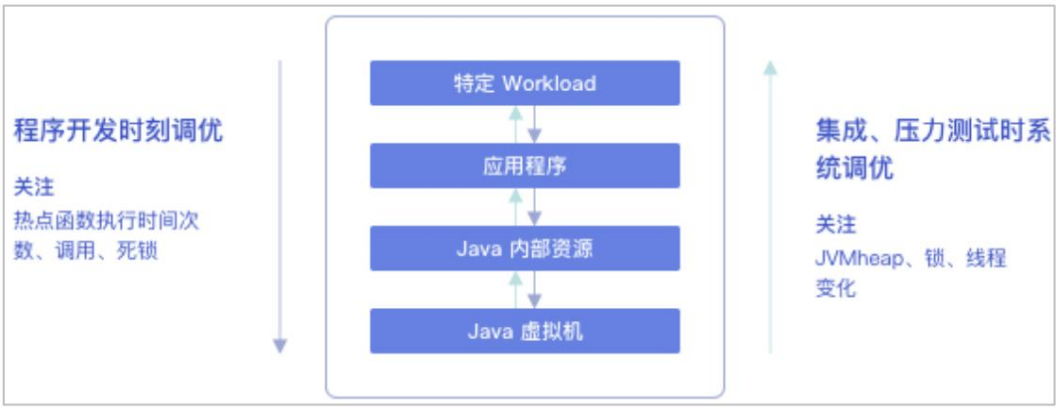


图3-29 Performance optimization perspective

3.3.5 System Diagnosis

The system diagnostic tool analyzes system performance metrics to identify anomalies such as memory leaks, memory overflows, and network packet loss, and provides optimization suggestions. The tool also supports load testing of the system, such as network I/O and storage I/O, to evaluate maximum system performance.

The system diagnostic tool supports the following functions:

Memory Diagnostics

Provides memory diagnostic capabilities to help users identify memory usage problems in applications and improve program reliability. Specifically, this includes memory leak diagnosis and memory out-of-bounds diagnosis.

Memory leak diagnosis

Analyze memory leaks in the application (including unreleased memory and abnormal memory release) to obtain specific leak information, and support the association of call stack information and source code.

Real-time tracking of memory consumption at the system layer, application layer (calling memory allocation functions), and allocator layer during application runtime.

Analyze the process memory state, system memory state, and call stack information when an OOM occurs.

Memory out-of-bounds diagnostics: Analyzes abnormal memory access points in the application, provides the abnormal access type and memory access information, and supports linking the call stack and source code.

Network I/O Diagnostics

The system performs network stress testing to determine the network's maximum capacity, providing fundamental reference data for network I/O performance optimization. It also diagnoses the network, pinpointing and resolving network I/O performance issues caused by network configuration problems and anomalies. Specific functions include:

Network testing: Obtain network operating status and maximum capacity, including connectivity, bandwidth, retransmission, packet loss rate, latency, jitter, etc., to assess network quality.

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Packet loss diagnosis: Identify network packet loss points, pinpoint the root cause of network packet loss, and provide repair suggestions.

Network packet capture: assists in network testing and packet loss diagnosis, and locates the root cause of network anomalies based on captured network message packets.

System load monitoring: Used to monitor system load during diagnostics, including CPU, memory, network I/O, hardware and software interrupts, etc., to understand whether there is insufficient system resources when network testing or network packet loss occurs.

Storage I/O Diagnostics

Stress testing of storage I/O yields the maximum capacity of storage devices, including throughput, IOPS, and latency, and this data is used to evaluate storage capabilities, providing fundamental reference data for storage I/O performance optimization.

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4 Kunpeng Instant Communication Platform Comprehensive Experiment

4.1 Introduction to the Kunpeng Instant Messaging Platform

4.1.1 Platform Function Introduction and Implementation Methods

4.1.1.1 Brief Introduction to the Requirements of the Kunpeng Instant Messaging Platform

Based on business needs, this experiment aims to build a comprehensive communication platform called "Kunpeng Instant Messaging Platform." The Kunpeng Instant Messaging Platform provides users with a platform for instant communication. Users can create conversations and communicate instantly with other users, sending text and images, viewing recent chat history, and searching for trending keywords in big data.

4.1.1.2 Functional Design of Kunpeng Real-Time Communication Platform

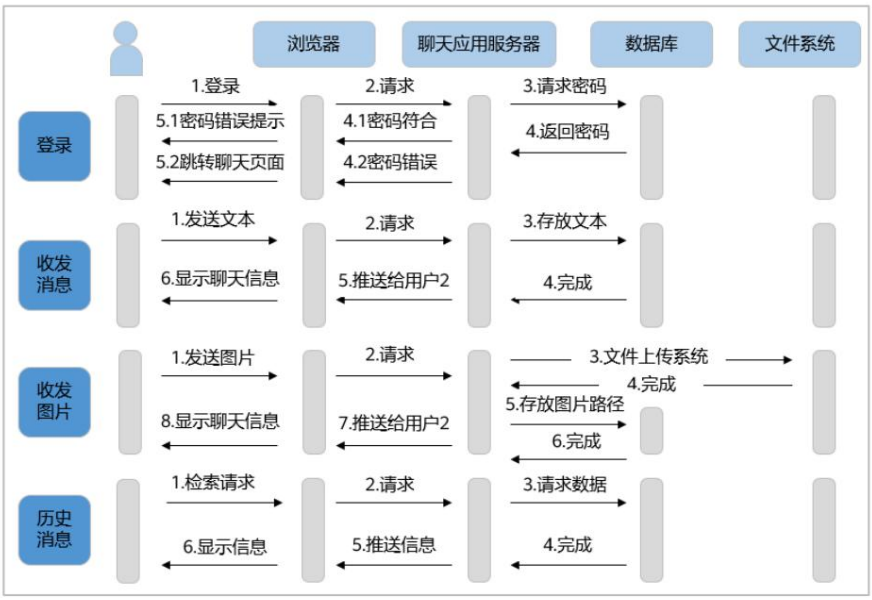
The core functions of the Kunpeng real-time communication platform are mainly divided into four aspects: login, real-time sending and receiving of messages, real-time sending and receiving of images, and querying historical information. All four functions are written in Java.

1. Login: Includes sub-functions such as login, request, and password verification.

2. Real-time message sending and receiving: Includes sub-functions such as sending text, storing text, pushing messages, and displaying chat information.

3. Real-time sending and receiving of images: Includes sub-functions such as sending images, uploading images to the file system, image push, and displaying chat information.

4. Query historical information: This includes sub-functions such as search request, requested data, push information, and displayed information.



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图4-1 Functional design

4.1.1.3 Real-time chat function implementation scheme: HTTP polling

There are multiple ways to implement the information push capability in the platform's chat function. Let's first understand the HTTP polling technology.

Polling refers to a process where the client sends HTTP requests to the server at regular intervals, and the server returns the latest data to the client upon receiving the requests. Client-side polling methods are generally divided into short polling and long polling.

ÿShort polling

The client sends a regular HTTP request to the server at regular intervals. The server checks if there is any data update on the current interface. If there is, the server returns the latest data to the client; otherwise, it prompts the client that there is no data update.

Short polling means that the client needs to send requests frequently, which consumes network traffic, and the server also needs to accept requests frequently, resulting in poor real-time message push performance.

ÿLong polling

The client sends an HTTP request to the server with a long network timeout period and does not actively disconnect the connection before the HTTP connection times out. If data is returned within the timeout period, the data is sent to the client and the connection is actively disconnected. If no data is updated during the waiting period, the connection is actively disconnected after the timeout. After that, the client will establish the same HTTP request again and repeat the above process.

Long polling significantly reduces the number of client requests, saving network bandwidth, but suspending requests for extended periods still leads to resource waste.

However, both short polling and long polling have some drawbacks.

First, the client needs to constantly send requests to the server, which consumes a lot of client resources, but not every request can get new information push.

Secondly, HTTP request and response messages need to contain long header information, of which only a small percentage is actually valid data, which also consumes additional bandwidth resources.

Finally, based on the principles of the HTTP protocol, communication can only be initiated by the client. The HTTP protocol cannot allow the server to proactively push information to the client. Therefore, if the server has continuous state changes, the client will experience a lag in obtaining updated data, which cannot guarantee the user experience.

4.1.1.4 Real-time chat functionality implementation scheme: WebSocket

The WebSocket protocol is a new network protocol based on TCP. It enables full-duplex communication between the browser and the server, allowing the server to proactively send information to the client, thus solving the problem of data retrieval delay in the HTTP protocol.

The following diagram illustrates the WebSocket communication process, which consists of three steps:

1. The client initiates an HTTP request, and after a three-way handshake, a TCP connection is established; 2. After

receiving the handshake request from the client, the server sends data;

3. After receiving the connection success message, the client begins full-duplex communication using the TCP transmission channel.

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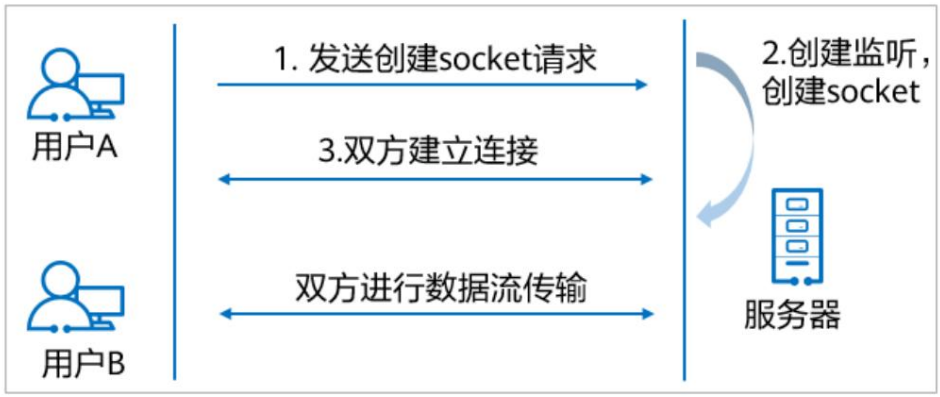


图4-2 **WebSocket** communication process

4.1.1.5 Implementation of Real-Time Chat Functionality on Instant Messaging Platform

Both WebSocket and HTTP are application-layer protocols based on TCP/IP, but WebSocket is a bidirectional communication protocol that simulates...

The Socket protocol allows for bidirectional sending and receiving of information. HTTP, on the other hand, is a one-way communication protocol, meaning requests can only be initiated by the client.

Furthermore, WebSocket and HTTP are related. During the WebSocket handshake, data is transmitted via HTTP. However, once established, the actual data transmission does not require the HTTP protocol.

4.1.1.6 Logical Architecture Design of Kunpeng Instant Messaging Platform

The platform's data flow is as follows: users access the Nginx web server through a client, Nginx is connected to an instant messaging platform, the platform's backend is written in Java, and the platform's data is uniformly stored in a MySQL database.

The backend program written in Java is the core of the platform's functionality. The following text will explain how the platform implements the hot word search function by connecting to big data components.

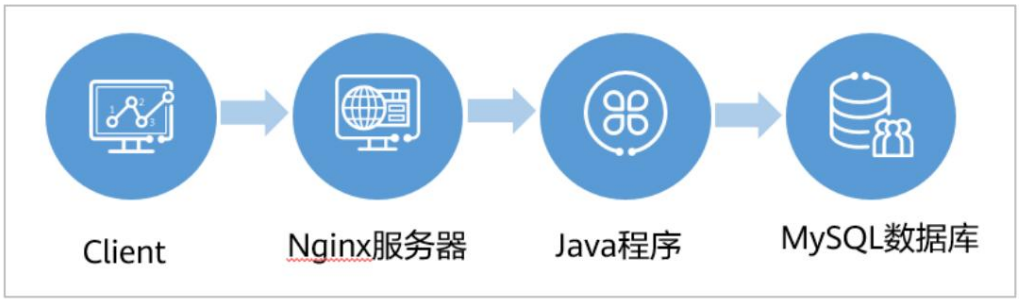


图4-3 Logical architecture design

4.1.1.7 Kunpeng Instant Messaging Platform Architecture

The platform's architecture mainly consists of five parts: Nginx, Java programs, MySQL, Hadoop, and Hive. The Java backend program is developed based on the Spring Boot and Mybatis frameworks.

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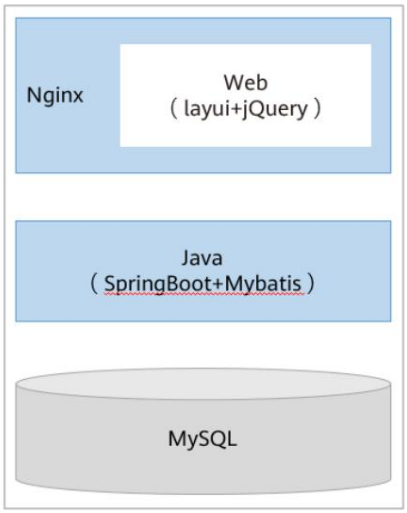


图4-4 Platform Architecture

ÿ Spring Boot Framework:

Spring Boot is a brand-new open-source framework provided by the Pivotal team on top of Spring. Its purpose is to simplify the setup and development process of Spring applications. Spring Boot eliminates a large number of XML configuration files and simplifies complex dependency management.

Spring Boot possesses all the excellent features of Spring. Whatever Spring can do, Spring Boot can do too, but it's simpler to use, more feature-rich, and more stable and robust. Therefore, we can use Spring Boot to quickly develop projects based on the Spring framework. With the rise of microservices technology in recent years, Spring Boot has also become a highly sought-after technology.

Spring Boot integrates a large number of commonly used third-party library configurations. These third-party libraries in Spring Boot applications are almost out-of-the-box with zero configuration. Most Spring Boot applications only require a very small amount of configuration code (based on Java configuration). When developing business code, we can easily focus on the business logic without having to pay too much attention to the framework configuration.

Currently , there

are many Java persistence layer framework products, the most common being Hibernate and MyBatis.

MyBatis was originally an open-source project called iBatis from Apache. In 2010, the project was moved from the Apache Software Foundation to Google Code and renamed MyBatis.

MyBatis provides a persistence layer framework that includes SQL Maps and Data Access Objects (DAOs). It eliminates almost all JDBC code and manual parameter setting, as well as result set retrieval. It uses simple XML or annotations for configuration and raw mapping, mapping interfaces and Java POJOs (Plain Old Java Objects) to records in the database.

Therefore, MyBatis, as an open-source, lightweight data persistence framework, has become an alternative to JDBC and Hibernate. Internally, it encapsulates JDBC, simplifying the complex processes of loading drivers, creating connections, and creating Statements, allowing developers to focus solely on the SQL statements themselves.

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4.1.1.8 Overview of Experimental Tasks

The experimental task is divided into four main modules: basic environment preparation, Kunpeng code migration, deployment of the real-time communication platform, and data processing optimization.

Basic Environment Preparation: Before starting the experiment, you need to set up the environment required for the subsequent experiments on a Huawei Cloud server, including Nginx, MySQL will facilitate subsequent experiments.

Kunpeng Code Migration: Use the Kunpeng Code Migration tool to migrate the source code of the JNA backend dependency library for instant messaging platforms and compile the source code.

Instant Messaging Platform Deployment: The deployment of the Kunpeng instant messaging platform is divided into front-end and back-end project deployment. The front-end is implemented using layui + jQuery, and the back-end is written in Java, based on the Spring Boot + Mybatis framework.

Data Processing Optimization: During the platform's data processing, based on big data solutions, Hadoop and Hive are deployed to complete a series of data analysis tasks, and performance optimization is performed for big data scenarios.

4.1.2 Introduction to MySQL 4.1.2.1

What is a Relational Database?

Definition of a relational database: A relational database is a database that uses a relational model to organize data. The relational model refers to a two-dimensional table model, and a relational database is a data organization composed of two-dimensional tables and the relationships between them.

Advantages of relational databases:

1. Easy to understand: The two-dimensional table structure is a concept that is very close to the logical world. The relational model is easier to understand than other models such as network and hierarchical models.

2. Easy to use: The universal SQL language makes it very convenient to operate relational databases.

3. Easy to maintain: Rich integrity (entity integrity, referential integrity and user-defined integrity) greatly reduces the probability of data redundancy and data inconsistency.

Commonly used relational databases:

OracleÿMySQLÿSQL ServerÿPostgreSQLÿGaussDBÿopenGauss ÿÿ

4.1.2.2 Introduction to MySQL

MySQL is a relational database that uses SQL to perform CRUD operations.

MySQL is an open-source and free database that is cross-platform, supports distributed systems, and works perfectly with web development languages such as PHP and Java, making it ideal for small and medium-sized enterprises as a web database (website database).

Advantages of MySQL

1. MySQL is an open-source database.

MySQL is an open-source database, meaning anyone can obtain its source code. This allows anyone to fix MySQL's flaws and use the database for any purpose. MySQL is a freely usable database.

2. MySQL's cross-platform advantages

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MySQL can run not only on Windows operating systems, but also on UNIX, Linux, and Mac OS. Because many websites choose UNIX and Linux as their server architecture, MySQL's cross-platform compatibility ensures its advantage in web applications. While Microsoft's SQL Server database is an excellent commercial database, it only runs on Windows operating systems. Therefore, MySQL's cross-platform compatibility is a significant advantage.

3. Powerful functions and easy to use

MySQL is a true multi-user, multi-threaded SQL database server. It can process large amounts of data quickly, efficiently, and securely. Compared to databases like Oracle, MySQL is very simple to use. MySQL's primary goals are speed, robustness, and ease of use.

4.1.3 Introduction to Nginx

4.1.3.1 Overview of Nginx

Nginx is a high-performance HTTP and reverse proxy server characterized by its low memory consumption and high concurrency capabilities.

Nginx has several powerful features, including forward proxy, reverse proxy, web server for static pages, load balancing, and separation of static and dynamic content.

1. Nginx can be used as a web server for static pages and also supports dynamic languages using the CGI protocol, such as Perl and PHP. However, it does not support Java. Java programs can only be executed by working with Tomcat. Nginx is specifically developed for performance optimization, with performance being its most important consideration. It can withstand high loads, and reports indicate that Nginx can support up to 50,000 concurrent connections.

2. Load balancing: Increase the number of servers and then distribute requests to each server, changing the situation where requests were concentrated on a single server to one where requests are distributed to multiple servers, thus distributing the load across different servers. This is what we call load balancing.

3. Separation of static and dynamic content: To speed up website parsing, dynamic and static pages can be parsed by different servers, thus reducing the load on a single server.

4.1.3.2 Reverse Proxy

A reverse proxy is a type of proxy server in computer networks. Based on a client's request, the server retrieves resources from one or more associated backend servers (such as web servers) and then returns these resources to the client. The client only knows the reverse proxy's IP address and is unaware of the server cluster behind the proxy server. The client can access the server without any configuration.

We only need to send the request to the reverse proxy server, which will then select the target server to retrieve the data and send it back to the client. In this case, the reverse proxy server and the target server appear to be the same server to the outside world, exposing only the proxy server address and hiding the real server IP address.

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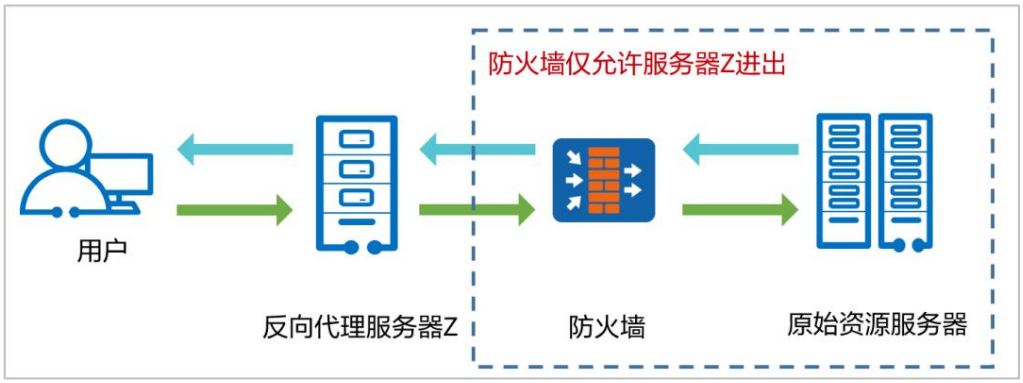


图4-5 Reverse Proxy

4.1.3.3 Forward Proxy

If we imagine the Internet outside the local area network (LAN) as a huge resource library, then clients on the LAN need to access...

Accessing the Internet requires a proxy server; this type of proxy service is called a forward proxy.

A forward proxy is a server (proxy server Z) located between the client [user A] and the origin server [server B]. To retrieve content from the origin server, user A sends a request to proxy server Z specifying the target (server B). Proxy server Z then forwards the request to server B and returns the retrieved content to the client. The client must undergo some special configuration to use a forward proxy.

The process of accessing a server through a proxy server is called forward proxying. It requires configuring a proxy server on the client side to access the specified website.

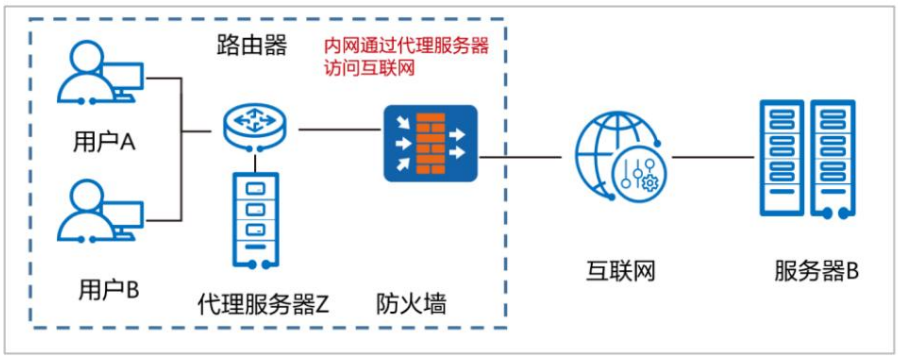


图4-6 Forward Proxy

4.2 Deployment of Basic Experimental Environment

4.2.1 Host Environment Selection

To optimize the performance of the instant messaging platform, a Kunpeng Elastic Cloud Server (ECS) was selected as the host. As a fundamental cloud service, the Kunpeng Elastic Cloud Server (ECS) is the most important service that users directly experience with Kunpeng. Users purchase a Kunpeng Elastic Cloud Server (ECS) and then add resources such as disks and networks to the cloud server, making it part of a development environment or production business cluster. The specifications of the Kunpeng Elastic Cloud Server selected in this experiment are as follows:

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Experimental Resources	Cloud server name	Configuration	**OS** version
Kunpeng Elastic Cloud Server (ECS) - Kunpeng		kc1.xlarge.2 4vCPUs | 8GiB	openEuler 22.03 64bit with ARM

4.2.2 Basic Environment Deployment

Basic environment deployment consists of two steps:

1. Deploy MySQL: MySQL serves as the database for the Kunpeng instant messaging platform, storing data generated by the platform, including user data.
   Messages, chat messages, etc.

2. Deploy Nginx: Nginx serves as the front-end server for the Kunpeng instant messaging platform, responsible for running the front-end applications.

4.2.3 MySQL Deployment Steps

Deploying MySQL involves four steps:

1. Configure the installation environment: Configure data disk, system users, etc.

2. Install dependencies: Obtain and install the dependencies required for MySQL to run.

3. Install MySQL: Configure and install MySQL using the installation package.

4. Run MySQL: Modify the configuration file to initialize the database and verify it.

4.2.4 Nginx Deployment Steps

The Nginx deployment process consists of three parts:

1. Compile and install: Obtain the dependency packages and source code, and compile the source code.

2. Configuration modification: Modify the configuration file, such as port number, running memory, and other information.

3. Run the test: Access Nginx via IP:PORT.

4.3 Kunpeng Code Migration

4.3.1 Introduction to JNA

JNA (Java Native Access) is an open-source Java framework developed by Sun Microsystems for calling native methods.

The technology is a framework built on the classic JNI foundation.

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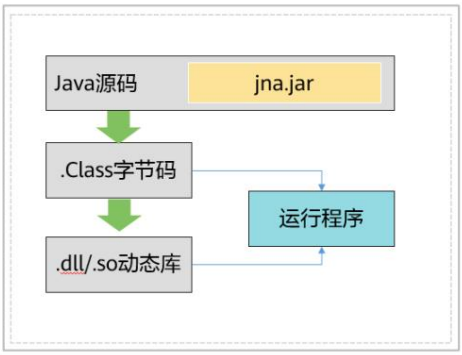


图4-7 Java framework

The backend program of the instant messaging platform calls a .so library function developed in C language. This function needs to be recompiled to run on the Kunpeng ECS server.

In the software field, a JAR file (Java Archive) is a software package file format typically used to aggregate a large number of Java class files, related metadata, and resource files (text, images, etc.) into a single file for developing Java platform applications or libraries. A JAR file is an archive file, built in ZIP format, and has the .jar file extension.

Java bytecode class files (.class) are "object files" generated by the Java compiler when compiling Java source files (.java). They are 8-bit binary stream files where data items are arranged sequentially from beginning to end without gaps. This makes the class file very compact and lightweight, allowing it to be quickly loaded into memory by the JVM and occupying less memory space (facilitating network transmission).

DLL (Dynamic Link Library) files, also known as "application extensions," are a type of software file. In Windows, many applications are not a single complete executable file; they are divided into relatively independent dynamic link libraries, or DLL files, placed in the system. When a program is executed, the corresponding DLL file is called.

Generally speaking, .so files are what are commonly referred to as dynamic link libraries, which are compiled from C or C++. Compared to Java...
Yes: It usually uses Class files (bytecode). Linux .so files cannot be run directly. Generally speaking, .so files are called shared libraries.

4.3.2 Task Flow

This step requires using the Kunpeng code migration tool to migrate the backend source code of the instant messaging platform and compile the source code.

The task process includes three steps:

1. Install the Kunpeng code migration tool.

2. Use tools for analysis and scanning: For the source code of the Kunpeng instant communication platform, use the Kunpeng code migration tool for analysis and scanning.
Describe the code and modify the source code based on the suggested changes.

3. Compile the source code: Compile the source code and generate a dynamic .so library for subsequent use by the instant messaging platform.

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4.4 Building the Kunpeng Instant Communication Platform

This step requires implementing the platform's core function, namely the chat function.

It consists of three steps:

1. Deploy the front-end project

The front-end is implemented using layui and jQuery and deployed on an Nginx server.

2. Deploy the backend project

The backend is implemented in Java, based on the Spring Boot + Mybatis framework. Maven is used for compilation and packaging.

3. Implementation of chat function

The main requirement is to implement the function of storing messages into the database, that is, to store message data into the messages table in MySQL.

4.5 Data Processing Based on Hive

4.5.1 Introduction to Big Data

4.5.1.1 Big Data Overview

In its annual Digital Universe research report, "Extracting Value from Chaos," IDC (International Data Corporation) defines big data as: Big data technology is a new generation of technologies and architectures designed to extract value from massive, diverse datasets through very rapid acquisition, discovery, and analysis, at an economically viable cost.

ÿValueÿÿ

The four Vs of big data mentioned above are Volume, Variety, Velocity, and Value.

1. Volume: The primary characteristic of big data is its large scale. With the development of the Internet, the Internet of Things, and mobile Internet technologies, all the trajectories of people and things can be recorded, resulting in explosive growth of data.

2. Variety. The wide range of data sources determines the diversity of data forms. Big data can be divided into three categories: first, structured data, such as financial system data, information management system data, and medical system data, characterized by strong causal relationships between data; second, unstructured data, such as videos, images, and audio, characterized by the absence of causal relationships between data; and third, semi-structured data, such as HTML documents, emails, and web pages, characterized by weak causal relationships between data. Statistics show that structured data currently accounts for more than 75% of the total amount of internet data, while the big data that generates value is often unstructured data.

3. Velocity: The speed of data growth and processing is a key indicator of the high speed of big data. Unlike traditional data carriers such as newspapers and letters, in the era of big data, the exchange and dissemination of big data are primarily achieved through the internet and cloud computing, resulting in extremely rapid data production and transmission. Furthermore, big data demands fast response times for data processing; for example, the analysis of hundreds of millions of data points must be completed within seconds. Data input, processing, and discarding must be effective immediately, with virtually no delay.

4. Value: The core characteristic of big data is value. In fact, the level of value density is inversely proportional to the total amount of data; that is, the higher the data value density, the smaller the total data volume, and vice versa. Any valuable data...

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Information extraction relies on massive amounts of basic data. However, an unresolved issue in the context of big data is how to use powerful machine algorithms to more quickly extract value from massive datasets.

4.5.1.2 Introduction to Big Data and Distribution of Component Relationships

The big data platform mainly provides four functions:

1. Data collection: Real-time data collection is performed using Kafka, and the data is aggregated into the platform;

2. Data storage: Massive data storage is accomplished through HDFS and HBase;

3. Data processing and computation: The system utilizes three processing engines—MapReduce (batch processing), Spark (in-memory computing), and Storm (real-time stream processing)—to process and compute data for various scenarios.

4. Data Application: Hive is used to build a data warehouse and perform SQL-like queries, ElasticSearch is used for real-time retrieval of massive document data, and Redis is used for in-memory queries of key-value data.

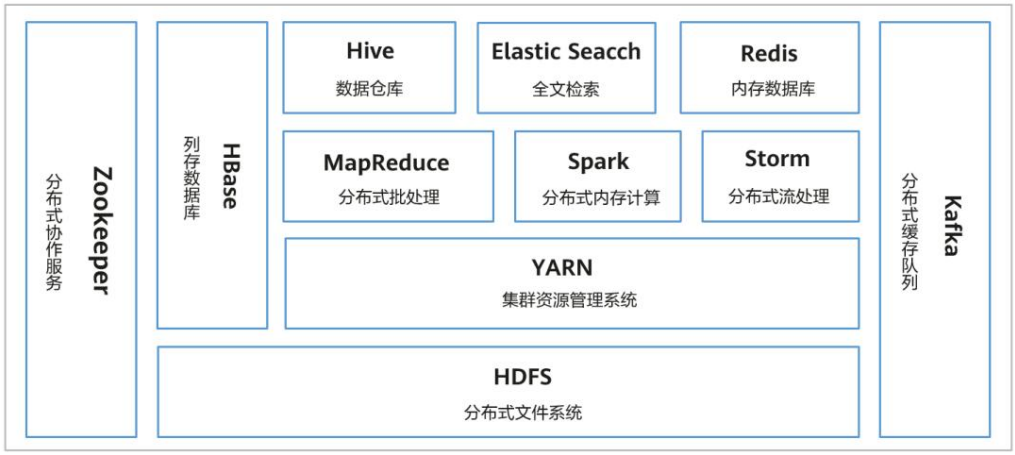


图4-8 Big data related components

The following is a detailed introduction to each component:

ÿ HDFS, or Hadoop Distributed File System for short, is a distributed file system. HDFS is a highly fault-tolerant system suitable for deployment on inexpensive machines. HDFS provides high-throughput data access, making it ideal for applications dealing with large-scale datasets.

HBase is the database for Hadoop, a distributed, scalable, big data storage solution. Designed for extremely large tables with billions of rows and millions of columns, it's a distributed database capable of random, real-time read/write access to large datasets. It provides storage capabilities similar to Google Bigtable, built on Hadoop and the Hadoop Distributed File System (HDFS). HDFS is a sub-project of the Apache Hadoop project, and HBase utilizes Hadoop HDFS as its file storage system. HBase resides in the structured storage layer, and Hadoop HDFS provides highly reliable underlying storage support for HBase. Except for some log files generated by HBase, all data files in HBase can be stored on the Hadoop HDFS file system.

Redis is a high-performance key-value storage system. Similar to Memcached, it supports a wider range of value types, including string, list, set, and sorted set.

The emergence of Redis has largely compensated for the shortcomings of key/value stores like memcached, and in some cases, it can serve as a good supplement to relational databases.

Machine Translated by Google

Kafka is a high-throughput distributed publish-subscribe messaging system that can handle all action stream data from consumer-scale websites and has become the preferred choice for big data systems between asynchronous and distributed messaging.

MapReduce is a programming model for parallel computation of large datasets (greater than 1TB). Data computed in a MapReduce program can come from multiple data sources, such as Local File System, HDFS, and databases. HDFS is the most commonly used, leveraging its high throughput to read and compute large amounts of data. After computation, the data can also be stored in HDFS.

Spark is a high-speed, general-purpose big data computing engine. It possesses the advantages of Hadoop MapReduce.
Unlike Hadoop, Spark allows intermediate outputs of a job to be stored in memory, eliminating the need to read and write to HDFS. Therefore, Spark is better suited for iterative MapReduce algorithms such as data mining and machine learning. Typically, data for Spark computations can come from multiple data sources, such as Local Files and HDFS. HDFS is the most commonly used, allowing users to read large amounts of data at once for parallel computation. After computation, data can also be stored in HDFS. It can be used with Hadoop and Apache Mesos, or independently. Storm , an open-source real-time data processing framework similar to Hadoop, is developed by Twitter.

Its simple programming model significantly reduces the difficulty of real-time processing and is one of the most popular stream processing frameworks today. Compared to other computing frameworks, Storm's biggest advantage is its millisecond-level low latency.

Hive is a data warehouse tool based on Hadoop. It can map structured data files to a database table and provide simple SQL query functionality. It can convert SQL statements into MapReduce tasks for execution. Its advantages include a low learning curve, the ability to quickly implement simple MapReduce statistics using SQL-like statements without developing dedicated MapReduce applications, making it very suitable for statistical analysis in data warehouses.

Elasticsearch can build full-text indexes (by breaking down the content of a text into several keywords and then creating an index based on those keywords). It builds an index. During a query, the index is searched based on keywords to find articles containing those keywords. It separates data and the index, sharding the index and distributing it across different nodes. The number of nodes can scale to hundreds, enabling real-time retrieval and processing of petabytes of structured or unstructured data. Simultaneously, sharding allows for replication to ensure data reliability, and the collaborative work of these shard replicas significantly improves retrieval performance. Furthermore, a simple RESTful API makes full-text search efficient and easy.

YARN is a new Hadoop resource manager, a general-purpose resource management system that provides unified resource management and scheduling for upper-layer applications, addressing the performance bottlenecks of the older MapReduce framework. Its basic idea is to separate resource management and job scheduling/monitoring functions into separate daemons.

ZooKeeper is a distributed application coordination service and an important component of Hadoop and HBase. It is a tool that provides consistency services for distributed applications, enabling nodes in a Hadoop cluster to coordinate with each other.
ZooKeeper is now an Apache top-level project, providing efficient, reliable, and easy-to-use coordination services for distributed systems.

4.5.1.3 Introduction to Hadoop

Hadoop is a software platform for analyzing and processing big data, consisting of four main modules:

Hadoop Distributed File System (HDFS): A distributed file system that runs on standard or low-end hardware. In addition to greater fault tolerance and native support for large datasets, HDFS also offers significantly higher data throughput than traditional file systems.

Yet Another Resource Negotiator (YARN): Manages and monitors cluster nodes and resource usage. It schedules jobs and tasks.

Machine Translated by Google

MapReduce: A framework that helps plan parallel computations on data. Map tasks extract input data and transform it into a dataset that can be computed on using key-value pairs. Reduce tasks use the output of the Map tasks to summarize the output and provide the desired results.

Hadoop Common: Provides common Java libraries that can be used on all modules.

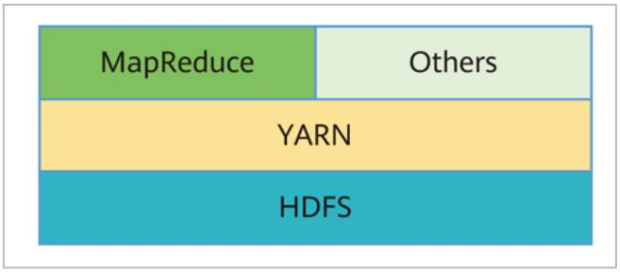


图4-9 Hadoop components

4.5.1.4 Introduction to Hive

Hive is a data warehouse infrastructure built on Hadoop. It provides a suite of tools for extract, transform, and load (ETL) data, a mechanism for storing, querying, and analyzing large-scale data stored in Hadoop. Hive defines a simple SQL-like query language called HiveQL, which allows users familiar with SQL to query data.

Hive's data computation relies on MapReduce, Spark, and Tez. Replacing the

original MapReduce with the new execution engine Tez has resulted in a significant performance improvement. Tez can convert multiple dependent jobs into a single job (requiring only one write to HDFS and fewer intermediate nodes), thereby greatly improving the performance of DAG jobs.

ÿ Hive Key Features:

1. Analysis and summarization of massive structured data.

2. Simplifies complex MapReduce task writing into SQL statements. 3. Flexible

data storage formats, supporting JSON, CSV, TEXTFILE, RCFILE, SEQUENCEFILE, and ORC (Optimized Row Columnar).

Hive Architecture:

1. User Interface: There are three main user interfaces: CLI, Client, and WebUI. CLI is the most commonly used. When CLI starts, it simultaneously starts a copy of Hive. Client is the Hive client, used by users to connect to the Hive Server. When starting Client mode, the node where the Hive Server is located needs to be specified, and the Hive Server needs to be started on that node.
WebUI accesses Hive through a browser.

2. Metadata Storage: Hive stores metadata in a database, such as MySQL or Derby. Metadata in Hive includes the table name, table columns and partitions and their attributes, table properties (whether it is an external table, etc.), and the directory where the table data is located.

4.5.1.5 Introduction to HiveQL

Hive provides HiveQL (Hive Query Language), a language similar to SQL, to manipulate structured data. Its basic principle is to automatically convert HiveQL into MapReduce or Spark tasks, thereby enabling the querying and analysis of massive amounts of data stored in the Hadoop cluster. This allows users who know SQL but do not know how to program MapReduce to use Hadoop for data processing.

Machine Translated by Google

HiveQL, or HQL for short, is a SQL dialect provided by Hive. Hive query operations strictly adhere to the Hadoop MapReduce job execution model. Hive translates the user's HiveQL statement into a MapReduce job through an interpreter and submits it to the Hadoop cluster. Hadoop monitors the job execution process and then returns the job results to the user.

HiveQL is a query language for analyzing and processing structured data in Hive's metadata store. It is very similar to SQL and highly scalable. It reuses familiar concepts from the relational database world, such as tables, rows, columns, and schemas, to simplify the learning process.

HiveQL has the following main characteristics:

HQL does not support row-level addition, modification, or deletion; all data is determined at the time of loading and cannot be changed.

Transactions are not supported .

Equi-joins are not supported; left join, right join, or inner join are generally used instead.

4.5.1.6 Big Data Computing Models, Problem Scenarios, and Representative Products

There are four main computing models for big data:

Batch processing

Batch processing computing mainly addresses the batch processing of large-scale data, which is also a very common data processing requirement in our daily data analysis work.

MapReduce is the most representative and influential big data batch processing technology. It can execute large-scale data processing tasks in parallel and is used for parallel computing of large datasets (greater than 1TB).

MapReduce greatly simplifies distributed programming by abstracting the complex parallel computing process running on large-scale clusters into two functions: Map and Reduce. Programmers can easily run their programs on distributed systems and perform computations on massive datasets even without knowledge of distributed parallel programming.

ÿAd -hoc analysis query

Suitable for the storage, management, and query analysis of large-scale data, representative products include Dremel, Presto, and Impala.

ÿ Stream processing

Streaming data is also an important data type in big data analytics.

Streaming data (or data stream) refers to a dynamic collection of data that is infinite in terms of time distribution and quantity. The value of the data decreases over time, so real-time computing is required to provide a response within seconds.

Stream processing can process continuously arriving streaming data from different data sources in real time, and provide valuable analytical results through real-time analysis.

Currently, many stream computing frameworks and platforms have emerged in the industry. The first category is commercial-grade stream computing platforms, including IBM InfoSphere Streams and IBM StreamBase. The second category is open-source stream computing frameworks, including Twitter Storm, Yahoo! S4 (Simple Scalable Streaming System), Spark Streaming, etc. The third category is stream computing frameworks developed by companies to support their own businesses, such as Facebook using a combination of Puma and HBase to process real-time data.

Image processing

In the era of big data, much big data is presented in the form of large-scale graphs or networks, such as social networks, the transmission routes of infectious diseases, and the impact of traffic accidents on road networks. In addition, much non-graph structured big data is often converted into graph models before processing and analysis.

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MapReduce, as a distributed computing framework with single input, two-stage, and coarse-grained data parallelism, often falls short when expressing multi-iteration, sparse, and fine-grained data, and is therefore unsuitable for solving large-scale graph computing problems.

Therefore, for computation of large graphs, a graph computation model is required, and many related graph computation products have emerged.

Pregel is a parallel graph processing system based on the BSP (Bulk Synchronous Parallel) model.

To address the challenges of distributed computation in large graphs, Pregel provides a scalable, fault-tolerant platform with a highly flexible API capable of describing various graph computations. Pregel is primarily used for graph traversal, shortest path calculation, and PageRank computation. Other representative graph computation products include Facebook's open-source implementation of

Pregel, Giraph, GraphX for Spark, and the graph data processing system PowerGraph.

4.5.1.7 The characteristics of big data parallel computing are naturally compatible with the Kunpeng multi-core architecture.

The Kunpeng multi-core architecture can improve the performance of big data platforms in terms of computing, storage, and networking from end to end. The following example illustrates this using the optimization of the MapReduce

process. ÿ Problems with MapReduce in traditional architectures: 1.

In Hive, reduce tasks often start too early, causing map resources to be occupied prematurely, resulting in overall performance deviations. Starting too late, on the other hand, prolongs the overall time, significantly impacting performance.

2. The default settings for MapReduce task, vcore, and mem may cause a mismatch between CPU and memory resources during actual operation, affecting overall performance.

3. Hive executes sequentially across multiple MapReduce tasks, impacting overall performance.

4. Excessive memory usage by a single map task during the map phase leads to the activation of spill operations, increasing disk pressure and reducing the efficiency of MR tasks.

Kunpeng Architecture Optimization Solution

1. Analyze use cases and rationally set the startup timing of Reduce Tasks to achieve a balance between computation and I/O, reducing end-to-end system overhead.

   This reduces system execution time and improves performance by approximately 10-20%.

2. Allocate the block size of DFS and the vcore of the MapReduce phase appropriately according to the application type (CPU-intensive/memory-intensive).

   MEM information enables resources to execute efficiently at each stage, improving performance by approximately 10-20%.

3. Analyze use cases, enable concurrency where there are no dependencies and concurrency is possible, fully decompose tasks, and set reasonable concurrency levels.

   This can be used for SQL queries with a large number of tasks, improving performance by approximately 5% per statement.

4. Set reasonable io sort mem and factor values to reduce disk spill operations during map operations, improving performance by about 5%.

4.5.1.8 Platform Data Analysis Flow Chart

For the Kunpeng instant messaging platform in this experiment, the platform data generated by the front-end application is stored in a MySQL database. When we want to perform data analysis based on the instant messaging data, we need to first transfer the data to Hadoop, and then use Hive to complete data processing and analysis, hot word retrieval and other operations.

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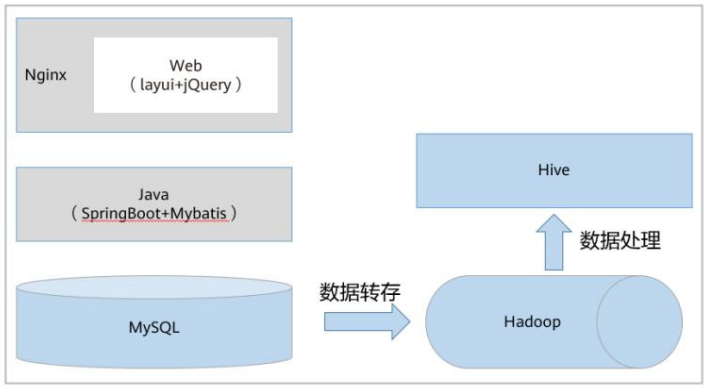


图4-10 Platform data analysis flow

4.5.2 Performance Tuning Approach

4.5.2.1 Reasons for optimization

ÿ Component parameters default values are conservative

To ensure compatibility with different environments, applications and operating systems set relatively low default values for performance-related parameters, which prevents them from maximizing the performance of cluster resources.

ÿ Rationally allocate resources to upstream and downstream components

Different components may be installed in the same big data cluster environment, and different components have different requirements for resources such as CPU, disk, and network, which need to be configured appropriately.

ÿPerformance bottlenecks vary depending on hardware configuration

Because hardware environments are often not standardized, when a hardware resource reaches its bottleneck ahead of schedule, targeted optimization is required based on the actual hardware configuration.

4.5.2.2 Common Optimization Approaches

Tuning strategies can be divided into two main categories: one is basic tuning operations, the purpose of which is to ensure that the cluster has better performance; the other is repeated resource monitoring, identifying bottlenecks, and optimizing actions to address specific problems and improve performance.

ÿEnsure testing pressure

When client-side load is insufficient to fully utilize the performance of a big data cluster, it is necessary to prioritize increasing client-side load.

ÿ Allocate physical resources

Based on the characteristics of the component, allocate as much physical resource (CPU, disk, memory, network, etc.) as possible to the component.

ÿ Monitor resource usage

Use performance monitoring tools to observe and record system status, such as CPU, disk, memory, network, application GC status, hot functions, etc.

ÿ Identify performance bottlenecks

Based on the characteristics of components and applications and monitoring data, performance bottlenecks can be identified. Bottlenecks may include physical resources, component parameters, testing tools, test networking, JVM, locks, etc.

ÿImplementation and optimization

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Optimize accordingly based on the identified bottlenecks. However, optimization methods may not always be effective, and it is necessary to further determine whether the bottleneck has been identified and whether the optimization methods are correct.

4.5.2.3 Introduction to Common Tuning Issues

The optimization of big data can be divided into three levels: application level, hardware level, and client level.

Common application-level issues include low CPU utilization and memory exhaustion despite abundant CPU resources; hardware-level issues include high network I/O utilization and excessive memory usage; if component parameters are optimized but performance is still poor, the problem may be with the client.

Hadoop consists of three core modules: HDFS, Yarn, and MapReduce, which are responsible for distributed storage, resource allocation and management, and distributed computing, respectively. There are also corresponding approaches to optimizing these three modules.

The Hadoop-HDFS module is responsible for distributed storage. To optimize this module, we can consider ensuring that the disk performance of DataNodes is consistent as much as possible, and optimize both disk I/O and network I/O.

The Hadoop-Yarn module is responsible for resource allocation and management. To optimize this module, we should provide the CPU and memory resources of the slaves to Yarn for management and use as much as possible, and adjust the container memory size appropriately according to the data volume of the application to make full use of all CPU cores and give full play to the advantages of Kunpeng's multi-core architecture.

The Hadoop MapReduce module is responsible for distributed computing. For tuning this module, the number of containers running simultaneously in the cluster is determined after the Yarn parameters are set. Appropriately adjust the number of Map and Reduce tasks and the start time of Reduce tasks to ensure that the total number of tasks can cover all CPU cores and avoid resource waste.

4.5.2.4 Performance Localization: Problem Localization Process

Performance localization requires the use of a series of resource monitoring tools:

nmon: Supports collecting data on the usage of various resources of the entire machine, such as CPU, disk, network, and memory, over a period of time.

perf: Get the call status within a specified process, the CPU resource consumption of each thread, and support the generation of flame graphs.

Dstat: Monitors overall system performance information, including CPU, disk, network, paging, etc.

top: Monitors the CPU and memory resource consumption of processes and the entire machine, and supports viewing the usage of each CPU core.

iostat: Monitors the number of read/write operations, data volume, and utilization rate of each disk.

sar: Monitors the number of network I/O reads and writes and the amount of data for each network interface card.

jstat: A JVM statistics and monitoring tool that monitors the resources and performance of Java applications, including heap size and application GC status.

Java VisualVM: A graphical Java program performance analysis tool that can monitor application performance consumption, GC status, thread heap information, etc.

Based on the data provided by the performance localization tool, we can pinpoint the problem and determine which part of the system the issue originates from.

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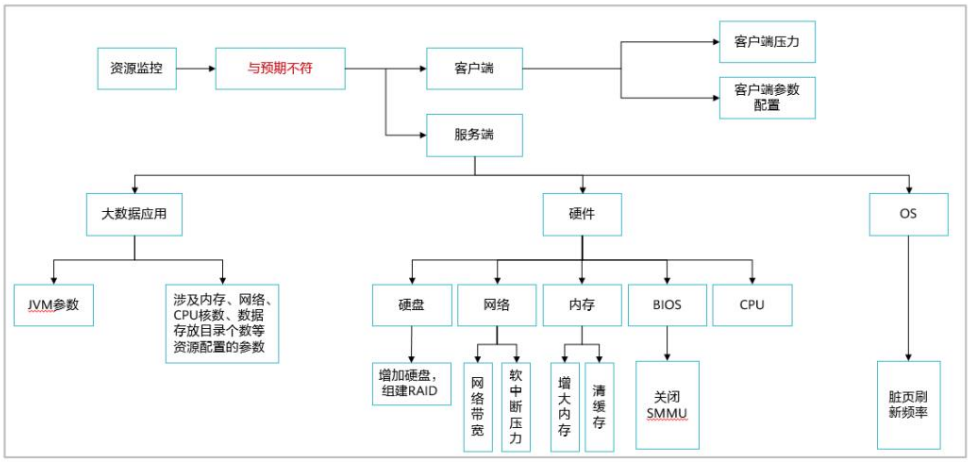


图4-11 Problem localization process

4.5.2.5 Performance Positioning: Tuning Process

For different performance analysis results, we can take corresponding optimization actions, as shown in the figure below.

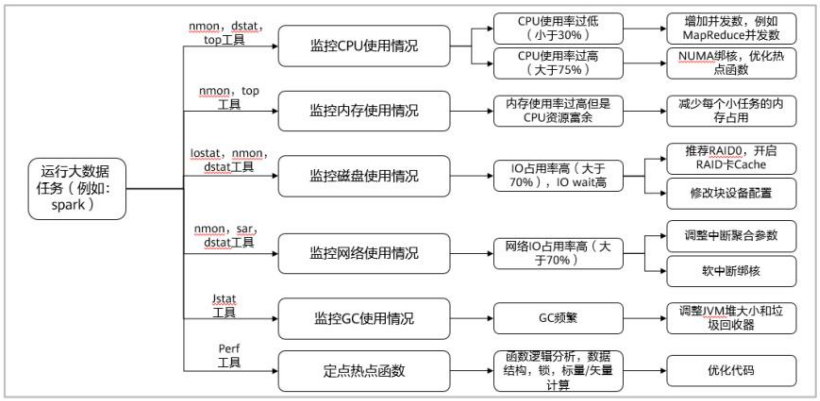


图4-12 Optimization process

4.5.3 Introduction to Experimental Procedure

4.5.3.1 Task Flow

The following describes the optimization task process, which consists of four steps:

1. Department: Hadoop, Hive

Hadoop, as a big data analytics tool, provides storage and computation for massive amounts of data. Hive is a data warehouse tool based on Hadoop that maps structured data files to a database table and provides SQL query functionality to facilitate data analysis.

2. Use Hive for data analysis

Practice writing HQL to complete data analysis tasks, using the data analysis test questions provided in the lab manual.

3. Deploy Kunpeng performance analysis tools

Follow the steps to install the Kunpeng performance analysis tool.

4. Optimize using Kunpeng performance analysis tools

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Use the Kunpeng performance analysis tool to scan the system, modify the optimization items according to the optimization suggestions, and verify them.

4.5.3.2 Hadoop and Hive Deployment Steps

Hadoop Deployment Steps

1. Deployment environment configuration;

2. Install Hadoop; 3. Modify

environment variables and configuration files;

4. Start and verify Hadoop.

Hive Deployment Steps

1. MySQL configuration;

2. Install Hive;

3. Modify environment variables and configuration files;

4. Start and verify Hive.

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5 Huawei Kunpeng Solution

5.1 Overview of Huawei Kunpeng Solution

5.1.1 Overview of Huawei Kunpeng Solution

Currently, the world has entered the digital economy era, where data is a key production factor. The booming development of digital applications has spurred new changes in data resources, including explosive growth in data volume, evolution of data structure diversity, ubiquitous distribution of data resources, and a continuous decline in the cost per bit of data. These changes in data elements have triggered the evolution of computing technology. Heterogeneous, high-performance, ubiquitous, collaborative, green, and inclusive characteristics have become the hallmarks of "new computing" in the digital economy era, bringing new opportunities for the global computing industry. The Kunpeng computing industry is committed to building a globally competitive computing industry cluster. By providing the industry with hardware and software infrastructure based on Kunpeng processors, it efficiently meets the market's demand for "new computing," supporting the digital transformation of society as a whole. Simultaneously, the implementation of the Kunpeng computing industry will drive the clustered development of the computing industry chain, promote collaborative innovation, stimulate local investment and employment, and form new momentum for the development of the digital economy.

China is accelerating the construction of new infrastructure, guided by the new development philosophy, driven by technological innovation, and based on information networks, to create an upgraded, integrated, and innovative infrastructure system that meets the needs of high-quality development. Various provinces are successively issuing policies and plans to accelerate the digital transformation of their industries. As the supporting foundation for the development of the digital economy, the computing industry will usher in tremendous development opportunities in this wave.

The rapid development of ICT technologies is accelerating innovation in computing applications. Faced with the diversification of application scenarios and data types in areas such as autonomous driving, cloud gaming, smart manufacturing, artificial intelligence, and the Internet of Things, traditional single architectures are struggling to meet data processing and user experience requirements, posing new challenges to computing platforms and driving the development of diverse computing architectures. Among these, RISC general-purpose architecture processors, represented by ARM, possess significant advantages in this era of diversified computing scenarios. For example, in distributed databases, big data, and...
In high-concurrency applications such as web front-ends, ARM architecture processors with more cores per chip offer better concurrent processing efficiency compared to traditional processors. Furthermore, the vast majority of mobile terminals use ARM architecture processors, and edge-cloud homogeneity provides developers with convenience in writing and optimizing code across the entire ecosystem, while also reducing performance losses and potential vulnerability risks caused by heterogeneous environment development. With the rise of edge computing and the rapid development of 5G technology, cloud-edge-device collaboration will become mainstream, further highlighting the collaborative advantages of ARM computing platforms. In the field of high-performance computing, diverse computing platforms represented by ARM and RISC-V are also playing an increasingly important role. For example, the EU's EPI (European Processor Initiative) project aims to build a domestically developed supercomputer with a core ARM architecture and RISC-V accelerator chips capable of handling billions of computers; Japan's "Fugaku" supercomputer system uses a domestically developed ARM architecture processor, becoming the world's first TOP500 champion supercomputer based on ARM chips. In the future, the significant differences between different application areas will give rise to more segmented ecosystems. Combining the load characteristics of upper-layer applications, allocating the most efficient underlying computing resources, and maximizing the utilization of computing resources will become the development direction of the computing industry.

The key to the success of new computing development lies in building a complete ecosystem. The Kunpeng computing industry, based on the Kunpeng processor, encompasses the entire industry chain, from underlying hardware and basic software to upper-level industry applications, including fundamental hardware and software infrastructure, industry applications, and services. Looking at the overall Kunpeng computing industry ecosystem, on the hardware side, it revolves around the Kunpeng processor and includes components such as the Ascend AI chip, smart network interface card (NIC) chip, baseboard management controller (BMC) chip, solid-state drives (SSDs), RAID cards, and motherboards.

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This includes complete systems such as personal computers, servers, and storage. In terms of basic software, it covers operating systems, virtualization software, databases, middleware, storage software, big data platforms, data protection, and cloud services. Regarding industry applications, the Kunpeng computing ecosystem covers various sectors including government, finance, telecommunications, energy, and large enterprises, providing comprehensive, complete, and integrated IT solutions.

As a member of the Kunpeng computing industry, Huawei focuses on innovation in computing architecture, R&D of processors and open-source basic software, and Huawei Cloud services, and is committed to promoting the development of the Kunpeng ecosystem. Through strategic, long-term R&D investment, Huawei attracts outstanding talent and advanced technologies from the global computing industry, continuously promotes the innovative development of full-stack computing technologies, and accelerates the construction of a global open-source system and industry standards for diverse computing. Huawei promotes the development of the Kunpeng computing industry based on a strategy of "hardware openness, software open source, and enabling partners."

Solutions based on Huawei Kunpeng computing platform (cloud platform or physical platform) are divided into general solutions and industry solutions. General solutions include high-performance computing (HPC), big data infrastructure, distributed storage, and core enterprise applications; industry solutions include targeted solutions for industries such as telecom operators, government, and finance.



图5-1 Huawei Kunpeng Solution Overview

5.1.2 Competitiveness of Huawei Kunpeng Cloud Full-Stack Solution

Huawei's Kunpeng Cloud solution, through its full-stack infrastructure, cloud services, and long-term ecosystem investment, has achieved a full-stack Kunpeng Cloud ecosystem, full-stack Kunpeng Cloud services, and full-stack Kunpeng Cloud infrastructure.

It is compatible with various Kunpeng cloud services, including ECS, BMS, containers, RDS, Redis, etc., covering the entire stack of solutions from chips, infrastructure, operating systems to scenario-based solutions.

Kunpeng's full-stack solution has penetrated deep into various industries, establishing close partnerships with over 50 industry ISVs.

5.1.3 Taishan servers cover mainstream application scenarios

Taishan servers offer advantages such as high performance, multi-core high concurrency, high throughput, and native homogeneous computing power. They are suitable for efficiently accelerating applications such as big data, distributed storage, native applications, high-performance computing, and databases, aiming to meet the diverse computing and green computing needs of data centers. Taishan server solutions can cover mainstream data center application scenarios.

Over the past three years, Huawei has collaborated with ISVs across various industries to incubate solutions and has conducted in-depth optimizations for advantageous scenarios, providing data center infrastructure and services based on the Kunpeng processor to a wide range of industry customers, including government, finance, telecom operators, power, and internet companies.

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5.2 Huawei Kunpeng General Solution

5.2.1 High-Performance Computing (HPC) Solutions

5.2.1.1 Overview of High-Performance Computing (HPC) Solutions

In the era of industrial electronic information technology, high-performance computing has evolved from single-machine processing to clustering. Enterprises mainly rely on building their own high-performance computing clusters, but this presents numerous challenges in enterprise production and usage:

The construction of high-performance computing clusters is a capital-intensive process, with high construction costs and long cycles. It often fails to meet the flexible business needs of enterprises, resulting in waste or shortage of resources.

The maintenance work is arduous and stressful. Cluster maintenance requires enterprises to invest a lot of resources and manpower, which prevents enterprises from fully focusing on their own business.

In the traditional self-built cluster approach, enterprises need to build their own data centers, which incurs high operating costs, including water, electricity, and system maintenance fees.

Hardware equipment is updated and iterated rapidly, computing power is expanding rapidly, planning cannot keep up with changes, new technologies are emerging one after another, equipment is aging severely, and the rapid pace of hardware updates makes it impossible to quickly adapt to new business needs.

With the rise of the industrial cloud era, PCs, workstations, and cluster machines are all being cloudified, and the development of high-performance computing cloudification is an inevitable trend. The advent of cloud computing enables all data and computing to be completed in the cloud, so that enterprise data does not have to be stored in the cloud, internal collaboration is more efficient, enterprise heavy assets and maintenance work are reduced, and customers can focus on their own business development.

HPC (High Performance Computing) is a computer cluster system that connects multiple computer systems together through various interconnection technologies. It utilizes the combined computing power of all connected systems to handle large-scale computing problems, and is therefore often referred to as a high-performance computing cluster.

High-performance computing cloud solutions (HPC Cloud) utilize cloud deployment to provide an efficient, reliable, flexible, and secure computing service; they can provide excellent computing services for scenarios such as industrial design and massive data processing; help customers reduce TCO, have short deployment cycles, low costs, shorten product launch cycles, and enhance enterprise product competitiveness.

Huawei's HPC solution is mainly divided into two layers:

1. Huawei mainly provides the IaaS resources required for computing, including computing, storage, and network resources.

2. The software used in the cluster scheduling layer varies slightly in different application scenarios. For example, PBS is more commonly used in industrial simulation scenarios, while open-source software Sge and Slurm are more commonly used for gene sequencing.

The core value and advantages of the solution are as follows:

1. Short deployment cycle and low cost.

2. Flexible deployment, avoiding waste or shortage of resources.

3. Public cloud solutions have low maintenance costs.

4. Meet the requirements of rapid technological development.

The solution targets memory-sensitive applications and applications involving integer calculations (or single-precision calculations), with typical application scenarios as follows:

Scientific computing, weather forecasting, environmental forecasting, industrial simulation, gene sequencing, energy exploration, animation rendering, and financial analysis, among others.

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5.2.1.2 Advantages of High-Performance Computing (HPC) Solutions

ÿHigh performance

Based on the Kunpeng computing platform, the Kunpeng 920 processor has up to 64 cores and 8 memory channels, making it suitable for memory-intensive and integer computing-based applications. Huawei smart network cards, SSDs, and Atlas accelerator cards accelerate high-performance computing applications and improve customer application performance.

High energy efficiency

Board-level liquid cooling, water flow between memory modules, enhanced heat exchange through turbulence, and full liquid-cooled cabinet cooling result in a PUE as low as 1.05, reducing customer management and maintenance costs.

ÿSafe and reliable

We develop our own chips, operating systems, MPI, compilers, and mathematical libraries, master key core technologies, provide sustainable product supply capabilities, comply with business environment requirements, proactively protect information security, and ensure that customers can deploy applications with confidence.

ÿOpen Ecosystem

We embrace open collaboration, working with ISVs, partners, developers, and industry alliances to build a full-stack ecosystem that is compatible with mainstream enterprise application software, supports rapid and smooth migration of a wide range of applications, and reduces customer migration costs.

5.2.1.3 HPC solutions offer high cost-effectiveness.

On the public cloud, HPC can be deployed on demand, enabling rapid deployment and scaling, accelerating product time-to-market and shortening research cycles. Public cloud HPC can be rented on demand, at low cost, lowering the barrier to entry for small and medium-sized customers. Applications such as gene sequencing, fluid mechanics, and meteorological environmental protection primarily involve memory-sensitive applications and integer computing (or single-precision computing), making them suitable for Kunpeng Cloud HPC application scenarios.

5.2.1.4 Other Advantages and Characteristics

ÿ Shared by multiple organizations and regional tenants

Dynamic application and sharing: computing nodes are applied for and released on demand.

Rent on demand to avoid over-investment and redundant construction;

User isolation measures ensure security;

Data is centralized and shared across organizations and regions.

Self -service capability

Automatically deploy virtual machines and cloud-based bare metals, automatically create clusters, and perform long-term automatic status monitoring;

Various HPC application templates are initialized and imported, and MPI libraries, compilation libraries, and optimization configurations are deployed in the VM template.

ÿ Flexible deployment methods

Shorter construction time, near-unlimited infrastructure;

Virtual machines/cloud-based bare metal and various compute/storage instances are available for flexible selection.

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5.2.1.5 Huawei Kunpeng's Innovative Technologies for HPC

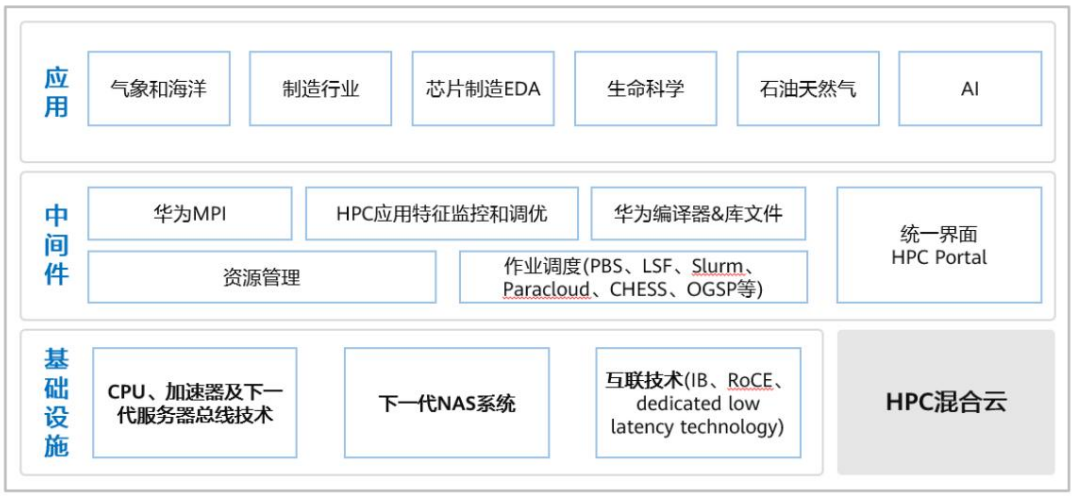


图5-2 Huawei Kunpeng **HPC** Solution

Huawei Kunpeng HPC Hybrid Cloud Solution incorporates many technological innovations.

At the infrastructure level: it uses servers equipped with CPUs from the Kunpeng series as the host environment, with multiple built-in accelerators, supports next-generation server bus technology, and storage devices support the construction of next-generation NAS systems. It supports interconnection of multiple hosts and supports interconnection technologies such as IB, RoCE, and specific low-latency technologies.

At the middleware level: it provides unified Huawei MPI support, a unified operating interface HPC Portal, rich Huawei compilers and library files, and the Kunpeng Development Kit (Devkit) to support application development, compilation, and debugging. Based on cluster management and scheduling components, it enables flexible resource management and performance job scheduling (components include PBS, LSF, Slurm, Paracloud, and CHESS).
OGSP, etc.), HPC application feature monitoring and optimization, etc.

At the application level: HPC solutions based on Kunpeng infrastructure and with comprehensive platform component functions can be applied to many application scenarios such as meteorology and oceanography, manufacturing industry, chip manufacturing EDA, life sciences, oil and gas, and AI.

5.2.1.6 Huawei's self-developed Kunpeng (ARM) HPC solution accelerates scientific research and innovation.

High-performance computing clusters are widely used in education and scientific research to accelerate exploration in cutting-edge fields such as high-energy physics, astronomy, geophysics, quantum chemistry, and molecular dynamics. Through mathematical modeling, quantitative analysis methods, and the use of computers to analyze and solve costly experiments and unsimulable experimental environments in traditional experimental science, they facilitate rapid iteration and verification of cutting-edge theoretical science, becoming one of the most powerful tools for improving the research level of university research teams. Huawei Kunpeng HPC solutions, with their advantages of high performance, high energy efficiency, and independent controllability, contribute to the rapid development of HPC in education and science.

5.2.1.7 Meteorological HPC Kunpeng Cloud Solution

High-performance computing in meteorological scenarios presents the following challenges:

ÿ Huge computational load

Mesoscale weather forecasting models (MM5, WRF, GRAPES, and AREMS) require an astonishing amount of computation. Furthermore, the nature of weather forecasting necessitates extremely high real-time performance, requiring models to complete calculations within a specified timeframe (generally no more than 2 hours). On the other hand, the demand for higher forecast accuracy is constantly increasing; current forecast accuracy has improved from hundreds or tens of kilometers to just a few kilometers, significantly increasing the computational burden on these models. Doubling the forecast accuracy increases the required computation by 16 times. This computational demand of numerical weather prediction simply cannot be met by a single CPU or ordinary computer.

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Success requires the use of parallel computing. On one hand, this involves parallelizing the model forecasting software through message passing or shared memory; on the other hand, it necessitates high-performance parallel computers. Currently, most mesoscale weather forecasting models have already achieved parallelization, such as...

MM5 and WRF support both MPI message passing parallelism and OpenMP shared memory parallelism, and also support...

A hybrid operation mode of MPI+OpenMP. GRAPES only supports MPI parallel mode, while AREMS has also completed the MPI parallelization work.

ÿExtremely dense communication

Because these models are parallel software, and generally employ finite difference grid-based parallel computation, the communication volume between CPUs is substantial when running mesoscale weather forecasting models, placing extremely high demands on communication performance. For example, the communication in MM5 and WRF includes inter-domain communication between the parent and nested domains, as well as communication between different data partitions within each domain. Therefore, high-performance computers require high-performance communication networks.

ÿHighly real-time, scheduled operation

The very nature of weather forecasting dictates its high real-time requirements. Furthermore, forecasting systems must operate automatically at fixed times and locations, without human intervention. They typically run in 2-4 fixed time slots each day, with each slot lasting approximately 2 hours. Since these operations occur at the same times daily, it is crucial to ensure that these models complete their calculations on time.

The main mode is characterized by its computational complexity .

From a software processing flow perspective, it is generally divided into preprocessing, main mode, and post-processing. Preprocessing includes downloading data and data assimilation, while post-processing mainly refers to graphical processing to generate products. Preprocessing and post-processing generally do not place high demands on the computer. The main mode is the core of the entire system and also where the main computational load is concentrated; this part has very high requirements for computer performance.

Huawei Cloud HPC solution delivers HPC resources as a service to users, addressing the large-scale computing needs of scenarios such as meteorological assimilation and massive data processing. Huawei Cloud HPC solution is optimized for HPC workloads, enabling on-demand rental of HPC resources, helping customers reduce TCO, improve the efficiency and accuracy of weather forecasts, and accelerate the transformation of meteorological services towards intelligence, cloudification, and digitalization. It supports hybrid deployment of x86, GPU, FPGA, and Kunpeng, with a 46% increase in memory bandwidth based on high-performance Kunpeng. Simultaneously, it supports unified management and orchestration of various resources, including physical resources, multiple types of bare metal, virtual machines, container resources, and public cloud, achieving "one cloud for one meteorological service."

5.2.1.8 Kunpeng Accelerates the Performance of CAE/CFD, Meteorological, and Gene Simulation Applications

The life sciences field widely utilizes neuroscience, genomics, epigenetics, proteomics, and other methods for scientific research. This, combined with mathematical analysis and simulation, and bioinformatics, enables precise understanding and prediction of the behavior of entire living systems. This requires the analysis and mining of massive amounts of genomic data, necessitating high-performance computing clusters to meet the demands. In the biopharmaceutical industry and neuroscience, molecular modeling, data management, drug molecule design, gene sequencing, cryo-electron microscopy, and other processes are fully autonomous and controllable based on Huawei Kunpeng HPC solutions, significantly accelerating the efficiency of new drug development and saving computation time.

ÿGene sequencing

Gene sequencing technology is a new type of gene detection technology that can analyze and determine the full gene sequence from blood or saliva to predict the likelihood of developing various diseases. It has been widely used in various fields, both medical and non-medical, such as prenatal testing, genetic testing, and targeted cancer therapy. High-throughput, low-cost sequencing technology has brought gene sequencing into every household.

Depending on the research objective, gene data analysis may involve various computational instances, such as I/O-intensive, CPU-intensive, and memory-intensive, requiring the analysis and mining of massive amounts of gene data. Currently, over 50 applications, including GATK, Ont-tombo, and HISAT2, have completed Kunpeng adaptation.

Cryo - electron microscopy

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Cryo-electron microscopy (cryo-EM) is a biophysical technique that rapidly freezes biomolecules in glassy ice within milliseconds, collects two-dimensional projections of these macromolecules using a low-temperature transmission electron microscope, and then reconstructs their three-dimensional fine structures. It is computationally intensive, involving image data correction, effective macromolecular particle identification, particle clustering, spatial construction, and data conversion; a single data processing job can often take several days. It also requires high-speed I/O, necessitating the rapid loading of large files containing hundreds of gigabytes of data. Software such as Relion and Eman2 have been adapted for the Kunpeng platform, offering superior performance.

5.2.1.9 Kunpeng HPC Open Source Ecosystem

OpenHPC is a community-driven FOSS (Free and open source software) toolkit for Linux-based HPC. OpenHPC has no special hardware requirements.

OpenHPC provides an integrated and tested collection of software components, along with supported standard Linux distributions, for building fully functional computing clusters. The components cover the entire HPC software ecosystem, including configuration and system management tools, resource management, I/O services, development tools, numerical libraries, and performance analysis tools.

5.2.1.10 Kunpeng HPC: Verified Platform and Applications

HPC is a jewel at the pinnacle of computing, used in everything from weather forecasting and large aircraft manufacturing to COVID-19 vaccine development, impacting scientific progress and national welfare. Kunpeng HPC has already completed platform verification in multiple research fields, including meteorology and oceanography, manufacturing simulation, life sciences, and basic scientific research.

5.2.2 Big Data Infrastructure Solutions

5.2.2.1 Types of Big Data Technologies and Representative Scenarios

There are four main computing models for big data:

Batch processing

Batch processing computing mainly addresses the batch processing of large-scale data, which is also a very common data processing requirement in our daily data analysis work.

MapReduce is the most representative and influential big data batch processing technology. It can execute large-scale data processing tasks in parallel and is used for parallel computing of large datasets (greater than 1TB).

MapReduce greatly simplifies distributed programming by abstracting the complex parallel computing process running on large-scale clusters into two functions: Map and Reduce. Programmers can easily run their programs on distributed systems and perform computations on massive datasets even without knowledge of distributed parallel programming.

ÿAd -hoc analysis query

Suitable for the storage, management, and query analysis of large-scale data, representative products include Dremel, Presto, and Impala.

ÿ Stream processing

Streaming data is also an important data type in big data analytics.

Streaming data (or data stream) refers to a dynamic collection of data that is infinite in terms of time distribution and quantity. The value of the data decreases over time, so real-time computing is required to provide a response within seconds.

Stream processing can process continuously arriving streaming data from different data sources in real time, and provide valuable analytical results through real-time analysis.

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Currently, many stream computing frameworks and platforms have emerged in the industry. The first category is commercial-grade stream computing platforms, including IBM InfoSphere Streams and IBM StreamBase. The second category is open-source stream computing frameworks, including Twitter Storm, Yahoo! S4 (Simple Scalable Streaming System), Spark Streaming, etc. The third category is stream computing frameworks developed by companies to support their own businesses, such as Facebook using a combination of Puma and HBase to process real-time data.

Image processing

In the era of big data, much big data is presented in the form of large-scale graphs or networks, such as social networks, the transmission routes of infectious diseases, and the impact of traffic accidents on road networks. In addition, much non-graph structured big data is often converted into graph models before processing and analysis.

MapReduce, as a distributed computing framework with single input, two-stage, and coarse-grained data parallelism, often falls short when expressing multi-iteration, sparse, and fine-grained data, and is therefore unsuitable for solving large-scale graph computing problems.

Therefore, for computation of large graphs, a graph computation model is required, and many related graph computation products have emerged.

Pregel is a parallel graph processing system based on the BSP (Bulk Synchronous Parallel) model.

To address the challenges of distributed computation in large graphs, Pregel provides a scalable, fault-tolerant platform with a highly flexible API capable of describing various graph computations. Pregel is primarily used for graph traversal, shortest path calculation, and PageRank computation. Other representative graph computation products include

Facebook's open-source implementation of Pregel, Giraph, GraphX for Spark, and the graph data processing system PowerGraph.

5.2.2.2 Supports commercial use of various big data platforms, including open source – Open ecosystem

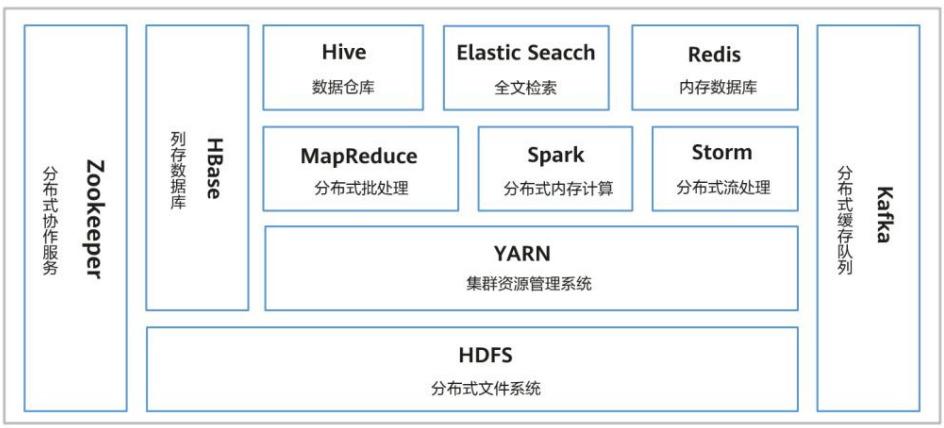
The big data platform mainly provides four functions:

1. Data collection: Real-time data collection is performed using Kafka, and the data is aggregated into the platform;

2. Data storage: Massive data storage is accomplished through HDFS and HBase;

3. Data processing and computation: The system utilizes three processing engines—MapReduce (batch processing), Spark (in-memory computing), and Storm (real-time stream processing)—to process and compute data for various scenarios.

4. Data Application: Hive is used to build a data warehouse and perform SQL-like queries, ElasticSearch is used for real-time retrieval of massive document data, and Redis is used for in-memory queries of key-value data.



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图5-3 Big Data Component Relationship Distribution

The following is a detailed introduction to each component:

HDFS, or Hadoop Distributed File System for short, is a distributed file system. HDFS is a highly fault-tolerant system suitable for deployment on inexpensive machines. HDFS provides high-throughput data access, making it ideal for applications dealing with large-scale datasets.

HBase is the database for Hadoop, a distributed, scalable, big data storage solution. Designed for extremely large tables with billions of rows and millions of columns, it's a distributed database capable of random, real-time read/write access to massive datasets. It provides storage capabilities similar to Google Bigtable, built on Hadoop and the Hadoop Distributed File System (HDFS). HDFS is a sub-project of the Apache Hadoop project, and HBase utilizes Hadoop HDFS as its file storage system. HBase resides in the structured storage layer, and Hadoop HDFS provides highly reliable underlying storage support for HBase. Except for some log files generated by HBase, all data files in HBase can be stored on the Hadoop HDFS file system.

Redis is a high-performance key-value storage system. Similar to Memcached, it supports a wider range of value types, including string, list, set, and sorted set (zset). The emergence of Redis largely compensates for the shortcomings of key/value stores like Memcached, and in some situations, it can serve as a valuable supplement to relational databases.

Kafka is a high-throughput distributed publish-subscribe messaging system that can handle all action stream data from consumer-scale websites and has become the preferred choice for big data systems between asynchronous and distributed messaging.

MapReduce is a programming model for parallel computation of large datasets (greater than 1TB). Data computed in a MapReduce program can come from multiple data sources, such as Local File System, HDFS, and databases. HDFS is the most commonly used, leveraging its high throughput to read and compute large amounts of data. After computation, the data can also be stored in HDFS.

Spark is a high-speed, general-purpose big data computing engine. It possesses the advantages of Hadoop MapReduce, but unlike Hadoop, Spark allows intermediate job outputs to be stored in memory, eliminating the need to read and write to HDFS. Therefore, Spark is better suited for iterative MapReduce algorithms such as data mining and machine learning. Typically, data for Spark computations can come from multiple data sources, such as local files and HDFS. HDFS is the most commonly used, allowing users to read large amounts of data at once for parallel computation. After computation, data can also be stored in HDFS. It can be used with Hadoop and Apache Mesos, or independently.

Storm is an open-source real-time data processing framework similar to Hadoop, developed by Twitter. Its simple programming model significantly reduces the difficulty of real-time processing and it is one of the most popular stream processing frameworks today. Compared to other computing frameworks, Storm's biggest advantage is its millisecond-level low latency.

Hive is a data warehouse tool based on Hadoop. It can map structured data files to a database table and provide simple SQL query functionality. It can convert SQL statements into MapReduce tasks for execution. Its advantages include a low learning curve, the ability to quickly implement simple MapReduce statistics using SQL-like statements without developing dedicated MapReduce applications, making it well-suited for statistical analysis in data warehouses.

Elasticsearch can build full-text indexes (breaking down text content into keywords and creating an index based on those keywords; queries then search the index using those keywords to find articles containing them). It separates data from the index, sharding the index and distributing it across different nodes. The number of nodes can scale to hundreds, enabling real-time retrieval and processing of petabyte-scale data.

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Structured or unstructured data. Sharding allows for backup of copies to ensure data reliability, and the collaborative work of each shard greatly improves retrieval performance. Furthermore, a simple RESTful API makes full-text search efficient and easy.

YARN is a new Hadoop resource manager, a general-purpose resource management system that provides unified resource management and scheduling for upper-layer applications, addressing the performance bottlenecks of the older MapReduce framework. Its basic idea is to separate resource management and job scheduling/monitoring functions into separate daemons.

ZooKeeper is a distributed application coordination service and a crucial component of Hadoop and HBase. It's a tool that provides consistency services for distributed applications, enabling nodes within a Hadoop cluster to coordinate with each other. ZooKeeper is now an Apache top-level project, providing efficient, reliable, and easy-to-use coordination services for distributed systems.

5.2.2.3 Industry Technology Trends - The Path of Hadoop Storage and Compute Separation

Hadoop was designed from the outset as a compute-in-memory system. Its initial design philosophy was that storage remained stationary while computation moved freely. Yarn, responsible for scheduling, would distribute computational tasks to the instances where the data resided, a key difference from traditional centralized storage. Why was Hadoop designed with this compute-in-memory coupling? In 2006, server bandwidth was only 100Mb/s to 1Gb/s, while HDD (hard disk) throughput was only 50MB/s. This bandwidth was far from sufficient for data transmission, making network bottlenecks particularly pronounced. The only solution was to distribute computational tasks to the locations where the data resided.

The saying "what is divided for a long time must unite, and what is united for a long time must divide" applies here. With the development of cloud computing technology, databases are also embracing the cloud-native era. In the current environment that increasingly emphasizes cloud-native technologies, the separation of storage and compute has become an inevitable trend. "Storage-compute separation" as an architectural concept is gradually becoming known and used in enterprise project development. However, with the reshaping of enterprise IT architecture brought about by digital transformation, storage-compute separation technology will gradually fade into history. Storage and compute refer to the combination of storage and computing. Computing usually refers to the computing power unit composed of CPU and memory, while storage refers to persistent data storage units. In enterprise production practice, there is no unchanging architecture, only an unchanging business-centric architectural mindset.

Bandwidth speeds, especially within IDC (Internet Data Center) server rooms, have increased from 1000mps, 2000mps, 10000mps, and even 100000mps, representing a 100-fold or higher increase in network bandwidth. The network is no longer the bottleneck. However, disk throughput performance hasn't seen a significant improvement, only about doubling (100MB/s). This hardware change has led to changes in software architecture. Efficient compression algorithms and columnar storage have further reduced I/O pressure, gradually shifting the bottleneck for big data from I/O to CPU. Simultaneously, as cluster sizes grow larger, the uneven utilization of storage and computing resources, and the limitations in system selection, are becoming increasingly apparent.

As the volume of accumulated data and the volume of big data business increase, the cost of data storage and processing is rising, leading to greater investment in enterprise data infrastructure. Simultaneously, big data processing components are numerous, each using different data processing formats. For example, data lakes and data warehouses use different formats, making data storage complex. Systems often need to store the same data multiple times to handle different scenarios, requiring extensive data copying and format conversion between different components, consuming significant resources. After more than a decade of big data development, with the emergence of massive workloads and big data use cases, the scale of single Hadoop clusters has increased, and multiple Hadoop clusters need to support different business needs simultaneously. Therefore, in a storage and compute coupled architecture, big data clusters will face the following challenges:

High cost

The demands for computing power and storage in business operations are unbalanced, and their growth rates are also uneven. When expanding capacity, both computing and storage need to be expanded simultaneously, which often results in wasted computing power.

Low resource utilization

Because multiple Hadoop clusters handle different workloads, the system load fluctuates with changes in business needs, resulting in peaks and troughs. However, the in-memory computing architecture means that the resources of each cluster are completely independent and isolated, and cannot be shared (in cross-industry in-memory computing architectures).

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The average resource utilization of Hadoop clusters is below 25%. High-density storage and computing power enhancement are largely ineffective; data skew occurs, and computing tasks may not be effectively scheduled to the instances where the data resides.

ÿ Difficulty in operation and maintenance

As business complexity increases and new services are launched more rapidly, the demands on server resource allocation also increase. A diverse range of server models increases maintenance difficulty, leading to higher data center space consumption and energy costs. Instances must balance compute and storage, limiting machine selection. Scaling down is complex, requiring data migration across instances, making elastic scaling impossible.

In traditional Hadoop architectures, compute and storage resources are tightly coupled in a certain ratio, requiring system expansion to be done only with an increase in the number of nodes, leading to wasted memory or disk space. "Storage-compute separation" not only saves costs but also allows resources to scale elastically according to business needs, allocating resources on demand, reducing system deployment and expansion costs, and resolving the problem of uneven resource utilization.

Improve resource utilization

"In-memory computing" architectures need to consider up to seven dimensions: CPU, memory, storage capacity/IOPS/bandwidth, and network IO/bandwidth. Insufficient resources in any one of these dimensions will prevent applications from meeting their requirements. "In-memory computing separation," on the other hand, breaks these seven dimensions down into two domains: computing and storage, reducing selection complexity. Separate computing and storage resource pools are built to improve resource utilization. Expansion allows for on-demand scaling of both computing and storage, saving on investment.

High availability

External storage arrays inherently possess excellent reliability designs, such as RAID redundancy, silent data verification, and a two-site, three-center architecture, ensuring reliability levels at least two orders of magnitude higher than servers. Therefore, storage should be a "long-term" shared storage solution. Servers have relatively weaker reliability; with "storage-compute separation," even if a server fails, the globally shared data allows for rapid business recovery without data replication, achieving overall high availability of the distributed database.

ÿHigh performance

By adopting a "store-compute separation" architecture, unnecessary overhead can be avoided by sharing a single data set globally, thus improving overall database performance. For example, in semi-synchronous storage, adding a slave node causes a 10% performance decrease. With "store-compute separation," semi-synchronous technology is no longer needed. External storage arrays offer deeper performance optimization; all-flash storage, especially all-NVMe storage, provides an exceptional performance experience. A shared storage resource pool not only improves resource utilization but also leverages the varying performance peaks and valleys of different applications at different times, distributing all I/O across all disks to smooth out performance spikes and valleys, achieving optimal performance.

5.2.2.4 Fully Distributed Storage - A single set of data supports multiple protocols, simplifying access and usage processes.

The unified compute and storage deployment model, limited to a single HDFS interface, is increasingly struggling to cope with the diverse data access needs of applications. Using multiple storage systems across different ecosystems can lead to data fragmentation, and inconsistent management tools increase operational workload. Against this backdrop, Huawei Data Lake provides multi-protocol access and interoperability capabilities, ensuring data is stored only once, avoiding multiple data copies, improving analytical efficiency, and saving on data storage costs.

5.2.2.5 Cloud-Native Era - Separation of Compute and Storage Simplifies Big Data Development

In big data development, if the form of integrated computing and storage is still used, the overall management becomes very complex.

After separating compute and storage, compute nodes become cloud-native, enabling automatic elastic scaling based on compute volume; storage expands seamlessly according to usage, without affecting business operations; resource utilization is higher, and cluster management is easier.

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5.2.2.6 BigData Pro Storage-Compute Separation Solution

Huawei Cloud has launched the BigData Pro big data solution based on Kunpeng, which uses the high-efficiency computing power of Kunpeng multi-core as its foundation, surrounds a data lake with OBS as storage, and adopts a storage-compute separation architecture to create a brand-new cloud-based big data solution.

BigData Pro has the following advantages: 1. Based on the

high-efficiency computing power of Kunpeng multi-core, it supports deployment methods such as elastic cloud servers, bare metal, and cloud containers;

2. Centered around a data lake with OBS as the storage, computing and storage are deployed separately, and computing and storage are connected using a high-speed cloud network to create a brand-new cloud-based big data solution.

Traditional big data computing and storage solutions, with their fixed resource allocation of computing servers and local storage disks, are constrained by hardware specifications. When computing bottlenecks occur, both computing and storage must be expanded simultaneously. However, when business growth is not proportional, this leads to idle and wasted resources. Statistics show a significant difference in computing and storage utilization rates; the average computing resource utilization rate of big data clusters is less than 50%, and the storage space utilization rate is even lower, less than 33%.

The storage-compute separation architecture allows for flexible configuration of computing and storage resources, enabling independent and elastic scaling.

3. By leveraging the multi-protocol access capabilities provided by OBS, such as HDFS, POSIX, and S3, data silos are broken down, and big data application systems can directly read and write cloud storage without protocol conversion.

4. A unified data lake enables zero data migration throughout the entire process, from data generation to data analysis to final aging and archiving.

5.2.2.7 Computing - Kunpeng Boosts ETL Performance by 30% with Multi-Core Computing Power

The Kunpeng architecture boasts 30% higher multi-core computing power than its x86 counterparts, better meeting the needs of high-concurrency scenarios such as big data and HPC. A single Kunpeng multi-core computing machine can reach up to 128 cores, with a maximum network forwarding performance of 9 million RPS, and supports high-speed cloud storage and ARM Native architecture.

5.2.2.8 Storage - 300MB per stream, multi-protocol interoperability, ultra-large total bandwidth, EB data volume

BigData Pro's data lake storage uses Huawei's commercial storage clusters, easily supporting scales up to the EB level.

Data lake storage improves big data performance and is compatible with the scaling of computing clusters. Its advantages are as follows:

1. Data persistence of 99.995%, high availability of 99.995%, and multiple Availability Zones (AZs) for active-active deployment.

2. Use as needed, expand storage seamlessly, and eliminate the hassle of manual expansion.

3. OBS single-stream (single file) upload and download speeds can reach up to 300MB/s.

5.2.2.9 Deployment Solution - Hadoop/Spark Integration with OBS: The OBSFileSystem plugin,

based on the OBS service, provides HDFS FileSystem semantics, enabling big data computing components such as MapReduce, Hive, and Spark to interface with OBS object storage services, providing a "data lake" storage for big data computing.

Scenario 1: Huawei Cloud Big Data Services MRS and DLI already include the OBSFileSystem plugin in their versions, so you can use OBS as big data storage without downloading or installing it.

Scenario 2: For self-built big data scenarios such as CDH, HDP, and open-source Hadoop, it is necessary to download, install, and configure them.

The OBSFileSystem plugin uses OBS as a big data storage solution.

5.2.2.10 Typical Big Data Customer Cases

ÿEfficient utilization of resources

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1. Computing resource utilization increased by 75%;

2. Storage resource utilization increased by 140%.

Extreme performance

1. Kunpeng's multi-core computing power and high concurrency performance;

2. Performance in big data and distributed scenarios is comparable to x86;

3. Some components can be improved by 10%-20%.

ÿSafe and controllable

1. Domestically developed chips;

2. Chip-level security;

3. Independent and controllable.

5.2.3 Kunpeng Cloud Phone Solution

5.2.3.1 What is a cloud phone?

A cloud phone (CPH) is a cloud server virtualized from Huawei Cloud bare metal servers, running a native Android operating system and possessing virtual phone functionality. Simply put, a cloud phone = cloud server + Android OS. Users can remotely control the cloud phone in real time to run Android apps in the cloud; they can also leverage the cloud phone's computing power to efficiently build applications such as cloud gaming, mobile office, and live entertainment.

As a new type of service, cloud phones serve as a significant extension and complement to traditional physical phones, and can be used in applications such as...
Applications can be used in scenarios such as APP simulation testing, cloud gaming, live streaming entertainment, and mobile office, enabling mobile applications to not only run on physical phones but also run intelligently in the cloud.

For internet industry scenarios such as APP simulation testing, the processing efficiency of a single mobile phone is very limited. By using cloud phones, the costs of manual operation and equipment procurement and maintenance can be significantly reduced.

Because cloud phones run their application data in the cloud, they offer a more secure and efficient mobile office solution for industries with high information security requirements, such as government and finance. Employees log into office systems using cloud phones, separating public and private data. At the same time, enterprises can also intelligently manage cloud phones, reducing costs and increasing efficiency while ensuring greater information security.

Exploring new possibilities in the gaming and live streaming industries

Cloud phones can also provide entirely new interactive experiences for industries such as gaming and live streaming, opening up new business models and market opportunities. Taking cloud gaming as an example, because the game content actually runs on a virtual phone in the cloud, it can be pre-installed, deployed, and dynamically loaded. Therefore, for the end player, the game can be played instantly without downloading, significantly increasing player conversion rates. At the same time, it allows users with low-to-mid-range mobile phones to smoothly run large-scale mobile games, expanding the user base for gaming.

5.2.3.2 Typical Application Scenarios of Huawei Cloud Kunpeng Cloud Phone

ÿCloud gaming scenario

Cloud gaming, a popular development trend in the gaming industry, provides players with a gaming service that eliminates the need for downloads and leverages the limitations of mobile devices through video streaming. It encompasses both PC and mobile game streaming. Cloud phones, as cloud-based emulated mobile phones, can capitalize on the isomorphic nature of mobile game commands to host game applications in the cloud.

The architecture is described below:

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1. The mobile game app is installed on a cloud phone. The audio and video of the cloud phone are streamed and encoded and output to the client for display. At the same time, the app receives operation commands from the client to control the game on the cloud phone.

2. The login server cluster adopts a load balancing and elastic scaling design, which can easily cope with ultra-large-scale concurrent scenarios.

3. Cloud phones can be deployed in various central nodes and edge clouds, effectively reducing latency in user interaction and achieving the best experience and optimal bandwidth cost-effectiveness.

ÿ APP simulation test scenario

Mobile phones are typically designed for personal use, but as mobile applications become more numerous and widespread, businesses also need to run a large number of mobile apps in specific scenarios to automate or enable intelligent functions. This necessitates a large number of emulated mobile phones to support the operation of these apps.

The architecture is described below:

1. Mobile Apps: Apps are installed on cloud phones. The enterprise automatically controls the phone to run one or more apps through pre-programmed scripts. Through human-like script operations, a variety of application scenarios can be realized.

2. Enterprises can build their own cloud phone management and maintenance platform, marketing system, or automated script platform in the elastic cloud server and control them separately through two independent VPCs.

3. Applications (APPs) in cloud phones can be centrally stored through object storage, saving a significant amount of network bandwidth consumption during application installation or updates.

4. A wide range of security and monitoring services can provide customers' business systems with comprehensive and secure protection measures to ensure the stable operation of their businesses.

ÿLive streaming entertainment scenarios

Live streaming entertainment is an innovative application scenario for cloud phones. By broadcasting the phone screen to multiple participants, it provides a multi-person interactive application scenario, improving user experience and live streaming effects.

The architecture is described below:

1. Mobile applications/games are installed on cloud phones, merging one or more mobile screens and outputting them to an encoding server for integrated encoding. The screens are then copied and streamed to multiple clients (PCs, mobile phones, tablets, etc.) for display. At the same time, the cloud phone receives operation commands from one or more clients.

2. The login server cluster adopts a load balancing and elastic scaling design, which can easily cope with ultra-large-scale concurrent scenarios.

Mobile office scenarios

With the widespread adoption of mobile applications, more and more enterprises are starting to access office work via mobile devices. However, this also brings risks to enterprise data security. While purchasing customized secure phones can enhance security, it still cannot prevent the leakage of sensitive data. Cloud-based mobile office applications can keep core enterprise data in the cloud, while only allowing authorized employees to view the phone screen.

The architecture is described below:

1. After enterprise applications are uploaded to object storage, they are installed in batches on cloud phones. The audio and video of the cloud phones are streamed and encoded and output to the client for display. At the same time, the client's operation commands are received to control the applications on the cloud phones. Enterprise data stays in the cloud, which is more secure and reliable.

2. The login/customer business server cluster adopts a load balancing and elastic scaling design, which can easily cope with ultra-large-scale concurrent scenarios.

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3. Cloud phones can be deployed in various central nodes and edge clouds, effectively reducing latency in user interaction and achieving the best experience and optimal bandwidth cost-effectiveness.

5.2.3.3 Comparative Analysis of Common Cloud Phone Solutions

ÿ X86 emulator

Advantages: x86 servers are readily available, have low technical barriers to entry, and offer high resource availability;

The business system architecture is flexible and can draw on mature product applications in the market to build a unique competitive advantage.

It allows for flexible configuration of simulated mobile phone specifications, offering a high degree of freedom.

Pain points: Additional instruction translation is required, resulting in performance loss of up to several times, and the cost-effectiveness of cluster size is very low;

The simulation fidelity and application compatibility are poor, and the application compatibility issues are difficult to resolve.

Implementations based on various external open-source or non-commercial simulator software are difficult to guarantee in terms of stability.

ÿReal Machine Farm

Advantages: It is basically the same as the real device, with high simulation accuracy and good application compatibility;

The technology has a low barrier to entry and is easy to build.

Pain point: Manual soldering and complex wiring make it difficult to guarantee product quality and result in very poor stability;

The second-hand mobile phone market changes rapidly, and the availability of phones with the corresponding designs is extremely poor.

ÿMobile Development Board Farm

Advantages: It is basically consistent with the hardware architecture of the real machine, with high simulation accuracy and good application compatibility;

Leveraging the highly integrated design of mobile phone chips, it offers high cost-effectiveness in gaming industry applications.

Pain point: It cannot achieve performance higher than that of a single mobile phone chip, making it difficult to meet the needs of high-performance application scenarios. It can only be based on the performance of a single mobile phone and then cut down to the nearest integer.

Long delivery cycles and unstable supply chains;

Designs based on consumer electronics circuits are insufficient to meet the service reliability and large-scale cluster management requirements of public clouds.

Huawei TaiShan Server

Advantages: Cloud phone specifications can be flexibly adjusted, with no upper limit to performance;

It can easily meet the ever-increasing demands of apps on mobile phone specifications and performance;

Server-based hardware design is better suited to enterprise-level reliability, stability, and cluster deployment capabilities;

It supports native ARM instructions, offering good compatibility and high performance.

Pain point: The hardware and software design of ARM-based servers is highly complex, and the technical threshold for building them on your own is very high;

The ecosystem applications are not yet comparable to x86, and the GPU support is not yet complete, requiring promotion from major manufacturers;

In the early stages of hardware development, the iteration speed was fast, with an average of one generation every two years, and the iteration speed was higher than the three-year availability period.

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5.2.3.4 Huawei Cloud Kunpeng Cloud Phone Technical Architecture

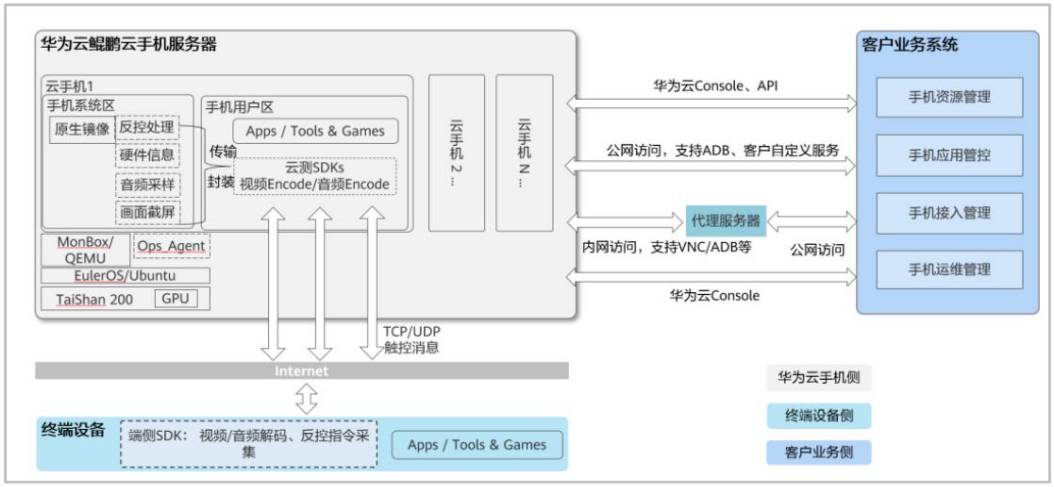


图5-4 Huawei Cloud Kunpeng Cloud Phone Technical Architecture

The cloud phone is implemented based on Huawei's TaiShan ARM server, integrating multiple high-performance professional GPUs to provide professional graphics and image processing capabilities. The TaiShan server runs EulerOS as the host OS, and within the host OS, containers are generated using the self-developed MonBox technology. These containers run the open-source AOSP system, thus virtualizing multiple cloud phones. Because both the TaiShan server and the phone system are based on the ARM architecture, the computational cost of instruction set conversion is reduced, resulting in a better user experience.

Cloud phones provide video, audio, and touch SDKs. Customers can develop corresponding apps based on their terminal devices to obtain audio and video from the cloud phone; they can also collect touch commands, such as touchscreen, swipe, and click commands, and then execute them on the cloud phone.

On the business side, customers can manage resources, applications, operations and maintenance, and access for cloud phones through the management console, API, ADB port, and other custom ports.

Resource Management: Purchase and query cloud phones.

Application management: push, install, uninstall, etc. of cloud mobile applications.

ÿOperations and Maintenance: Restart, reset, shut down, and power on cloud phones.

ÿAccess Management: Cloud Phone Access Authentication.

5.2.3.5 Native Applications - Edge-Cloud Homogeneous Advantages, 80% Performance Improvement

Huawei Cloud phones support direct migration of binary source code to the cloud without the need to recompile the code;

Huawei Cloud Phone boasts exceptional compatibility, achieving emulator-level compatibility for the x86 + Intel Houdini architecture and full device-level compatibility for the ARM-Based architecture.

5.2.3.6 Significant performance improvements from the self-developed kbox software simulation architecture

Based on the self-developed Kbox software, the density of cloud phones on a single server has been increased by 100%, the end-to-end cloud phone access latency has been continuously optimized, and the network access traffic of cloud phones has been reduced iteratively.

Kbox is Huawei's self-developed high-density container simulation cloud phone solution. Based on the Android framework, it enables functions such as hardware simulation and GPU device pass-through, thus enabling Android cloud phones.

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Kbox primarily implements hardware emulation functions such as vInput, sensors, GPS, and Mock (IMEI/WIFI), as well as GPU device passthrough, enabling Android cloud phone container solutions. Kbox consists of two parts: binary code and open-source patches.

The binary includes: vInput (touchscreen input module), Sensors (sensor-related simulation), GPS (GPS-related simulation), IMEI/WIFI Mock (simulation), Gralloc (off-screen rendering), and HWC (image compositing).

Demo: MediaFramework, Vold, adbd, Init, netd. Based on the open-source Android system code, incremental patches are provided for customers' reference.

5.2.3.7 Innovative instruction stream separation rendering technology brings high-definition image quality to large screens.

Huawei's proprietary AVS3 video encoding technology can significantly reduce bandwidth consumption for cloud rendering, while cloud phones' innovative instruction stream separation rendering technology brings high-definition picture quality to large screens.

Disconnection and Reconnection – Empty Rendering Technique: The client and server maintain the same OpenGL state machine. When the connection is lost and reconnected, the server synchronizes the complete state machine to the client and re-establishes the connection.

Fast loading – Vertex texture caching technology: By extracting the feature values of vertex textures, it identifies and eliminates duplicate vertex texture data.

Network stabilization – Data mapping technology: Establishes a map mapping between resources allocated by the server and resources managed by the client to achieve unidirectional transmission of control command flow and avoid a significant drop in frame rate when network latency increases.

Dynamic compression of command streams: lossless compression of control command streams, lossy compression of texture data streams, and residual data transmission of vertex data streams.

OpenGL ES 3.0: Improves the OpenGL library API adaptation layer, enabling games that support OpenGL ES 3.x.

5.2.3.8 Typical Customer Cases for APP Simulation Testing Scenarios

Huawei Cloud phones have successfully delivered value to a gaming company. This gaming company, which focuses on the research, development, publishing, and operation of mobile games, is committed to building a world-leading cultural and creative digital content platform through the synergy of its capabilities, resources, and capital.

Before using Huawei Cloud phones, the following pain points existed:

1. Traditional server deployments suffer from insufficient performance and complex management;

2. Mobile phone farms are costly and difficult to upgrade;

3. The development board has poor stability and high cost;

4. A low-cost, easy-to-manage, upgradeable, stable and reliable system is needed for large-scale game package testing.

Huawei Cloud Phone has gained customer approval in these four aspects:

1. Native architecture processor, guaranteeing performance;

2. Server clusters ensure system reliability and stability;

3. Fully cloud-based resources are easy to manage;

4. Test execution time is precise and controllable, plan management is effective, test data reports can be collected with one click, and efficiency is high.

The end customer received the following benefits:

The failure rate of the test machine was reduced from 60% to 1%; the available test time was increased from 10 hours/day to 24 hours/day; the test scale was increased from hundreds of concurrent nodes to thousands of concurrent nodes; and the test data export time was reduced from hours to minutes.

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6 Kunpeng Community

6.1 The Positioning and Role of the Kunpeng Community

6.1.1 The Positioning of the Kunpeng Community

The Kunpeng Community is a one-stop resource and service platform for Kunpeng developers' technical support and ecosystem enablement. It provides comprehensive software resources, professional skills, technical support, ecosystem policies and product solutions in the Kunpeng field, and works with community participants to build a comprehensive community based on the Kunpeng computing industry.

6.1.2 The Role of the Kunpeng Community

One-stop technical support platform:

The Kunpeng Community provides partners and developers with a one-stop technical support platform, including the latest industry news, resource and tool downloads, a technical support center, learning and online experiments, and other content. The platform will provide users with comprehensive technical support.

A window for showcasing the Kunpeng computing industry to the outside world:

The Kunpeng Community serves as a window to showcase the booming development of the Kunpeng computing industry. It will present users with the latest developments and future trends in the Kunpeng computing industry, enabling them to better understand its future direction and prospects.

Kunpeng's industrial layout, ecosystem partner gathering, and industrial progress promotion:

The Kunpeng Community promotes Kunpeng's industrial layout, ecosystem partner aggregation, and industrial progress. It provides users with the latest Kunpeng products and solutions, as well as the latest developments and collaborative achievements of Kunpeng's industrial ecosystem partners, enabling users to better understand Kunpeng's industrial layout and development progress.

6.2 Introduction to the various sections of the Kunpeng Community

6.2.1 Product Segment

The product section includes an overview of the Kunpeng computing industry, Kunpeng processors, Kunpeng server motherboards and complete machine products, openEuler, openGauss, openLooKeng, cloud services and solutions, product documentation, and other content.

The Kunpeng processor is a server processor independently developed by Huawei, featuring advantages such as multi-core processing and high concurrency. Kunpeng server motherboards and complete server products are server hardware products based on the Kunpeng processor, including various models with different specifications.

Kunpeng Cloud Service is a cloud computing service based on the Kunpeng architecture provided by Huawei Cloud, including various cloud service products such as cloud servers, cloud storage, and cloud databases. Additionally, based on the Kunpeng computing platform, the product section provides developers with technical documentation for application software porting, deployment, and optimization scenarios.

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图6-1 Product Segment

6.2.2 Solutions Section

The solutions section provides industry overviews, requirements and challenges analysis for different industry applications, and offers relevant case studies for developers to learn from and draw inspiration from.

6.2.3 Developer Section

The developer section includes the Kunpeng Development Kit (DevKit), Kunpeng Application Enabler Kit (BoostKit), Kunpeng HPC, openEuler development resources, openGauss development resources, development resources, resource downloads, compatibility queries, developer documentation, developer programs, and more, providing developers with a one-stop learning support platform.



图6-2 Developer section

6.2.4 Partner Section

The Kunpeng Ecosystem Partner Network is a series of ecosystem partner programs launched by Huawei around the Kunpeng industry technology roadmap. By providing resources and support in learning, building, listing, and sales, it is committed to helping Kunpeng ecosystem partners build industrial competitiveness, connect with customers to create business opportunities, and grow together and create the future in an open, innovative, collaborative, and mutually beneficial ecosystem.

6.2.5 University Section

University talent development communities can connect universities and enterprises, link teaching elements, and bridge time and space, providing educators and students with teaching resources related to the Kunpeng ecosystem, teacher training, teaching practice, intelligent base-based communities, and virtual teaching and research rooms, creating a dedicated learning and exchange space.

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In addition, to bring together the Kunpeng, Ascend, and Euler ecosystems and jointly build a talent foundation for the development of the digital economy, various talent development programs are offered, including but not limited to the Kunpeng Elite Internship Program and the openEuler & openGauss Talent Development Acceleration Program.

6.2.6 Kunpeng Solutions Market Segment

This is a one-stop platform for showcasing and interacting with software and hardware products, solutions, partners, and expert technical services. Here, users can learn about and view selected solutions powered by Kunpeng, and begin exploring Kunpeng partner solutions based on Huawei's technology stack, products, software, and industry dimensions.

6.2.7 Industry Information Section

This section provides access to various events and news related to Huawei Kunpeng computing. Examples include the Huawei Connect Conference Kunpeng Zone, the Huawei Partner & Developer Conference Kunpeng Zone, the China Information Technology Innovation Competition Kunpeng Track, and the Huawei Developer Conference Kunpeng Zone. It also showcases the latest official Kunpeng updates and news, providing users with firsthand information about Kunpeng.

6.2.8 Support and Service Section

Based on computing products and solutions, experts from various fields use professional tools to provide excellent end-to-end professional services, accelerating the realization of users' business value; face-to-face instruction from senior lecturers enables computing talent development and helps enterprises achieve digital transformation.

6.3 Kunpeng Developers - Kunpeng Development Kit (DevKit)

Aimed at the entire R&D workflow, it improves application migration and optimization efficiency, accelerates native development, and provides development enablement tools covering all aspects of code migration, development debugging, compilation, testing, optimization, and diagnostics. This facilitates developers in quickly creating high-performance software compatible with Kunpeng. You

can view the latest updates on the Kunpeng Development Kit at the following link: https://www.hikunpeng.com/document/detail/zh/kunpengdevps/wtsnew/wtsnew-kunpengdevps.html



图6-3 Kunpeng Development Kit **DevKit**

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6.3.1 Kunpeng Code Migration Tool

The code migration tool can scan and automatically analyze the software to be migrated, providing a migrationability assessment report; it can also perform source code analysis on the software to be migrated, accurately locate the code that needs to be migrated, and provide user-friendly migration guidance or one-click code replacement; it also supports refactoring x86 software packages into Kunpeng platform software packages, one-click migration of specialized software, and other Kunpeng affinity analyses.

6.3.2 Kunpeng Development Framework

The Kunpeng Development Framework is a tool that enables users to develop on Kunpeng servers. It fully leverages the various computing power types of the Kunpeng platform and superior performance third-party components, providing capabilities such as Kunpeng engineering wizards, heuristic programming, and code affinity checks. It allows for one-click import of Kunpeng acceleration libraries and rapid construction of Kunpeng application software frameworks, helping developers to more easily develop Kunpeng applications and enabling efficient innovation.
The latest developments of the Kunpeng Development Framework can be viewed through the following link.

6.3.3 Kunpeng Compilation Toolchain

Bisheng Compiler: Developed based on the open-source LLVM and optimized and improved, it also supports the Fortran language front-end and is a high-performance compiler for the Kunpeng platform.

Bisheng JDK: Based on OpenJDK, Bisheng JDK is a stable, reliable, high-performance, and easy-to-test JDK for Java developers and users. Bisheng JDK has been deeply optimized and enhanced for the Kunpeng platform.

GCC for openEuler: GCC for openEuler is a compiler toolchain (including compiler, assembler, and linker) developed based on the open-source GCC. It is released as open source by the openEuler community and its binary package is provided free of charge through the Kunpeng community. It supports multiple processor architectures, including ARM and x86.

6.3.4 Cloud Testing Service

It provides developers with compatibility testing, reliability testing, security testing, functional testing, and performance testing services based on the Kunpeng platform, helping users quickly identify and locate problems in application runtime.

6.3.5 Kunpeng Performance Analysis Tool

The performance analysis tool supports system performance analysis, Java performance analysis, and system diagnostics on the Kunpeng platform. It provides a comprehensive overview of the system and performance data collection and analysis capabilities for common application scenarios, and offers optimization suggestions based on an expert tuning system. It also provides a tuning assistant to guide users in quickly optimizing system performance.

6.3.6 Huawei Dynamic Binary Translation Tool (ExaGear)

ExaGear is a self-developed dynamic binary translation tool by Huawei. It translates x86 application instructions into ARM64 instructions at runtime and executes them, thereby enabling Linux x86 applications to run on ARM64 servers without recompilation. This helps customers quickly migrate Linux x86 applications without source code to ARM servers and ensures stable and reliable operation.

6.4 Kunpeng Developers - Kunpeng Application Enablement Kit BoostKit

The Kunpeng Application Enablement Suite BoostKit is based on full-stack optimization of hardware, basic software, and application software, providing high-performance open-source components, basic acceleration packages, and application acceleration packages to enable ultimate application performance.

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图6-4 Kunpeng Application Enablement Suite **BoostKit**

6.4.1 Big Data Scenarios

Focusing on challenges such as low query efficiency and difficulty in performance optimization for big data, this product provides open-source enablement and tuning of big data components, basic acceleration packages such as IO intelligent prefetching, application acceleration packages such as Spark algorithm acceleration libraries, and the open-source openLooKeng query engine to improve the efficiency of big data analysis.

6.4.2 Distributed Storage Scenarios

Focusing on the challenges of low Ceph performance and high cost, we provide Ceph parameter optimization, MD5 digest algorithm, IO passthrough and other features to improve system performance, and reduce storage costs through compression algorithms, Zlib compression, EC Turbo and other features to provide a cost-effective storage solution.

6.4.3 Database Scenarios

Addressing challenges such as low query efficiency in open-source MySQL OLAP and lock performance issues in OLTP scenarios, this software provides basic acceleration packages for MySQL AP performance acceleration and TP lock performance optimization. It deeply optimizes OLAP query analysis efficiency and OLTP online transaction processing capabilities, fully leveraging multi-core computing power.

6.4.4 Virtualization Scenarios

Focusing on pain points such as low virtualization performance, high network overhead, and severe resource fragmentation, this guide provides virtualization open-source enablement and tuning, improves system performance through features such as V-Turbo and OVS acceleration, reduces resource fragmentation using features such as NUMA memory interleaving, and fully leverages the multi-core advantages of Kunpeng.

6.4.5 ARM Native Scenarios

Supports seamless cloud migration of mobile applications; the complete solution includes hardware, Host OS, ExaGear, and virtualization solution (Kbox).

Guest OS, End-to-Cloud Collaboration Engine.

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6.4.6 Web Application Scenarios

Focusing on issues such as HTTPS connection performance for web applications and availability of open-source web components, this guide provides an enablement and optimization guide for open-source web components. By enabling the Kunpeng RSA acceleration engine, it helps customers improve website security and achieve the ultimate user experience for HTTPS access.

6.4.7 CDN

Focusing on issues such as low CDN node throughput and availability of open-source components, this document provides an enablement and optimization guide for mainstream CDN open-source software. By enabling the Kunpeng RSA acceleration engine and providing NUMA optimization and other methods, it fully leverages the advantages of the Kunpeng multi-core architecture to achieve higher throughput and lower latency.

6.4.8 HPC High-Performance Computing Scenarios

Based on the Danube Suite, it provides efficient resource management and scheduling capabilities for large-scale HPC clusters, meeting the requirements of cluster resource utilization, job scheduling throughput, and multi-cluster management in HPC scenarios.

6.5 Kunpeng Developer - Learning Column

6.5.1 Kunpeng Online Courses

Kunpeng Online Courses are a collection of teaching courses designed by the Kunpeng community and presented to Kunpeng software developers in a MOOC format. The courses aim to help developers understand the fundamentals of the Kunpeng system, hardware and software products, cloud services, and learn how to migrate x86 applications to the Kunpeng platform. Currently, the platform has over 110 courses available, covering Kunpeng DevKit, etc.
Kunpeng BoostKit, openEuler, openGauss, openLooKeng, HPC, and many other aspects.

6.5.2 Kunpeng Online Experiment

Kunpeng Online Experiments: Following a detailed experiment manual, users can complete the guided operations step-by-step on the online virtualization platform provided by the Kunpeng community, quickly experiencing online cloud services and implementing, debugging, and verifying cloud services in the cloud. Lightweight, easy to learn, and providing a good user experience, the experiment interface is shown in the following figure:

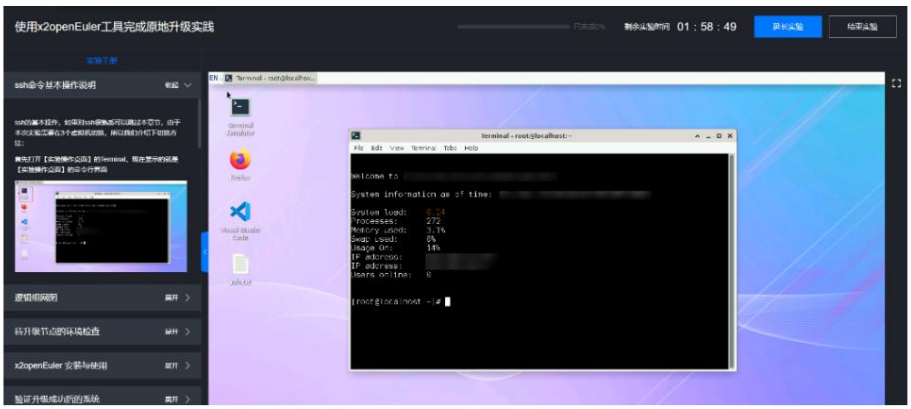


图6-5 Kunpeng Online Experiment Interface

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6.5.3 Kunpeng Authentication System

The Kunpeng certification system is a carefully designed tiered training and certification system tailored to different users and product categories, offering professional certification and micro-certification.

The Kunpeng application developer professional certification is currently divided into HCIA level and HCIP level. The HCIA level certification is for engineers who can deploy and migrate business applications on the Kunpeng computing platform, perform performance testing and optimization, and handle common problems encountered during application deployment. The HCIP level certification is for senior engineers who can develop applications based on the Kunpeng architecture and port advanced Kunpeng applications.

MicroCertification offers a one-stop online learning, experimentation, and exam service, allowing even beginners to learn cutting-edge technologies, quickly acquire scenario-based skills, and obtain authoritative certifications. The MicroCertification process is: online learning -> hands-on experiments -> purchase certification -> online exam -> obtain certificate. Upon completion of MicroCertification, you can enjoy related benefits.



图6-6 Kunpeng Certification System

6.6 Partners - Kunpeng Ecosystem Partner Network and Partner Program

6.6.1 Kunpeng Ecosystem Partners

The Kunpeng Eco-Partner Network (KEPN) is a series of ecosystem partner programs launched by Huawei around the Kunpeng industry technology roadmap. By providing resources such as learning, building, listing, and sales support, it is committed to helping Kunpeng ecosystem partners build industrial competitiveness, connect with customers to create business opportunities, and grow together and create the future in an open, innovative, collaborative, and mutually beneficial ecosystem.

6.6.2 Advantages of the Kunpeng Ecosystem Partner Network

Capability Enhancement: Receive exclusive technical and management training from ecosystem partners to rapidly improve their ability to build and innovate products and solutions around the Kunpeng full-stack technology, as well as their operational management capabilities related to enterprise growth;

Business Growth: Gain access to connect with Huawei customers and partners, and share the industry's digital business growth opportunities brought about by independent innovation technologies;

Marketing resources: Access to a series of marketing events, industry summits, developer technology salons, and special seminars hosted by Huawei and Kunpeng ecosystem partners;

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Industry talent: Gain access to Kunpeng full-stack industry technology talents carefully cultivated by high-quality partner universities across the country, and accelerate the construction of enterprise product and solution competitiveness;

6.6.3 Types of Kunpeng Ecosystem Partners

Kunpeng ecosystem partners are categorized into complete machine hardware partners, application software partners, basic software partners, all-in-one solution partners, and ecosystem operation partners. When applying to join, ecosystem partners can choose the appropriate type to obtain the corresponding status and support.

6.6.4 Kunpeng Ecosystem Partner Network Member Levels

Certified Level: This level applies to all partners. After completing identity registration and meeting the certification level requirements, partners can become certified level partners and enjoy certification level learning and training benefits;

Preferred Tier: This tier is for partners who hold a leading position in their industry or specialized field and have made significant contributions to the Kunpeng industry. In addition to the benefits of the Certified Partner tier, they will also receive more learning and training resources, including digital leadership training, NRE incentives, MDF incentives, and invitations to participate in annual company-wide flagship events.

Leading Tier: This tier applies to partners who hold a dominant position in the industry or specialized field and have made outstanding contributions to the Kunpeng industry. In addition to the benefits of the Preferred Partner tier, they will receive greater incentive resources, as well as benefits such as new product trials, roadmap sharing and synchronous development, and support services for building partner-specific capabilities.

Kunpeng Spreading Wings Partner Program:

The Kunpeng Wings Partner Program is an ecosystem partner program launched by Huawei around the Kunpeng computing platform. It aims to help partners integrate the full stack of technologies and brand resources of the Kunpeng computing industry, build product and solution competitiveness, connect customers with the Kunpeng ecosystem, and create unlimited business opportunities.

Euler Sailing Partner Program:

The Euler Sailing Partner Program is an ecosystem partner program launched by Huawei around openEuler. It aims to ensure the healthy and orderly development of the openEuler ecosystem, accelerate the development of products and solutions by ecosystem partners based on the openEuler series of products and components, and enable the development and commercial success of the partner ecosystem.

Kunpeng Scientific Research and Innovation Enabling Program:

The Kunpeng Scientific Research and Innovation Enablement Program is an ecosystem partner program launched by Huawei around the Kunpeng computing infrastructure. It aims to enable domestic universities and research institutes to carry out scientific and technological research and independent and reliable software development and technological breakthroughs based on the Kunpeng computing infrastructure.

6.7 Support and Information

Kunpeng Forum:

The Kunpeng Forum is a hub for technical experts, industry professionals, and enthusiastic developers to exchange ideas. These experts and masters sit down and discuss, sharing practical experience and technical perspectives. Currently, it has 12 channels covering technical knowledge and activities within the Kunpeng ecosystem, helping developers quickly broaden their technical horizons.

Kunpeng Blog:

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The Kunpeng Forum is a hub for technical experts, industry professionals, and enthusiastic developers to exchange ideas. These experts and masters sit down and discuss, sharing practical experience and technical perspectives. Currently, it has 12 channels covering technical knowledge and activities within the Kunpeng ecosystem, helping developers quickly broaden their technical horizons.

News and Information:

The latest updates and news released by the official Kunpeng team provide users with first-hand information about Kunpeng.

Developer Profile Monthly:

The Developer Resources Monthly Bulletin regularly summarizes the latest developments in enabling the Kunpeng ecosystem, high-quality and popular documents, and newly launched documents, and compiles popular cases in various technical fields, leading users to quickly understand the latest developments in developer resources.

Support and Services:

Based on computing products and solutions, experts from various fields use professional tools to provide excellent end-to-end professional services, accelerating the realization of users' business value; face-to-face instruction from senior lecturers enables computing talent development and helps enterprises achieve digital transformation.